

8.223: CLASSICAL MECHANICS II

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This document is a summary of course 8.223 Classical Mechanics II, taught in January 2026 by Professor Michael Williams of the Massachusetts Institute of Technology. This document was created for the sole purpose of exam review for this course; it uses the official course notes and external sources cited below.

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Outside Sources

- Landau, L. D. & Lifshitz, E. M. “Теоретическая Физика: Механика” 3rd Ed. (1975)

1 The Lagrangian

1.1 Generalized Coordinates

One of the fundamental concepts of Mechanics is that of the **particle**. This represents a body whose dimensions are negligible when its motion is described. Every particle has a position in space defined by a *position vector* $\mathbf{r}$, the vector originating from the origin of the chosen coordinate system to the position of the particle. We call its time derivative $\mathbf{v} = \dot{\mathbf{r}}$ the *velocity*¹ of the particle. The second time derivative $\mathbf{a} = \ddot{\mathbf{r}}$ is the *acceleration* of the particle.

It is clear that in order to define the positions of N particles, one needs N vectors $\mathbf{r}$ and thus $3N$ scalars. These are called the “degrees of freedom” of the system containing these N particles.

Definition 1.1.1 (Generalized Coordinate)

Let a system have s degrees of freedom. Any set of quantities $\{q_1, q_2, \dots, q_s\}$ that is sufficient to define the position of this system are *generalized coordinates* of the system. The set of time derivatives $\{\dot{q}_1, \dot{q}_2, \dots, \dot{q}_s\}$ is the set of *generalized velocities*.

Given the set of coordinates q and velocities $\dot{q}$, the accelerations $\ddot{q}$ are uniquely defined, and thus the motion can be found from the *equations of motion*, second-order differential equations that relate acceleration to velocity and position.

1.2 The Action

The phenomena observed in mechanical systems can be summarized by a single principle.

Axiom 1.2.1 (Principle of Least Action/Hamilton’s Principle)

Every mechanical system is characterized by a specific function $L(q, \dot{q}, t)$. The motion of the system must satisfy the following condition:

Let the position of the system at time t_1 be given by $q^{(1)}$; let the position of the system at time t_2 be similarly given by $q^{(2)}$. The system must move between the positions $q^{(1)}$ and $q^{(2)}$ such that the integral below takes a minimal value.

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad (1.2.1)$$

The function L is called the *Lagrangian* of the system, and the integral S is called the *action*.

Given this principle, it is possible to impose a mathematical constraint on the function L . First, we observe that adding a constant does not change whether a path is the one that minimizes S or not:

$$L' \equiv L + k \Rightarrow S' = \int_{t_1}^{t_2} L'(q, \dot{q}, t) dt = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt + k(t_2 - t_1) = S + k' \quad (1.2.2)$$

and therefore there is no dependence on x . Likewise, adding a total time derivative of a function $f(x, t)$ does not change the conclusion:

$$L' \equiv L + \frac{d}{dt}f(x, t) \Rightarrow S' = \int_{t_1}^{t_2} L'(q, \dot{q}, t) dt = \int_{t_1}^{t_2} L(q, \dot{q}, t) + \frac{d}{dt}f(x, t) dt = S + f(x, t) \Big|_{t_1}^{t_2} = S + k' \quad (1.2.3)$$

again a constant.

We can find a differential equation that governs the function L . We will vary the path $q(t)$ such that $q(t_1)$ and $q(t_2)$ do not change, but

$$q(t) \rightarrow q(t) + \delta q(t) \quad (1.2.4)$$

within the interval $[t_1, t_2]$. This change $\delta q(t)$ is called a *variation* of q ; it satisfies

$$\delta q(t_1) = \delta q(t_2) = 0 \quad (1.2.5)$$

¹Also called “vector velocity” in the literature.

We find the change in S caused by this variation:

$$\delta S = \int_{t_1}^{t_2} L(q + \delta q, \dot{q} + \delta \dot{q}, t) dt - \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \simeq \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt \quad (1.2.6)$$

If S is minimized, $\delta S = 0$, and therefore

$$\int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt = 0 \quad (1.2.7)$$

We integrate by parts, observing that $\delta \dot{q} = d(\delta q)/dt$:

$$\delta S = \left(\frac{\partial L}{\partial \dot{q}} \delta q \right) \Big|_{t_1}^{t_2} + \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q dt = 0 \quad (1.2.8)$$

The first term is zero because δq is zero at the boundaries (t_1, t_2) . Therefore, the remaining integral must be zero for all δq , and thus the only way this is possible is if

$$\boxed{\frac{\partial L}{\partial q} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right)} \quad (1.2.9)$$

If there are s degrees of freedom, it should be possible to vary each function $q_i(t)$ independently of the others, and thus we obtain s such equations.

Proposition 1.2.2 (Euler–Lagrange Equation)

Let a system have s degrees of freedom. Let L be the Lagrangian of this system. Then, for each generalized coordinate q_i with $i \in \{1 \dots s\}$, we have

$$\boxed{\frac{\partial L}{\partial q_i} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right)} \quad (1.2.10)$$

Due to the Principle of Least Action, the physics of a system does not change if the Lagrangian L is multiplied by a real constant, nor if we add a total time derivative to the Lagrangian, because if $L = L_0 + dF/dt$,

$$S - S_0 = \int_{t_1}^{t_2} \frac{dF}{dt} dt = F(q^{(2)}, t_2) - F(q^{(1)}, t_1) \quad (1.2.11)$$

which is a constant, and therefore gives zero under variation.

1.3 Lagrangian of a Particle

We can characterize the function L , the Lagrangian, by relying on certain basic principles and a few axioms.

Axiom 1.3.1 (Fundamental Principles)

1. There exist inertial reference frames – there always exists a reference frame in which the center of mass of a system has zero velocity.
2. Space and time are both homogeneous and isotropic.

In classical mechanics, we add the Galilean assumptions (which are unfortunately all false²):

Axiom 1.3.2 (Galilean Assumptions)

1. Time is absolute. Every elementary interaction is instantaneous.
2. Nature is deterministic. There therefore exist equations of motion that can be used to fully describe the behavior of a system.
3. There exist no fundamental constants of nature.

Consequently, we obtain four mathematical facts for the action of a particle in one dimension.

$$\begin{array}{ll} \text{(a) } \dot{x} \mapsto \dot{x} + v & \Rightarrow \delta S = ct \\ \text{(b) } x \mapsto x + \delta x & \Rightarrow \delta S = ct \end{array} \qquad \begin{array}{ll} \text{(c) } t \mapsto t + \delta t & \Rightarrow \delta S = ct \\ \text{(d) } x \mapsto -x \ \& \ \dot{x} \mapsto -\dot{x} & \Rightarrow \delta S = ct \end{array}$$

The first represents the equivalence of inertial reference frames; the second represents the homogeneity of space; the third represents the homogeneity of time; the fourth represents the isotropy of space. We observe that the Lagrangian can depend only on $\dot{x}^2$.

$$L = \gamma(m)\dot{x}^2 \Rightarrow L' = \gamma(m)(\dot{x} + v)^2 = \gamma(m)(\dot{x}^2 + 2\dot{x}v + v^2) \quad (1.3.1)$$

The last term is a constant, and the second-to-last term is the total derivative of $2vx$, so it can also be omitted. Because this choice of L does not depend directly on x or t , properties (b) and (c) hold. Because L is even in $\dot{x}^2$, property (d) also holds. Thus, all properties are satisfied.

Proposition 1.3.3 (Lagrangian of a Free Particle)

Let a free particle have mass m . The Lagrangian of this particle is given by

$$L = \frac{m\dot{x}^2}{2} \quad (1.3.2)$$

We will soon see why $\gamma(m) = m/2$.

By applying Lagrange's equation, we obtain simply Newton's First Law:

$$\frac{\partial L}{\partial x} = 0 \quad ; \quad \frac{\partial L}{\partial \dot{x}} = 2\gamma(m)\dot{x} \quad (1.3.3)$$

Therefore,

$$2\gamma(m)\ddot{x} = 0 \Rightarrow \ddot{x} = 0 \quad (1.3.4)$$

and thus the velocity of a free particle is constant.

Let us take two particles. In the absence of an interaction, the Lagrangians add; if there is an interaction, it is also taken into account:

$$L_{\text{tot}} = L_1 + L_2 + I(\mathbf{r}_1, \mathbf{r}_2) \quad (1.3.5)$$

We have identified that $L_1 = \gamma(m)\dot{r}_1^2$ and $L_2 = \gamma(m)\dot{r}_2^2$. The same constraints (a)–(d) must apply to L_{tot} . Therefore, we propose that I takes the form

$$I(\mathbf{r}_1, \mathbf{r}_2) = f(|\mathbf{r}_1, \mathbf{r}_2|) \quad (1.3.6)$$

for f a function that will depend on the type of interaction. This satisfies property (d), because the absolute value takes care of that; property (b) leads to

$$|x_1 - x_2| \mapsto |x_1 + \delta x - x_2 + \delta x| = |x_1 - x_2| \quad (1.3.7)$$

Thus δS is zero, satisfying property (b). For simplicity, let us remain in the one-dimensional case. Thus,

$$L = \gamma(m)\dot{x}_1^2 + \gamma(m)\dot{x}_2^2 - U(|x_1 - x_2|) \quad (1.3.8)$$

²It turns out that these assumptions are approximations of real life when $c \rightarrow \infty$ and $\hbar \rightarrow 0$.

Let us use Lagrange's equation:

$$\frac{\partial L}{\partial x_i} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) \quad (1.3.9)$$

We find that

$$\frac{\partial L}{\partial x_i} = -\frac{\partial U}{\partial x_i} \quad ; \quad \frac{\partial L}{\partial \dot{x}_i} = 2\gamma(m)\dot{x}_i \quad (1.3.10)$$

Therefore, Lagrange tells us that

$$-\frac{\partial U}{\partial x_i} = \frac{d}{dt} (2\gamma(m)\dot{x}_i) = 2\gamma(m)\ddot{x}_i \quad (1.3.11)$$

We recognize the form of Newton's Second Law; if $F_i \equiv -\frac{\partial U}{\partial x_i}$ and we set $\gamma(m) = \frac{m}{2}$,

$$\boxed{F_i = m\ddot{x}_i} \quad (1.3.12)$$

in Newton's words. We can also recover the Third Law, by observing that for this interaction,

$$F_1 = -\frac{\partial}{\partial x_1} U(|x_1 - x_2|) = \mp U' \quad ; \quad \boxed{F_2} = -\frac{\partial}{\partial x_2} U(|x_1 - x_2|) = \pm U' = \boxed{-F_1} \quad (1.3.13)$$

Proposition 1.3.4 (Newton's Laws)

Starting from the assumptions of homogeneity and isotropy of space and time, together with the non-relativistic (Galilean) assumptions, it is possible to recover the Three Laws of Newton from the Principle of Least Action (also called Hamilton's Principle).

1.4 Examples

1.4.1 A Mass and a Spring

Let us take a simple example, a block of mass m attached to a spring with constant k . The potential of a spring is

$$U = \frac{kx^2}{2} \quad (1.4.1)$$

if we take the origin ($x = 0$) to be where the spring exerts no force. Thus, the Lagrangian is

$$L = T - U = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2} \quad (1.4.2)$$

Lagrange's equation gives

$$\frac{\partial L}{\partial x} = \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \quad (1.4.3)$$

$$\Rightarrow -kx = m\ddot{x} \quad (1.4.4)$$

$$\Rightarrow \ddot{x} + \frac{k}{m}x = 0 \quad (1.4.5)$$

which is indeed Hooke's law.

1.4.2 A Sphere on a Ramp

Let us take a second example, a spherical ball that rolls without slipping on a ramp that can slide on a flat, frictionless surface, as shown below.

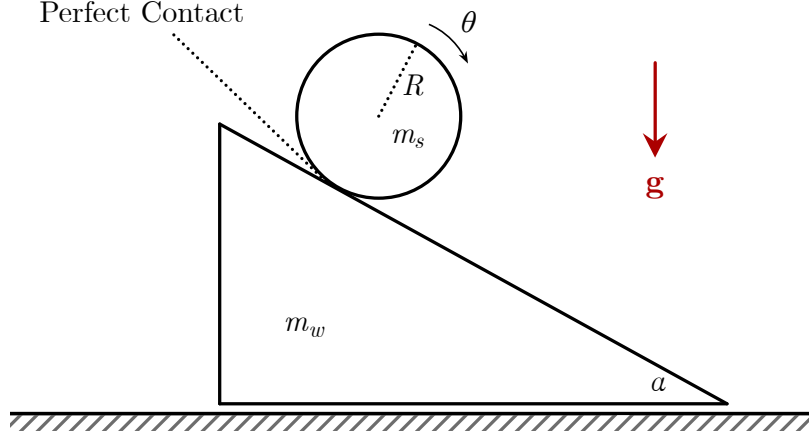


Figure 1. The Ramp and its Sphere.

This system is two-dimensional; we therefore have four degrees of freedom (by default). However, we have the constraint of perfect contact, which allows us to eliminate one degree of freedom of the sphere, expressing its Cartesian coordinates using the angle θ . Because the triangle does not change its vertical position, we can also eliminate the y_w degree of freedom.

We therefore have

$$\mathbf{r}_w = \langle x_w, 0 \rangle \quad ; \quad \mathbf{r}_s = \langle x_s, y_s \rangle \quad (1.4.6)$$

If we express x_s and y_s in terms of θ , we obtain

$$\mathbf{r}_w = \langle x, 0 \rangle \quad ; \quad \mathbf{r}_s = \langle x + R\theta \cos \alpha, -R\theta \sin \alpha \rangle \quad (1.4.7)$$

Thus, their time derivatives are

$$\dot{\mathbf{r}}_w = \langle \dot{x}, 0 \rangle \quad ; \quad \dot{\mathbf{r}}_s = \langle \dot{x} + R\dot{\theta} \cos \alpha, -R\dot{\theta} \sin \alpha \rangle \quad (1.4.8)$$

The kinetic energy of the sphere is

$$T_s = \frac{m_s \dot{\mathbf{r}}_s^2}{2} + \frac{I \dot{\theta}^2}{2} \quad (1.4.9)$$

with $I = \frac{2m_s R^2}{5}$; thus,

$$T_s = \frac{m_s}{2} \left(\dot{x}^2 + 2R\dot{x}\dot{\theta} \cos \alpha + R^2 \dot{\theta}^2 \cos^2 \alpha + R^2 \dot{\theta}^2 \sin^2 \alpha + \frac{2R^2}{5} \dot{\theta}^2 \right) \quad (1.4.10)$$

$$= \frac{m_s}{2} \left(\dot{x}^2 + 2R\dot{x}\dot{\theta} \cos \alpha + \frac{7}{5} R^2 \dot{\theta}^2 \right) \quad (1.4.11)$$

The kinetic energy of the triangle is

$$T_w = \frac{m_w \dot{x}^2}{2} \quad (1.4.12)$$

The potential energy is that of a uniform gravitational field, and thus

$$U = mg(\mathbf{x}_s \cdot \hat{\mathbf{y}}) = -m_s g R \theta \sin \alpha \quad (1.4.13)$$

The Lagrangian becomes

$$L = T - U = \frac{m_w \dot{x}^2}{2} + \frac{m_s}{2} \left(\dot{x}^2 + 2R\dot{x}\dot{\theta} \cos \alpha + \frac{7}{5} R^2 \dot{\theta}^2 \right) + m_s g R \theta \sin \alpha \quad (1.4.14)$$

We compute the partial derivatives:

$$\frac{\partial L}{\partial \theta} = m_s g R \sin \alpha \quad ; \quad \frac{\partial L}{\partial \dot{\theta}} = m_s \left(R \dot{x} \cos \alpha + \frac{7}{5} R^2 \dot{\theta} \right) \quad (1.4.15)$$

and also

$$\frac{\partial L}{\partial x} = 0 \quad ; \quad \frac{\partial L}{\partial \dot{x}} = m_w \dot{x} + m_s (\dot{x} + 2R\dot{\theta} \cos \alpha) \quad (1.4.16)$$

Thus, for θ we obtain

$$\cancel{m_s} g R \sin \alpha = \cancel{m_s} \left(R \ddot{x} \cos \alpha + \frac{7}{5} R^2 \ddot{\theta} \right) \quad (1.4.17)$$

$$\Rightarrow g R \sin \alpha = R \ddot{x} \cos \alpha + \frac{7}{5} R^2 \ddot{\theta} \quad (1.4.18)$$

$$\Rightarrow \ddot{\theta} \cdot \frac{7R^2}{5} = R(g \sin \alpha - \ddot{x} \cos \alpha) \quad (1.4.19)$$

For x we obtain

$$0 = m_w \ddot{x} + m_s (\ddot{x} + 2R\ddot{\theta} \cos \alpha) \Rightarrow \ddot{x} = -2m_s \frac{R\ddot{\theta} \cos \alpha}{m_s + m_w} \quad (1.4.20)$$

Thus,

$$\Rightarrow \ddot{\theta} \cdot \frac{7R^2}{5} = R \left(g \sin \alpha + \frac{2m_s}{m_s + m_w} R \ddot{\theta} \cos^2 \alpha \right) \quad (1.4.21)$$

$$\Rightarrow \ddot{\theta} \left(\frac{7R^2}{5} - \frac{2m_s}{m_s + m_w} R^2 \cos^2 \alpha \right) = R g \sin \alpha \quad (1.4.22)$$

$$\Rightarrow \ddot{\theta} = \frac{g \sin \alpha}{R \left(\frac{7}{5} - \frac{2m_s}{m_s + m_w} \cos^2 \alpha \right)} \quad (1.4.23)$$

This is the equation of motion for $\theta(t)$. It will be quadratic, with half of this value of $\ddot{\theta}$ as the coefficient of t^2 . Similarly, $\ddot{x}$ will be a constant, given by

$$\ddot{x} = -\frac{2m_s R \cos \alpha}{m_s + m_w} \cdot \frac{g \sin \alpha}{R \left(\frac{7}{5} - \frac{2m_s}{m_s + m_w} \cos^2 \alpha \right)} \quad (1.4.24)$$

and therefore $x(t)$ will also be quadratic.

1.4.3 Two Blocks and a String

We can make a third example. Consider two blocks with masses m_1 and m_2 . The block m_1 hangs from a string of length l attached to the other mass m_2 , which is on a table and is free to rotate around the string. Let us find the equations of motion of the masses.

The degrees of freedom of the masses are three: we describe the position of the mass m_2 using polar coordinates:

$$\mathbf{r}_2 = \langle r \cos \theta, r \sin \theta, 0 \rangle \quad (1.4.25)$$

The mass m_1 is described by

$$\mathbf{r}_1 = \langle 0, 0, z \rangle \quad (1.4.26)$$

The time derivatives are

$$\dot{\mathbf{r}}_1 = \langle 0, 0, \dot{z} \rangle \quad (1.4.27)$$

$$\dot{\mathbf{r}}_2 = \langle \dot{r} \cos \theta - r \sin \theta \dot{\theta}, \dot{r} \sin \theta + r \cos \theta \dot{\theta}, 0 \rangle \quad (1.4.28)$$

We compute the kinetic energy, and therefore we must square these vectors:

$$\dot{\mathbf{r}}_2^2 = \dot{r}^2 \cos^2 \theta - 2\dot{r}r \sin \theta \cos \theta \dot{\theta} + r^2 \sin^2 \theta \dot{\theta}^2 + \dot{r}^2 \sin^2 \theta + r^2 \cos^2 \theta \dot{\theta}^2 + 2\dot{r} \sin \theta r \cos \theta \dot{\theta} \quad (1.4.29)$$

$$= \dot{r}^2 + r^2 \dot{\theta}^2 \quad (1.4.30)$$

$$\dot{\mathbf{r}}_1^2 = \dot{z}^2 \quad (1.4.31)$$

We relate the degrees of freedom z and r in the following way:

$$-z + r = l \Rightarrow z = r - l \Rightarrow \dot{z} = \dot{r} \quad (1.4.32)$$

We therefore find that

$$\dot{\mathbf{r}}_1^2 = \dot{r}^2 \quad (1.4.33)$$

The kinetic energy of the first mass is therefore

$$T_1 = \frac{m_1 \dot{\mathbf{r}}_1^2}{2} = \frac{m_1}{2} \dot{r}^2 \quad (1.4.34)$$

For the second mass:

$$T_2 = \frac{m_2 \dot{\mathbf{r}}_2^2}{2} = \frac{m_2}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) \quad (1.4.35)$$

The potential energy is that of the hanging mass in a gravitational field, so $U = mg(r - l)$. Thus,

$$L = T_1 + T_2 - U = \frac{m_1}{2} \dot{r}^2 + \frac{m_2}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + mg(l - r) \quad (1.4.36)$$

We apply the Euler–Lagrange equation for both coordinates to obtain

$$\boxed{\ddot{r} = \frac{\dot{\theta}^2 r m_2 - m_1 g}{m_1 + m_2}} \quad ; \quad \boxed{\ddot{\theta} = -\frac{2\dot{\theta}\dot{r}}{r}} \quad (1.4.37)$$

2 Integrals of the Motion

2.1 Energy

Let a system have s degrees of freedom. Because the equations of motion arise from s second-order differential equations, $2s$ integration constants are required to find the exact equations of motion. Because time does not appear explicitly in the differential equations, the choice of the origin of the time axis is completely arbitrary, and thus we can eliminate one of the $2s$ constants. Therefore, there will be $2s - 1$ constants that remain fixed during the motion. These are the *integrals of motion*.

Of these $2s - 1$, there are 7 that are very important. We will find them one by one, starting from the principles of homogeneity and isotropy of space and time, beginning with the homogeneity of time. Given this fact, the Lagrangian L of a system cannot depend explicitly on time. Thus, the total time derivative of the Lagrangian is given by

$$\frac{dL}{dt} = \sum_i \frac{\partial L}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \frac{\partial L}{\partial t} \quad (2.1.1)$$

We apply Lagrange's equation, which tells us that

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \quad (2.1.2)$$

Thus,

$$\frac{dL}{dt} = \sum_i \dot{q}_i \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} + \sum_i \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i = \sum_i \frac{d}{dt} \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) \quad (2.1.3)$$

Therefore, rearranging,

$$0 = \sum_i \frac{d}{dt} \left(\dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \right) - \frac{dL}{dt} = \frac{d}{dt} \left(\sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L \right) \quad (2.1.4)$$

We call the quantity whose derivative we compute above the *energy* of the system.

Definition 2.1.1 (Energy of a System)

Let a system have s degrees of freedom. Let L be its Lagrangian. Suppose that there is no explicit time dependence in the expression of the Lagrangian, i.e. $\partial L / \partial t = 0$. The total energy, E , of this system is given by

$$E = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L \quad (2.1.5)$$

We can demonstrate several essential facts.

Proposition 2.1.2 (Conservation of Energy)

Let a system have s degrees of freedom. Let L be its Lagrangian. The energy E is therefore constant; that is, $\dot{E} = 0$.

Proof

Proved above. □

Proposition 2.1.3 (Form of the Energy)

Let a system have s degrees of freedom. Let L be its Lagrangian. Suppose that the system is subject only to conservative fields, such that L takes the form $L = T - U$. Then, the energy takes the form

$$E = T + U \quad (2.1.6)$$

Proof

If the field is conservative, the potential energy takes the form $U = U(\mathbf{x})$. Thus, the first term in E is

$$\sum_i \left(\frac{\partial L}{\partial \dot{x}_i} \right) \dot{x}_i = \sum_i \left(\frac{\partial}{\partial \dot{x}_i} \frac{m \dot{x}_i^2}{2} \right) \dot{x}_i = \sum_i m \dot{x}_i^2 = 2T \quad (2.1.7)$$

Thus,

$$\boxed{E = 2T - (T - U) = T + U} \quad (2.1.8)$$

as desired. $\square$

Proposition 2.1.4 (Non-Conservative Forces in the EoM)

Let $\{F_i^{(k)}\}$ be a set of non-conservative forces acting on a system with s degrees of freedom, such that the forces indexed by k correspond to the k -th coordinate. Then, if we know which coordinates they act on, it is possible to simply add them to the equations of motion (EoM) as follows:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_k} - \frac{\partial L}{\partial q_k} = \sum_i F_i^{(k)} \quad (2.1.9)$$

For example, consider a single-particle system attached to a spring (constant k) with one degree of freedom (x), such that the damping force is given by

$$F_{\text{am}} = -m\gamma\dot{x} - \alpha\dot{x}|\dot{x}| \quad (2.1.10)$$

Thus,

$$L_{\text{conservatif}} = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2} \quad (2.1.11)$$

which gives

$$m\ddot{x} + kx = -m\gamma\dot{x} - \alpha\dot{x}|\dot{x}| \Rightarrow m\ddot{x} + kx + m\gamma\dot{x} + \alpha\dot{x}|\dot{x}| = 0 \quad (2.1.12)$$

as desired.

2.2 Momentum

The second essential conserved quantity comes from the homogeneity of space. Consider an infinitesimal displacement ϵ ; we will obtain the necessary condition for the Lagrangian to be invariant.

Each coordinate of a particle in the system $\mathbf{r}_i$ will be displaced by ϵ such that

$$\mathbf{r} \mapsto \mathbf{r} + \epsilon \quad (2.2.1)$$

The variation in L will be

$$\delta L = \sum_a \frac{\partial L}{\partial \mathbf{r}_a} \cdot \delta \mathbf{r}_a = \epsilon \cdot \sum_a \frac{\partial L}{\partial \mathbf{r}_a} \quad (2.2.2)$$

Because ϵ was arbitrary, we have

$$\sum_a \frac{\partial L}{\partial \mathbf{r}_a} = \mathbf{0} \quad (2.2.3)$$

We apply Lagrange's equation for $\mathbf{r}_a$ as coordinates $\langle q_{a1}, q_{a2}, q_{a3} \rangle$:

$$\frac{\partial L}{\partial \mathbf{r}_a} = \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{r}}_a} \quad (2.2.4)$$

Thus,

$$\frac{d}{dt} \sum_a \frac{\partial L}{\partial \dot{\mathbf{r}}_a} = \mathbf{0} \quad (2.2.5)$$

We call the quantity differentiated above the (*translational*) *momentum*.

Definition 2.2.1 (Momentum)

Let a system have s degrees of freedom. Let L be its Lagrangian. The *momentum* of the system, $\mathbf{P}$, is given by

$$\mathbf{P} = \sum_a \frac{\partial L}{\partial \dot{\mathbf{r}}_a} \quad (2.2.6)$$

Proposition 2.2.2 (Conservation of Momentum)

Let a closed system have N particles. Let L be its Lagrangian. Let $\mathbf{P}$ be the total momentum of the system. In this case, $\dot{\mathbf{P}} = \mathbf{0}$.

Proof

Proved above. □

Proposition 2.2.3 (Form of the Momentum for a System of Particles)

Let a system have N particles (and thus $s = 3N$ degrees of freedom). Suppose there is no external field. The momentum $\mathbf{P}$ is given by

$$\mathbf{P} = \sum_a m_a \dot{\mathbf{r}}_a = \sum_a m_a \mathbf{v}_a \quad (2.2.7)$$

Proof

We apply the definition to the Lagrangian of a system of N free particles:

$$\mathbf{P} = \sum_a \frac{\partial L}{\partial \dot{\mathbf{r}}_a} = \sum_a \frac{\partial}{\partial \dot{\mathbf{r}}_a} \left(\sum_b \frac{m_b \dot{\mathbf{r}}_b^2}{2} + \cancel{U(\mathbf{r}_1, \dots, \mathbf{r}_N)} \right) = \sum_a m_b \dot{\mathbf{r}}_b \delta_{ab} = \boxed{\sum_a m_a \dot{\mathbf{r}}_a = \sum_a m_a \mathbf{v}_a} \quad (2.2.8)$$

as desired. □

Proposition 2.2.4 (Conservation of Momentum in an External Field)

Let there be a conservative external field that depends only on coordinates in the set A . Let a system have N particles and $s = 3N$ degrees of freedom. The components of the total momentum that are orthogonal to those in the set A will be conserved.

Proof

Along those axes, there will be no dependence of the Lagrangian on position. Thus, those components will be conserved. However, along the axes in the set A , the Lagrangian will change under a displacement in space, because there is a dependence of the potential energy and thus the Lagrangian has the form

$$U(\{\mathbf{r}_1 \cdot \hat{\mathbf{n}} \dots \mathbf{r}_s \cdot \hat{\mathbf{n}}\}) \quad (2.2.9)$$

for any $\hat{\mathbf{n}} \in A$. □

Proposition 2.2.5 (Change of Momentum under a Potential)

Let a system have N particles and $s = 3N$ degrees of freedom. Let an external field be expressed by a potential of the form

$$U = U(\mathbf{r}_1 \dots \mathbf{r}_N) \quad (2.2.10)$$

Then,

$$\dot{\mathbf{P}}_a = \mathbf{F}_a \quad (2.2.11)$$

where $\mathbf{F}_a$ is the force acting on the a -th particle.

Proof

We start with the definition of momentum:

$$\mathbf{P}_a = \frac{\partial L}{\partial \dot{\mathbf{r}}_a} \quad (2.2.12)$$

Thus,

$$\dot{\mathbf{P}}_a = \frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{r}}_a} = \frac{\partial L}{\partial \mathbf{r}_a} = -\frac{\partial U}{\partial \mathbf{r}_a} \equiv \mathbf{F}_a \quad (2.2.13)$$

as desired. $\square$

Definition 2.2.6 (Generalized Momentum)

Let a system have s degrees of freedom. Let $\{q_i\}$ be the set of generalized coordinates used in the description of the system. The *generalized momenta* are

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \quad (2.2.14)$$

Definition 2.2.7 (Generalized Force)

Let a system have s degrees of freedom. Let $\{q_i\}$ be the set of generalized coordinates used in the description of the system. The *generalized forces* are

$$F_i = \dot{p}_i = \frac{\partial L}{\partial q_i} \quad (2.2.15)$$

Let us create a method by which we can use these generalized momenta.

Definition 2.2.8 (Cyclic Coordinate)

Let L be the Lagrangian of a system with s degrees of freedom. The generalized coordinate q_i is said to be *cyclic* if

$$\frac{\partial L}{\partial q_i} = 0 \quad (2.2.16)$$

That is, the Lagrangian does not depend explicitly on the coordinate q_i .

Theorem 2.2.9 (Noether)

Let q_i be a cyclic coordinate of a system with s degrees of freedom and Lagrangian L . The generalized momentum associated with q_i , written p_i , is conserved.

Proof

We use the definition of the Generalized Force:

$$F_i = \dot{p}_i = \frac{\partial L}{\partial q_i} \quad (2.2.17)$$

The quantity on the right is zero by construction. Thus,

$$\dot{p}_i = 0 \Rightarrow p_i = \text{ct} \quad (2.2.18)$$

Therefore, p_i is conserved. $\square$

2.3 Center of Mass

If we change the inertial reference frame K to another inertial frame K' such that their relative velocity is $\mathbf{V}$, the velocities $\mathbf{v}_a$ and $\mathbf{v}'_a$ of the particles of a system expressed in the frames K and K' are related by

$$\mathbf{v}_a = \mathbf{v}'_a + \mathbf{V} \quad (2.3.1)$$

Thus, the momenta are related by

$$\mathbf{p}_a = \mathbf{p}'_a + \mathbf{V}m_a \quad (2.3.2)$$

and therefore

$$\mathbf{P} = \mathbf{P}' + \mathbf{V} \sum_a m_a \quad (2.3.3)$$

Suppose that one of the reference frames (let us choose K') is such that $\mathbf{P} = \mathbf{0}$. Then, the velocity $\mathbf{V}$ is given by

$$\mathbf{V} = \frac{\mathbf{P}}{\sum_a m_a} = \frac{\sum_a m_a \mathbf{v}_a}{\sum_a m_a} \quad (2.3.4)$$

Definition 2.3.1 (State of Rest)

Let K be a reference frame in which a system has momentum $\mathbf{P}$. If $\mathbf{P} = \mathbf{0}$, the system is said to be *at rest* with respect to the reference frame K .

Let $\mu \equiv \sum_a m_a$ be the total mass of the system. We can therefore use the expression for the velocity $\mathbf{V}$ to find the *center of mass* of the system, $\mathbf{R}$.

Definition 2.3.2 (Center of Mass)

Let K be a reference frame. Let a system have N masses with masses m_a and positions $\mathbf{r}_a$. The position of the *center of mass* is given by

$$\mathbf{R} \equiv \frac{\sum_a m_a \mathbf{r}_a}{\sum_a m_a} \quad (2.3.5)$$

In words, it is the weighted sum (with weights m_a) of the positions $\mathbf{r}_a$ of the particles.

Indeed, conservation of momentum is equivalent to the fact that the center of mass moves uniformly along a straight line in space. It is often convenient to choose a reference frame K in which the center of mass (and thus the system) is at rest.

We can express the energy of a system using the center of mass and its velocity $\mathbf{V} \equiv \dot{\mathbf{R}}$:

$$E = \frac{\mu V^2}{2} + E_i \quad (2.3.6)$$

where E_i is the *internal energy* of the system. We demonstrate this fact by the following procedure:

$$E = \sum_a \frac{m_a v_a^2}{2} + U \quad (2.3.7)$$

$$= \sum_a \frac{m_a (\mathbf{v}'_a + \mathbf{V})^2}{2} + U \quad (2.3.8)$$

$$= \frac{\mu V^2}{2} + \mathbf{V} \cdot \sum_a m_a \mathbf{v}'_a + \sum_a \frac{m_a (v'_a)^2}{2} + U \quad (2.3.9)$$

$$= E' + \mathbf{V} \cdot \mathbf{P}' + \frac{\mu V^2}{2} \quad (2.3.10)$$

If we take K' to be the reference frame in which the velocity of the center of mass is zero, i.e. the rest frame, we have $\mathbf{P}' = \mathbf{0}$ and therefore

$$E = E_i + \frac{\mu V^2}{2} \quad (2.3.11)$$

where E_i is thus the energy of the system in the rest frame.

2.4 Angular Momentum

We can continue with the isotropy of space. Consider an infinitesimal rotation $\delta\phi$. The magnitude of this vector is the size of the angle, in radians, and its direction is such that the right-hand rule is satisfied. Thus, for each particle at position $\mathbf{r}$ relative to the center of rotation (which we take as the origin of the coordinate system), its change in position is

$$\delta\mathbf{r} = \delta\phi \times \mathbf{r} \quad (2.4.1)$$

Thus, the change in velocity is

$$\delta\mathbf{v} = \delta\phi \times \mathbf{v} \quad (2.4.2)$$

for $\mathbf{v} = \dot{\mathbf{r}}$. Therefore, the change in the Lagrangian is

$$\delta L = \sum_a \left(\frac{\partial L}{\partial \mathbf{r}_a} \cdot \delta\mathbf{r}_a + \frac{\partial L}{\partial \mathbf{v}_a} \cdot \delta\mathbf{v}_a \right) = 0 \quad (2.4.3)$$

We recognize $\frac{\partial L}{\partial \mathbf{v}_a}$ as $\mathbf{p}_a$, and $\frac{\partial L}{\partial \mathbf{r}_a}$ as $\dot{\mathbf{p}}_a$, giving

$$\sum_a (\dot{\mathbf{p}}_a \cdot (\delta\phi \times \mathbf{r}_a) + \mathbf{p}_a \cdot (\delta\phi \times \mathbf{v}_a)) = 0 \quad (2.4.4)$$

We factor out $\delta\phi$:

$$\delta\phi \cdot \sum_a (\mathbf{r}_a \times \dot{\mathbf{p}}_a + \dot{\mathbf{r}}_a \times \mathbf{p}_a) = \delta\phi \cdot \frac{d}{dt} \sum_a (\mathbf{r}_a \times \mathbf{p}_a) = 0 \quad (2.4.5)$$

Because this holds for all $\delta\phi$, it must be that

$$\frac{d}{dt} \sum_a (\mathbf{r}_a \times \mathbf{p}_a) = 0 \quad (2.4.6)$$

The quantity differentiated is called the *angular momentum*.

Definition 2.4.1 (Angular Momentum)

Let a system have N particles. The *angular momentum* is given by

$$\mathbf{M} = \sum_a \mathbf{r}_a \times \mathbf{p}_a \quad (2.4.7)$$

Proposition 2.4.2 (Conservation of Angular Momentum)

Let a closed system have N particles. Let L be its Lagrangian. Let $\mathbf{M}$ be the total angular momentum of the system. In this case, $\dot{\mathbf{M}} = \mathbf{0}$.

Proof

Proved above. □

Proposition 2.4.3 (Value of the Angular Momentum)

Let a system have N particles. The value $\mathbf{M}$ of the angular momentum depends on the choice of origin of the coordinate system. In particular, let one origin be displaced by a vector distance $\mathbf{a}$ from another. Then,

$$\mathbf{M} = \mathbf{M}' + \mathbf{a} \times \mathbf{P} \quad (2.4.8)$$

for $\mathbf{P}$ the (translational) momentum.

Proof

We start from the definition:

$$\mathbf{M} = \sum_a \mathbf{r}_a \times \mathbf{p}_a = \sum_a \mathbf{r}'_a \times \mathbf{p}_a + \mathbf{a} \times \sum_a \mathbf{p}_a = \mathbf{M}' + \mathbf{a} \times \mathbf{P} \quad (2.4.9)$$

as desired. $\square$

We can now explore how the angular momentum changes when we change from one inertial reference frame K to another K' with a velocity $\mathbf{V}$ between the frames.

$$\mathbf{M} = \sum_a m_a \mathbf{r}_a \times \mathbf{v}_a = \sum_a m_a \mathbf{r}_a \times \mathbf{v}'_a + \sum_a m_a \mathbf{r}_a \times \mathbf{V} = \mathbf{M}' + \mu \mathbf{R} \times \mathbf{V} \quad (2.4.10)$$

where μ is the total mass of the system and $\mathbf{R}$ is the position of the center of mass (obviously a function of time). We can rearrange to obtain

$$\mathbf{M} = \mathbf{M}' + \mathbf{R} \times \mathbf{P} \quad (2.4.11)$$

In the spirit of internal energy, we denote $\mathbf{M}'$ as the internal angular momentum, noting that $\mathbf{R} \times \mathbf{P}$ arises from the motion of the center of mass, without which $\mathbf{P} = \mu \mathbf{V} = \mathbf{0}$.

2.5 Mechanical Similarity

We can infer essential properties of the equations of motion without finding them explicitly. We begin with a definition and a theorem.

Definition 2.5.1 (Homogeneous Function)

Let $f(x_1, x_2, \dots, x_s) : \mathbb{R}^s \rightarrow \mathbb{R}$ be a function. It is said to be *homogeneous* if there exists a real k such that

$$f(\alpha x_1, \alpha x_2, \dots, \alpha x_s) = \alpha^k f(x_1, x_2, \dots, x_s) \quad (2.5.1)$$

for any set of parameters $\{x_i\}$. The constant k is called the *degree of homogeneity*.

Theorem 2.5.2 (Euler, for Homogeneous Functions)

Let f be a function of s real variables that is homogeneous of degree k . Suppose that f is continuously differentiable on a subset of $\mathbb{R}^s$. Then, the following partial differential equation is satisfied on this subset:

$$kf(x_1, \dots, x_s) = \sum_{i=1}^s x_i \frac{\partial f}{\partial x_i}(x_1, \dots, x_s) \quad (2.5.2)$$

The converse is also true – any maximal^a continuously differentiable solution of this differential equation is a homogeneous function of degree k .

^aThis indicates that the solution cannot be extended to a larger domain.

Suppose that the Lagrangian of a system of N particles is given by

$$L = T - U(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad (2.5.3)$$

Suppose also that U is homogeneous of degree k :

$$U(\alpha \mathbf{r}_1, \dots, \alpha \mathbf{r}_N) = \alpha^k U(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad (2.5.4)$$

Suppose that all coordinates are scaled by a factor α , and that time is similarly scaled by a factor β :

$$\mathbf{r}_a \mapsto \alpha \mathbf{r}_a \quad ; \quad t \mapsto \beta t \quad (2.5.5)$$

Thus,

$$\mathbf{v}_a \mapsto \frac{\alpha}{\beta} \mathbf{v}_a \quad ; \quad T \mapsto \frac{\alpha^2}{\beta^2} T \quad (2.5.6)$$

The potential energy is multiplied by a factor α^k by construction. If

$$\frac{\alpha^2}{\beta^2} = \alpha^k \Rightarrow \beta = \alpha^{1-k/2} \Rightarrow \frac{\alpha}{\beta} = \alpha^{1-1+k/2} = \alpha^{k/2} \quad (2.5.7)$$

the total effect is merely a multiplication of the Lagrangian by a constant factor, which does not change the equations of motion. This indicates that the system is invariant under such a transformation. This corresponds to families of trajectories allowed as solutions of the equations of motion. In particular,

$$\frac{t'}{t} = \left(\frac{\ell'}{\ell}\right)^{1-k/2} \quad (2.5.8)$$

for ℓ a linear dimension of the trajectory. Consequently, the ratios of energies, velocities, and angular momenta can be expressed as functions of the ratio of linear dimensions ℓ'/ℓ :

$$\frac{v'}{v} = \left(\frac{\ell'}{\ell}\right)^{k/2} \quad ; \quad \frac{E'}{E} = \left(\frac{\ell'}{\ell}\right)^k \quad ; \quad \frac{M'}{M} = \left(\frac{\ell'}{\ell}\right)^{1+k/2} \quad (2.5.9)$$

We can now conclude certain facts.

- Let there be a system with small oscillations – $U \propto x^2$ and thus $k = 2$, which indicates that the period is independent of the amplitude:

$$\frac{t'}{t} = \left(\frac{\ell'}{\ell}\right)^{1-2/2} = 1 \quad (2.5.10)$$

and thus the time scales are always the same.

- Let there be a system immersed in a uniform field. In this case, $U \propto x$ and thus $k = 1$:

$$\frac{t'}{t} = \left(\frac{\ell'}{\ell}\right)^{1-1/2} = \sqrt{\frac{\ell'}{\ell}} \quad (2.5.11)$$

Thus, for projectiles, the time to impact is proportional to the square root of the maximum altitude.

- For a Newtonian or Coulomb force law, $U \propto x^{-2}$, thus $k = -2$, which gives

$$\frac{t'}{t} = \left(\frac{\ell'}{\ell}\right)^{1-(-2)/2} = \left(\frac{\ell'}{\ell}\right)^{3/2} \iff t^2 \propto \ell^3 \quad (2.5.12)$$

which is exactly Kepler's Third Law.

Theorem 2.5.3 (Virial)

Let there be a system of N particles with kinetic energy T and potential energy U , where U is homogeneous of degree k . Then,

$$2\overline{T} = k\overline{U} = \overline{\sum_a^N \mathbf{r}_a \cdot \frac{\partial U}{\partial \mathbf{r}_a}} \quad (2.5.13)$$

Proof

The kinetic energy T is a quadratic function (and thus homogeneous of degree 2) of the velocities; therefore, Euler's Theorem gives

$$\sum_a \mathbf{v}_a \cdot \frac{\partial T}{\partial \mathbf{v}_a} = 2T \quad (2.5.14)$$

If we replace $\partial T / \partial \mathbf{v}_a$ with $\mathbf{p}_a$, we have

$$2T = \sum_a \mathbf{p}_a \cdot \mathbf{v}_a = \frac{d}{dt} \left(\sum_a \mathbf{p}_a \cdot \mathbf{r}_a \right) - \sum_a \mathbf{r}_a \cdot \dot{\mathbf{p}}_a \quad (2.5.15)$$

We recall the definition of the average value of a function f :

$$\bar{f} = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^{\tau} f(t) dt \quad (2.5.16)$$

It is clear that

$$\exists F(t) : \mathbb{R}^+ \rightarrow \mathbb{R} : \exists B \in \mathbb{R} : \forall t \in \mathbb{R}^+ : |F(t)| < B : \quad f(t) = \frac{dF}{dt} \quad (2.5.17)$$

implies

$$\bar{f} = \lim_{\tau \rightarrow \infty} \frac{1}{\tau} \int_0^{\tau} f(t) dt = \lim_{\tau \rightarrow \infty} \frac{F(\tau) - F(0)}{\tau} = 0 \quad (2.5.18)$$

Suppose that the system remains in a finite region of space with finite velocities. Then the function $\sum \mathbf{p}_a \cdot \mathbf{r}_a$ is bounded, and the average value of the first term on the right-hand side of (2.5.15) is 0. We replace $\dot{\mathbf{p}}_a$ with $-\partial U / \partial \mathbf{r}_a$ to obtain

$$2\bar{T} = \overline{\sum_a \mathbf{r}_a \cdot \frac{\partial U}{\partial \mathbf{r}_a}} = k\bar{U} \quad (2.5.19)$$

using Euler's Theorem. □

We can continue, knowing that $\bar{T} + \bar{U} = \bar{E} = E$:

$$\bar{U} = \frac{2E}{k+2} \quad ; \quad \bar{T} = \frac{kE}{k+2} \quad (2.5.20)$$

with k again the degree of homogeneity of U . For small oscillations, $k = 2$ gives

$$\bar{U} = \frac{2E}{4} = \frac{E}{2} \quad ; \quad \bar{T} = \frac{2E}{4} = \frac{E}{2} \quad (2.5.21)$$

and thus the energy is shared equally between potential and kinetic energy. Mathematically,

$$\bar{U} = \bar{T} \quad (2.5.22)$$

For Newtonian interactions with $k = -1$,

$$\bar{U} = \frac{2E}{1} = 2E \quad ; \quad \bar{T} = \frac{-E}{1} = -E \quad (2.5.23)$$

The energy of the system is therefore negative, with $-\bar{U} = 2\bar{T}$ and $\bar{T} = -E$. Clearly, $\bar{T} \geq 0$. This corresponds to the fact that the motion of such a system takes place in a finite region of space only if the total energy of the system is negative.

2.6 Approximations to SHOs

It is possible to take complicated systems, find a stable equilibrium point about a coordinate, and linearize the differential equation to model the behavior of small oscillations as that of a simple harmonic oscillator (SHO).

Definition 2.6.1 (SHO)

The *Simple Harmonic Oscillator* (SHO) is a system of a mass m described by a differential equation of the following form,

$$\ddot{x} + \omega_0^2 x = 0 \quad (2.6.1)$$

where x describes the difference between the position of the mass as a function of time and its equilibrium position.

Proposition 2.6.2 (Modeling an Arbitrary System as an SHO)

Let a system have s degrees of freedom, such that there exists a state of the system for which $\ddot{q}_1 = \dot{q}_1 = 1$. Let $V(q_1) \equiv U(q_1, q_2, \dots, q_s)$ such that $V' = \partial U / \partial q_1$, etc. It is possible to model this system as an SHO about a stable equilibrium $\{q_0 \equiv q_1^{(0)}, q_2^{(0)}, \dots\}$ such that $V'(q_0) = 0$ and $V''(q_0) > 0$ for oscillations of the following form with the following bound on the amplitude:

$$q(t) \simeq q_0(t) + A \exp \left(i \left(\sqrt{\frac{V''(q_0)}{m}} \right) t \right) \quad ; \quad A \ll 2 \left| \frac{V''(q_0)}{V'''(q_0)} \right| \quad (2.6.2)$$

See the third example. The frequency of the oscillations is given by ω_0 , i.e. the square root of the coefficient of x in the linearized equation.

2.7 Examples**2.7.1 Damped Harmonic Oscillator**

Consider the Lagrangian

$$L = e^{\gamma t} \left(\frac{m\dot{x}^2}{2} - \frac{kx^2}{2} \right) \quad (2.7.1)$$

Let us apply the Euler–Lagrange equation:

$$\frac{\partial L}{\partial x} = -e^{\gamma t} kx \quad ; \quad \frac{\partial L}{\partial \dot{x}} = e^{\gamma t} m\dot{x} \quad (2.7.2)$$

Thus,

$$e^{\gamma t} kx + e^{\gamma t} m\ddot{x} + e^{\gamma t} \gamma m\dot{x} = 0 \quad (2.7.3)$$

We rearrange:

$$\ddot{x} + \gamma\dot{x} + \frac{k}{m}x = 0 \quad (2.7.4)$$

This is clearly the damped harmonic oscillator. We see that because $\partial L / \partial t$ is not zero, energy is not conserved. In fact, given this, the energy of the system cannot be calculated using the definition given above.

2.7.2 Two Blocks - Review

Let us return to the example of the previous section, in which two blocks of masses m_1 and m_2 are such that the block m_1 hangs from a string of length l attached to the other mass m_2 , which is on a table and is free to rotate around the string.

We recall the positions and velocities, and the equations of motion:

$$\mathbf{r}_2 = \langle r \cos \theta, r \sin \theta, 0 \rangle \quad ; \quad \mathbf{r}_1 = \langle 0, 0, z \rangle \quad (2.7.5)$$

The time derivatives are

$$\dot{\mathbf{r}}_1 = \langle 0, 0, \dot{z} \rangle \quad (2.7.6)$$

$$\dot{\mathbf{r}}_2 = \langle \dot{r} \cos \theta - r \sin \theta \dot{\theta}, \dot{r} \sin \theta + r \cos \theta \dot{\theta}, 0 \rangle \quad (2.7.7)$$

The Lagrangian is

$$L = \frac{m_1}{2} \dot{r}^2 + \frac{m_2}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + mg(l - r) \quad (2.7.8)$$

The (coupled) equations of motion are

$$\ddot{r} = \frac{\dot{\theta}^2 r m_2 - m_1 g}{m_1 + m_2} \quad ; \quad \ddot{\theta} = -\frac{2\dot{\theta}\dot{r}}{r} \quad (2.7.9)$$

We want to eliminate the term containing $\dot{\theta}$ from $\ddot{r}$ in order to decouple the system of differential equations. We observe that the coordinate θ is cyclic, because it does not appear in the expression for L . Therefore, the momentum associated with θ is conserved by Noether's Theorem. Thus, we compute this generalized momentum:

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m_2 r^2 \dot{\theta} \quad (2.7.10)$$

We can therefore substitute this quantity into the equation for $\ddot{r}$:

$$\dot{\theta} = \frac{p_\theta}{m_2 r^2} \implies \ddot{r} = \frac{\left(\frac{p_\theta}{m_2 r^2}\right)^2 r m_2 - m_1 g}{m_1 + m_2} = \frac{p_\theta^2 r^{-3} m_2^{-1} - m_1 g}{m_1 + m_2} \quad (2.7.11)$$

which is now an equation involving only r and not θ . The value of p_θ can be found from the initial conditions, namely

$$p_\theta = m_2 r_0^2 \dot{\theta}_0 \quad (2.7.12)$$

We also observe that this quantity is in fact the angular momentum of the system:

$$\mathbf{M} = \sum_a m_a \mathbf{r}_a \times \dot{\mathbf{r}}_a \quad (2.7.13)$$

The second mass has a nonzero angular momentum, because for the first mass, $\dot{\mathbf{r}}_1 \parallel \mathbf{r}_1$ and thus their cross product is zero. Thus,

$$\mathbf{M} = m_2 \mathbf{r}_2 \times \dot{\mathbf{r}}_2 = m_2 \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ r \cos \theta & r \sin \theta & 0 \\ \dot{r} \cos \theta - r \sin \theta \dot{\theta} & \dot{r} \sin \theta + r \cos \theta \dot{\theta} & 0 \end{vmatrix} = m_2 r^2 \dot{\theta}^2 \hat{\mathbf{z}} \quad (2.7.14)$$

2.7.3 Oscillation Frequency

Consider the same system. Let us find the frequencies of small oscillations about an equilibrium point, defined such that

$$\ddot{r} = 0 \quad (2.7.15)$$

Assume for simplicity that the masses are identical ($m \equiv m_1 = m_2$). From the equation of motion,

$$\ddot{r} = \frac{p_\theta^2}{2m^2 r_0^3} - \frac{g}{2} = 0 \implies \frac{p_\theta^2}{2m^2 r_0^3} = \frac{g}{2} \implies r_0 = \sqrt[3]{\frac{p_\theta^2}{m^2 g}} \quad (2.7.16)$$

Now, suppose that we have small oscillations, such that

$$r(t) = r_0 + \delta r(t) \quad (2.7.17)$$

with $\delta r(t) \ll r_0$. Thus,

$$\dot{r} = \dot{\delta r} \quad ; \quad \ddot{r} = \ddot{\delta r} \quad (2.7.18)$$

We seek to transform the equation for $\ddot{\delta r}$ so as to arrive at

$$\ddot{\delta r} = -\omega^2 \delta r \quad (2.7.19)$$

with $\omega^2 > 0$ of course.

2.7.4 The Virial Theorem for Estimating the Mass of a Group of Galaxies

For a gravitational system,

$$2\overline{T} = -\overline{U} \quad (2.7.20)$$

Let us use the fact that for a group of galaxies, even though it is impossible to measure the behavior over a period (which is on the order of billions of years), we can apply thermodynamic/statistical reasoning which says that the current state is indeed a fairly homogeneous equilibrium distribution, and thus one simply needs to average over the current state instead of taking a time average.

We can estimate that the mass of each galaxy is roughly the same, that there is a uniform density, and that the true speed satisfies

$$v^2 = 3v_{\text{doppler}}^2 \quad (2.7.21)$$

where v_{doppler} comes from the line-of-sight component, using the Doppler effect. For roughly $N \sim 1000$ galaxies, we obtain

$$2\overline{T} = \sum_i \overline{m_i v_i^2} \sim \frac{M}{N} \sum v_i^2 = M\overline{v}^2 \quad (2.7.22)$$

with M the total mass of the group. For a sphere, the gravitational potential energy is

$$\overline{U} \sim \frac{3}{5} \frac{GM^2}{R} \quad (2.7.23)$$

with R the radius of the sphere. Thus,

$$M \sim \frac{5R\overline{v}^2}{3G} \quad (2.7.24)$$

which gives the surprising result that galaxy velocities are too fast, and thus motivates the existence of dark matter.

A similar calculation for an individual galaxy gives a similar result, which shows that dark matter is present not only at the level of groups of galaxies but also within a galaxy itself.

3 Gravitational Systems

3.1 Non-Inertial Frames

Suppose that the Lagrangian of a one-particle system in an external field is

$$L_0 = \frac{m\mathbf{v}_0^2}{2} - U \quad (3.1.1)$$

in an inertial frame. In a non-inertial frame, the principle of least action still holds, and thus

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = \frac{\partial L}{\partial \mathbf{r}} \quad (3.1.2)$$

is still true. However, the form of L is no longer that of L_0 . The relation between the frames is

$$\mathbf{v}_0 = \mathbf{v}' + \mathbf{V}(t) \quad (3.1.3)$$

Thus,

$$L' = \frac{m\mathbf{v}'^2}{2} + m\mathbf{v}' \cdot \mathbf{V} + \frac{m\mathbf{V}^2}{2} - U \quad (3.1.4)$$

The function $\mathbf{V}^2(t)$ is a function of t and thus is a total time derivative, and it can be eliminated. Thus,

$$L' = \frac{m\mathbf{v}'^2}{2} + m\mathbf{v}' \cdot \mathbf{V}(t) - U \quad (3.1.5)$$

We expand the middle term:

$$m\mathbf{V}(t) \cdot \mathbf{v}' = m\mathbf{V} \cdot \frac{d\mathbf{r}'}{dt} = \frac{d}{dt} (m\mathbf{V} \cdot \mathbf{r}') - m\mathbf{r}' \cdot \frac{d\mathbf{V}}{dt} \quad (3.1.6)$$

The first term of the rearrangement can be eliminated, being a total time derivative. Let $\mathbf{W} = \dot{\mathbf{V}}$. Thus,

$$L' = \frac{m(\mathbf{v}')^2}{2} - m\mathbf{W}(t) \cdot \mathbf{r}' - U \quad (3.1.7)$$

Thus, Lagrange's equation becomes

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = \frac{\partial L}{\partial \mathbf{r}} \quad \Rightarrow \quad m \frac{d\mathbf{v}'}{dt} = -\frac{\partial U}{\partial \mathbf{r}'} - m\mathbf{W}(t) \quad (3.1.8)$$

Thus, a fictitious force appears, namely $-m\mathbf{W}(t)$.

Now consider a reference frame having not only a variable relative velocity $\mathbf{V}(t)$, but also a variable angular velocity $\boldsymbol{\Omega}(t)$. In this case,

$$\mathbf{v}' = \mathbf{v} + \boldsymbol{\Omega} \times \mathbf{r} \quad (3.1.9)$$

We use $\mathbf{r}$ and not $\mathbf{r}'$ because they coincide. The Lagrangian becomes

$$L = \frac{mv^2}{2} + m\mathbf{v} \cdot \boldsymbol{\Omega} \times \mathbf{r} + \frac{m(\boldsymbol{\Omega} \times \mathbf{r})^2}{2} - m\mathbf{W} \cdot \mathbf{r} - U \quad (3.1.10)$$

We cannot eliminate the term with $\boldsymbol{\Omega}^2$ because there is a dependence on position, and thus the term is not a total time derivative. The equation of motion is found from the Euler–Lagrange equation:

$$dL = m\mathbf{v} \cdot d\mathbf{v} + m d\mathbf{v} \cdot \boldsymbol{\Omega} \times \mathbf{r} + m d\mathbf{r} \cdot \mathbf{v} \times \boldsymbol{\Omega} + m(\boldsymbol{\Omega} \times \mathbf{r}) \times \boldsymbol{\Omega} \cdot d\mathbf{r} - m\mathbf{W} \cdot d\mathbf{r} - \frac{\partial U}{\partial \mathbf{r}} \cdot d\mathbf{r} \quad (3.1.11)$$

$$\Rightarrow \frac{\partial L}{\partial \mathbf{v}} = m\mathbf{v} + m\boldsymbol{\Omega} \times \mathbf{r} \quad (3.1.12)$$

$$\Rightarrow \frac{\partial L}{\partial \mathbf{r}} = m\mathbf{v} \times \boldsymbol{\Omega} + m(\boldsymbol{\Omega} \times \mathbf{r}) \times \boldsymbol{\Omega} - m\mathbf{W} - \frac{\partial U}{\partial \mathbf{r}} \quad (3.1.13)$$

Thus,

$$m\ddot{\mathbf{r}} = -\frac{\partial U}{\partial \mathbf{r}} - m\mathbf{W} + m\mathbf{r} \times \dot{\boldsymbol{\Omega}} + 2m\dot{\mathbf{r}} \times \boldsymbol{\Omega} + m\boldsymbol{\Omega} \times (\mathbf{r} \times \boldsymbol{\Omega}) \quad (3.1.14)$$

The blue term is the same as before, the fictitious force. The red term appears only when the rotation is non-uniform. The green term is called the *Coriolis Force* – it depends on the velocity $\dot{\mathbf{r}}$, like a dissipative force. The purple term is called the *centrifugal force*, which lies in the plane normal to the axis of $\boldsymbol{\Omega}$, pointing away from the axis. Its magnitude is

$$|F_c| = m\rho\Omega^2 = \frac{mv_c^2}{\rho} \quad (3.1.15)$$

where ρ is the distance of the particle in question from the axis, and $v_c \equiv \rho\Omega$ is the tangential speed.

In the special case where $\mathbf{W} = 0$ and $\dot{\boldsymbol{\Omega}} = 0$, the Lagrangian is

$$L = \frac{mv^2}{2} + m\mathbf{v} \cdot \boldsymbol{\Omega} \times \mathbf{r} + \frac{m}{2}(\boldsymbol{\Omega} \times \mathbf{r})^2 - U \quad (3.1.16)$$

Thus,

$$m\ddot{\mathbf{r}} = -\frac{\partial U}{\partial \mathbf{r}} + 2m\dot{\mathbf{r}} \times \boldsymbol{\Omega} + m\boldsymbol{\Omega} \times (\mathbf{r} \times \boldsymbol{\Omega}) \quad (3.1.17)$$

The particle's energy is obtained by substituting the momentum $\mathbf{p}$, given by

$$\mathbf{p} = \frac{\partial L}{\partial \dot{\mathbf{r}}} = m\mathbf{v} + m\boldsymbol{\Omega} \times \mathbf{r} \quad (3.1.18)$$

into $E = \mathbf{p} \cdot \dot{\mathbf{r}} - L$:

$$E = \frac{m(v^2 - (\boldsymbol{\Omega} \times \mathbf{r})^2)}{2} + U \quad (3.1.19)$$

From this there results a *centrifugal potential energy*, given by

$$U_{\text{centrifuge}} = -\frac{m}{2}(\boldsymbol{\Omega} \times \mathbf{r})^2 \quad (3.1.20)$$

We can express these results using the angular momentum $\mathbf{M}$.

Proposition 3.1.1 (Transformation of Integrals of Motion to a Non-Inertial Frame)

Let K_0 be an inertial reference frame. Let K' be a non-inertial reference frame that rotates uniformly about the frame K_0 such that velocities transform as

$$\mathbf{v}_0 = \mathbf{v} + \boldsymbol{\Omega} \times \mathbf{r} \quad (3.1.21)$$

Let a system have Lagrangian L_0 in the frame K_0 .

- The momentum $\mathbf{p}$ of each particle is invariant: $\mathbf{p} = \mathbf{p}_0$.
- The angular momentum $\mathbf{M}$ of each particle is also invariant: $\mathbf{M} = \mathbf{M}_0$.
- The energy transforms according to $E = E_0 - \mathbf{M} \cdot \boldsymbol{\Omega}$.

Proof

The invariance of momentum follows from equation (3.1.18). The invariance of angular momentum is trivial once the translational one is established. The energy is expressed in the stated way when we observe that

$$E = \frac{m(v^2 - (\boldsymbol{\Omega} \times \mathbf{r})^2)}{2} + U \quad (3.1.22)$$

$$= \frac{m((\mathbf{v}_0 - \boldsymbol{\Omega} \times \mathbf{r})^2 - (\boldsymbol{\Omega} \times \mathbf{r})^2)}{2} + U \quad (3.1.23)$$

$$= \frac{mv_0^2}{2} - m\mathbf{v}_0 \cdot \boldsymbol{\Omega} \times \mathbf{r} + U \quad (3.1.24)$$

$$= E_0 - m\mathbf{r} \times \mathbf{v}_0 \cdot \boldsymbol{\Omega} \quad (3.1.25)$$

$$= E_0 - \mathbf{M} \cdot \boldsymbol{\Omega} \quad (3.1.26)$$

as desired. This transformation does not change when we describe a system of N particles instead of just one. $\square$

3.2 Lagrange Points

We can apply a non-inertial framework to find the famous Lagrange points, the points in a two-body system where a small mass m satisfies $\ddot{r} = \dot{r} = \ddot{\theta} = \dot{\theta} = 0$.

Consider a reference frame rotating with the Earth and the Sun such that their positions do not change in this frame. Then, the Lagrangian of the mass m is given by

$$L = \frac{m}{2} (\dot{r}^2 + r^2(\dot{\theta} + \Omega)^2) - U \quad (3.2.1)$$

Kepler's Third Law (see Eq. 2.5.12) allows us to find Ω :

$$\Omega^2 = \frac{GM}{R^3} \quad (3.2.2)$$

We use the Euler-Lagrange Equation to obtain the equations of motion:

$$m(\ddot{r} - r\dot{\theta}^2) = mr\Omega^2 + 2mr\dot{\theta}\Omega - \frac{\partial U}{\partial r} \quad (3.2.3)$$

$$m(r^2\ddot{\theta} + 2r\dot{r}\dot{\theta}) = -2mr\dot{r}\Omega - \frac{\partial U}{\partial \theta} \quad (3.2.4)$$

We seek points such that

$$\ddot{r} = \dot{r} = \ddot{\theta} = \dot{\theta} = 0 \quad (3.2.5)$$

Thus, the equations of motion simplify:

$$\frac{\partial U}{\partial r} = mr\Omega^2 \quad (3.2.6)$$

$$\frac{\partial U}{\partial \theta} = 0 \quad (3.2.7)$$

The potential U of a mass m in the gravitational field of $m_\odot$ (the mass of the Sun) and m_e (the mass of the Earth) is given by

$$U = -\frac{Gmm_\odot}{r_\odot} - \frac{Gm_em}{r_e} \quad (3.2.8)$$

where r_e and $r_\odot$ are the distances from the mass m to the Earth and the Sun, respectively. We prepare some useful expressions. Let $M \equiv m_e + m_\odot$. Let R be the distance between the Sun and the Earth. We therefore have

$$r_{\odot,e}^2 = \left(r \cos \theta \pm \frac{Rm_{e,\odot}}{M} \right)^2 = r^2 + R^2 \frac{m_{e,\odot}^2}{M^2} \pm 2R \frac{m_{e,\odot}}{M} r \cos \theta \quad (3.2.9)$$

Thus,

$$2r_{\odot,e} \frac{\partial r_{\odot,e}}{\partial \theta} = \mp 2R \frac{m_{e,\odot}}{M} r \sin \theta \quad (3.2.10)$$

$$\Rightarrow \frac{\partial r_{\odot,e}}{\partial \theta} = \mp \frac{Rm_{e,\odot} r \sin \theta}{Mr_{\odot,e}} \quad (3.2.11)$$

We begin with the partial derivative of U with respect to θ . Thus,

$$\frac{\partial U}{\partial \theta} = \frac{\partial U}{\partial r_e} \frac{\partial r_e}{\partial \theta} + \frac{\partial U}{\partial r_\odot} \frac{\partial r_\odot}{\partial \theta} \quad (3.2.12)$$

$$= -Gm \left(-\frac{m_e}{r_e^3} \cdot \left(-2R \frac{m_\odot}{M} r \sin \theta \right) - \frac{m_\odot}{r_\odot^3} \cdot \left(2R \frac{m_e}{M} r \sin \theta \right) \right) \quad (3.2.13)$$

$$= -\frac{2GmRr \sin \theta}{M} \left(\frac{m_e m_\odot}{r_e^3} - \frac{m_\odot m_e}{r_\odot^3} \right) \quad (3.2.14)$$

$$= 0 \quad (3.2.15)$$

Clearly, this requires $r_e = r_\odot$ if $\sin \theta \neq 0$, or else $\sin \theta = 0$ for any $r_e, r_\odot$.

We proceed to the equation with r , preparing

$$2r_{\odot,e} \frac{\partial r_{\odot,e}}{\partial r} = 2r \pm 2R \frac{m_{e,\odot}}{M} \cos \theta \quad (3.2.16)$$

$$\Rightarrow \frac{\partial r_{\odot,e}}{\partial r} = \frac{Mr \pm Rm_{e,\odot} \cos \theta}{Mr_{\odot,e}} \quad (3.2.17)$$

Thus,

$$\frac{\partial U}{\partial \theta} = \frac{\partial U}{\partial r_e} \frac{\partial r_e}{\partial r} + \frac{\partial U}{\partial r_{\odot}} \frac{\partial r_{\odot}}{\partial r} \quad (3.2.18)$$

$$= -Gm \left(-\frac{m_e}{r_e^3} \cdot \left(\frac{Mr - Rm_{\odot} \cos \theta}{M} \right) - \frac{m_{\odot}}{r_{\odot}^3} \cdot \left(\frac{Mr + Rm_e \cos \theta}{M} \right) \right) \quad (3.2.19)$$

$$= Gmr \left(\frac{m_e}{r_e^3} + \frac{m_{\odot}}{r_{\odot}^3} \right) \quad (3.2.20)$$

$$= mr\Omega^2 \quad (3.2.21)$$

Kepler's Third Law tells us that

$$\frac{\mathcal{G}M}{R^3} = \mathcal{G} \left(\frac{m_{\odot}}{r_{\odot}^3} + \frac{m_e}{r_e^3} \right) \quad (3.2.22)$$

Take the case where $a \equiv r_e = r_{\odot}$:

$$\frac{M}{R^3} = \frac{m_{\odot} + m_e}{a^3} \Rightarrow a = R \quad (3.2.23)$$

Thus, we are constrained to have the Lagrange points such that $r_e = r_{\odot} = R$ at the intersection of the circle of radius $Rm_{\odot}/M$ around the origin with the line $x = R(1/2 - m_e/M)$.

There are three other Lagrange points, along the axis satisfying $\sin \theta = 0$. It is easier to find them using forces: one simply has to balance the forces of the Sun and the Earth such that their sum is the centripetal force required to maintain the orbit:

$$m\Omega^2 r = G \frac{mm_e}{r_e^2} \hat{r}_e + G \frac{mm_{\odot}}{r_{\odot}^2} \hat{r}_{\odot} \quad (3.2.24)$$

These give the points L_1 (between the Sun and the Earth), L_2 (beyond the Earth), and L_3 (beyond the Sun, opposite the Earth).

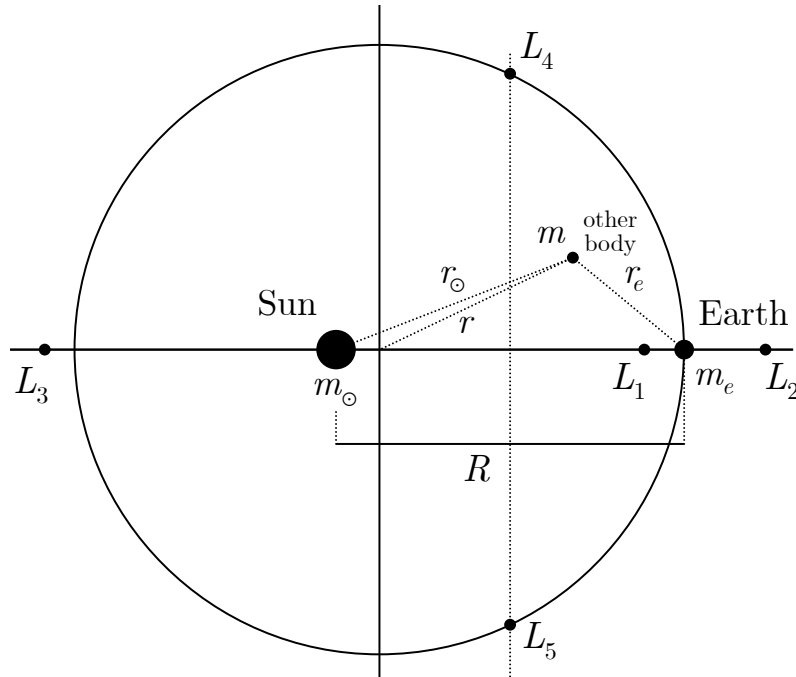


Figure 2. Lagrange Points.

3.3 Reduced Mass

Consider the two-particle problem³. We know that the potential energy depends only on the distance between the particles (see the first section of the text). Thus, the Lagrangian takes the form

$$L = \frac{m_1 \dot{\mathbf{r}}_1^2}{2} + \frac{m_2 \dot{\mathbf{r}}_2^2}{2} - U(|\mathbf{r}_1 - \mathbf{r}_2|) \quad (3.3.1)$$

Let $\mathbf{r} \equiv \mathbf{r}_1 - \mathbf{r}_2$ be the vector from mass 2 to mass 1. Choose the origin of the coordinate system to be the center of mass, such that

$$m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 = \mathbf{0} \quad (3.3.2)$$

Thus, we can express $\mathbf{r}_1$ and $\mathbf{r}_2$ in terms of $\mathbf{r}$ and their respective masses:

$$\mathbf{r}_1 = \mathbf{r} \frac{m_2}{m_1 + m_2} \quad ; \quad \mathbf{r}_2 = \mathbf{r} \frac{m_1}{m_1 + m_2} \quad (3.3.3)$$

Substituting this into the Lagrangian gives

$$L = \frac{m \dot{\mathbf{r}}^2}{2} - U(r) \quad (3.3.4)$$

where m is the *reduced mass* of the system. We observe that this is identical to the Lagrangian of a mass in a radial field, which depends only on the distance from the origin (in this case r).

Definition 3.3.1 (Reduced Mass)

Let a two-body system have masses m_1 and m_2 . The *reduced mass* is defined as

$$m \equiv \frac{m_1 m_2}{m_1 + m_2} \quad (3.3.5)$$

Once the trajectory $\mathbf{r}(t)$ is determined, the trajectories $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ can be easily found from the formulas in (3.3.3).

3.4 Motion in a Radial Field

We begin with a definition.

Definition 3.4.1 (Radial Field)

A *radial field* is a field whose potential depends only on the distance r from a so-called “central” point as follows:

$$U = U(r) \quad (3.4.1)$$

The force is therefore given by

$$\mathbf{F} = -\frac{\partial U}{\partial \mathbf{r}} = -\frac{dU}{dr} \hat{\mathbf{r}} \quad (3.4.2)$$

We know that the angular momentum $\mathbf{M}$ of a system immersed in such a field is conserved if it is computed about the center of the field, because such a field is invariant under rotations about the center. The angular momentum of a particle a is

$$\mathbf{M}_a = \mathbf{r}_a \times \mathbf{p}_a \Rightarrow \mathbf{M}_a \cdot \mathbf{r}_a = 0 \quad (3.4.3)$$

Thus, because $\mathbf{M}_a$ is conserved, the direction of $\mathbf{r}_a$ is constrained to remain in the plane perpendicular to $\mathbf{M}_a$ ⁴.

The Lagrangian of a particle in such a system is given by

$$L = \frac{m(\dot{r}^2 + r^2 \dot{\phi}^2)}{2} - U(r) \quad (3.4.4)$$

³in English: Two-Body Problem.

⁴Thus, astronomical objects that break apart via an internal mechanism form a disk due to conservation of angular momentum.

It is possible to reduce a two-mass system with masses m_1 and m_2 in the center-of-mass frame to the problem of the motion of a single mass μ , the reduced mass, in a radial field given by

$$U(r) = -G \frac{\mu(m_1 + m_2)}{r} \quad (3.4.5)$$

with r the distance between the masses. This yields an identical Lagrangian, with $m \rightarrow \mu$ and $U(r)$ the one above.

We observe that the coordinate ϕ is cyclic, because the Lagrangian does not depend on it explicitly; thus,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}} = \frac{\partial L}{\partial \phi} = 0 \quad (3.4.6)$$

and therefore $\frac{\partial L}{\partial \dot{\phi}}$ is an integral of motion. In particular, this expression is the generalized momentum associated with ϕ . This generalized momentum is in fact the particle's angular momentum:

$$p_\phi = m r^2 \dot{\phi} = M_z = \text{ct} \quad (3.4.7)$$

We can understand a geometric property of the motion using this constant. Let df be the unit of area such that

$$df = \frac{r^2}{2} d\phi \quad (3.4.8)$$

Thus,

$$M_z = 2m\dot{f} \quad (3.4.9)$$

The term $\dot{f}$ is called the *sectorial velocity* – the fact that M_z is constant gives us Kepler's Second Law: equal areas are swept out over equal time intervals.

Because there is no explicit time dependence in the Lagrangian, energy is conserved:

$$\dot{E} = 0 \quad ; \quad E = T + U = \frac{m(\dot{r}^2 + r^2\dot{\phi}^2)}{2} + U(r) \quad (3.4.10)$$

Using the angular momentum $M = |\mathbf{M}|$:

$$E = \frac{m\dot{r}^2}{2} + \frac{M^2}{2mr^2} + U(r) \quad (3.4.11)$$

We write $\dot{r}$:

$$\dot{r} = \frac{dr}{dt} = \sqrt{\frac{2}{m}(E - U(r)) - \frac{M^2}{m^2 r^2}} \quad (3.4.12)$$

We write the time t as a function of r :

$$t = \int \frac{1}{\sqrt{\frac{2}{m}(E - U(r)) - \frac{M^2}{m^2 r^2}}} dr + t_0 \quad (3.4.13)$$

We write $d\phi = \frac{M dt}{mr^2}$ to find $\phi(r)$, the equation of the trajectory:

$$\phi = \int \frac{M}{r^2 \sqrt{2m(E - U(r)) - \frac{M^2}{r^2}}} dr + \phi_0 \quad (3.4.14)$$

We note that the angle ϕ is a monotone function of time, because (3.4.7) indicates that the derivative cannot change sign. We can therefore construct an effective potential energy, such that the denominator of the integral under the square root is $2m(E - U_{\text{eff}})$:

$$U_{\text{eff}} = U(r) + \frac{M^2}{2mr^2} \quad (3.4.15)$$

Clearly, the values of r such that

$$E = U_{\text{eff}} = U(r) + \frac{M^2}{2mr^2} \quad (3.4.16)$$

will be the bounds on the distance from the center, because at those points we have $\dot{r} = 0$. This allows us to determine whether the trajectory is closed: if an angle commensurable⁵ with 2π is swept out between the bounds on r , the trajectory will be closed.

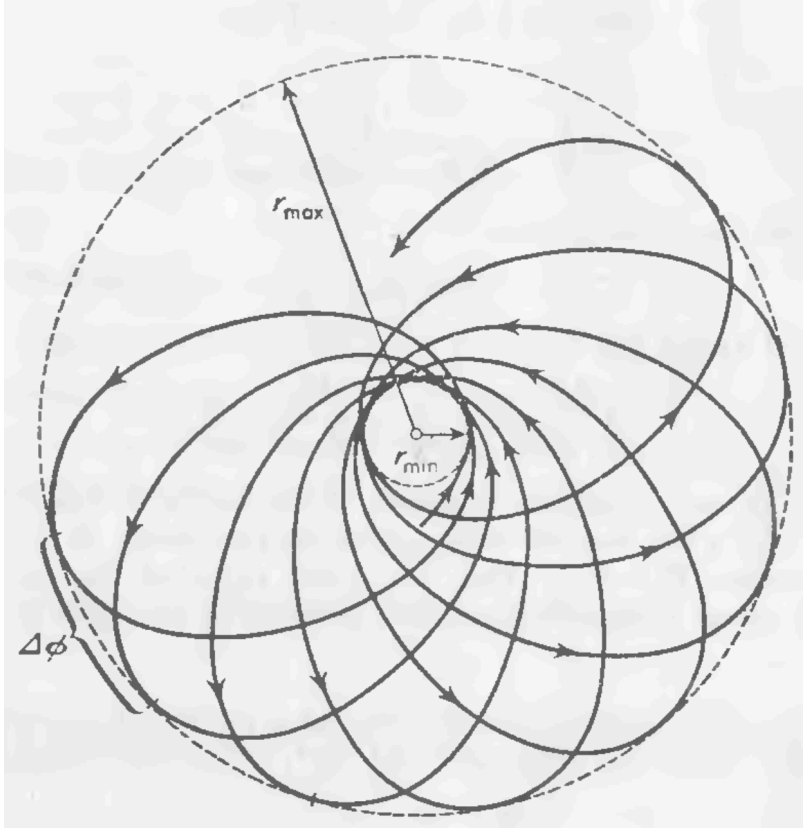
The swept angle is given by

$$\Delta\phi = 2 \int_{r_{\min}}^{r_{\max}} \frac{M/r^2}{\sqrt{2m(E - U(r)) - M^2/r^2}} dr \quad (3.4.17)$$

There must exist integers m, n such that

$$\Delta\phi = 2\pi \frac{m}{n} \quad (3.4.18)$$

if the trajectory is closed. If the trajectory is not closed, we obtain a motion as shown below:



Theorem 3.4.2 (Bertrand)

The *only* potentials $U(r)$ such that there exist m, n as above are those having one of the following two forms:

$$U(r) \propto \frac{1}{r} \quad \text{where} \quad U(r) \propto r^2 \quad (3.4.19)$$

⁵having a rational division (in this case by 2π)

Proof

The equation of motion of a particle of mass m in a central field is given by

$$m\ddot{r} - mr\omega^2 = m\ddot{r} - \frac{M^2}{mr^3} = -\frac{\partial U}{\partial r} \quad (3.4.20)$$

with $\omega \equiv \dot{\phi}$. We are allowed a change of variables from ϕ to t such that

$$\frac{d}{dt} = \frac{M}{mr^2} \frac{d}{d\phi} \quad (3.4.21)$$

which gives

$$\mathcal{M} \cdot \frac{M}{mr^2} \frac{d}{d\phi} \left(\frac{M}{mr^2} \frac{dr}{d\phi} \right) - \frac{M^2}{mr^3} = -\frac{\partial U}{\partial r} \quad (3.4.22)$$

Let us change variables such that $u \equiv 1/r$ and multiply by mr^2/M^2 :

$$\frac{mr^2}{M^2} \frac{M^2}{mr^2} \frac{d}{d\phi} \left(\frac{1}{r^2} \frac{dr}{d\phi} \right) - \frac{mr^2}{M^2} \frac{M^2}{mr^3} = -\frac{mr^2}{M^2} \frac{\partial U}{\partial r} \quad (3.4.23)$$

$$\Rightarrow \frac{d}{d\phi} \left(\frac{1}{r^2} \frac{dr}{d\phi} \right) - \frac{1}{r} = -\frac{mr^2}{M^2} \frac{\partial U}{\partial r} \quad (3.4.24)$$

The change of variables requires

$$\frac{dr}{d\phi} = -\frac{1}{u^2} \frac{du}{d\phi} \quad (3.4.25)$$

and thus

$$\frac{d}{d\phi} \left(\frac{1}{r^2} \frac{dr}{d\phi} \right) = \frac{d}{d\phi} \left(\frac{1}{r^2} \left(-\frac{1}{u^2} \frac{du}{d\phi} \right) \right) \quad (3.4.26)$$

$$= -\frac{d^2 u}{d^2 \phi} \quad (3.4.27)$$

Thus, the equation becomes

$$\Rightarrow \frac{d^2 u}{d^2 \phi} + u = \frac{m}{M^2 u^2} \frac{\partial U}{\partial r} \quad (3.4.28)$$

Similarly,

$$\frac{\partial U}{\partial r} = \frac{\partial U(1/u)}{\partial u} \frac{\partial u}{\partial r} = -\frac{\partial U(1/u)}{\partial u} \frac{1}{r^2} = -u^2 \frac{\partial U(1/u)}{\partial u} \quad (3.4.29)$$

and thus

$$\Rightarrow \frac{d^2 u}{d^2 \phi} + u = -\frac{m}{M^2} \frac{\partial U(1/u)}{\partial u} \quad (3.4.30)$$

This is called *Binet's equation*. We will now prove the conditions required for a stable, closed, non-circular orbit. Let $J(u)$ be such that

$$J(u) \equiv \frac{d^2 u}{d\phi^2} + u = -\frac{m}{M^2} \frac{d}{du} U \left(\frac{1}{u} \right) = -\frac{m}{M^2 u^2} f \left(\frac{1}{u} \right) \quad (3.4.31)$$

for f the radial force. For a purely circular trajectory, $\frac{d^2 u}{d\phi^2}$ must be zero, which gives

$$u_0 = J(u_0) = -\frac{m}{M^2 u_0^2} f \left(\frac{1}{u_0} \right) \quad (3.4.32)$$

for $u_0 = 1/r_0$. Consider a perturbation $\eta \equiv u - u_0$ of a circular orbit, such that $J(u)$ can be approximated by a Taylor series:

$$J(u) \approx J(u_0) + \eta J'(u_0) + \frac{\eta^2}{2} J''(u_0) + \dots \quad (3.4.33)$$

We substitute, factoring out the constant term:

$$\frac{d^2\eta}{d\phi^2} + \eta = \eta J'(u_0) + \frac{\eta^2}{2} J''(u_0) + \dots \quad (3.4.34)$$

We rewrite as

$$\frac{d^2\eta}{d\phi^2} + \beta^2\eta = \frac{\eta^2}{2} J''(u_0) + \dots \quad (3.4.35)$$

for $\beta^2 \equiv 1 - J'(u_0)$. The case $\beta = 0$ corresponds to the circular orbit. For small perturbations, we ignore the right-hand side of the equation and obtain

$$\eta(\phi) = h_1 \cos(\beta\phi) \quad (3.4.36)$$

for h_1 an integration constant. For a closed orbit, β must be rational, following the reasoning of (3.4.18). β must also not vary with r , because the possible values of β are totally disconnected from one another. Thus,

$$J'(u_0) = \frac{2}{u_0} \left(\frac{m}{M^2 u_0^2} f\left(\frac{1}{u_0}\right) \right) - \left(\frac{m}{M^2 u_0^2} f\left(\frac{1}{u_0}\right) \right) \frac{1}{f\left(\frac{1}{u_0}\right)} \frac{d}{du_0} f\left(\frac{1}{u_0}\right) \quad (3.4.37)$$

$$= -2 + \frac{u_0}{f\left(\frac{1}{u_0}\right)} \frac{d}{du_0} f\left(\frac{1}{u_0}\right) = 1 - \beta^2 \quad (3.4.38)$$

This holds for any value of u_0 , so

$$\frac{df}{dr} = (\beta^2 - 3) \frac{f}{r} \quad (3.4.39)$$

Thus, the function f admits a power law of the form

$$f(r) = -\frac{k}{r^{3-\beta^2}} \quad (3.4.40)$$

Thus,

$$J(u) = \frac{mk}{M^2} u^{1-\beta^2} \quad (3.4.41)$$

This gives

$$J''(u_0) = -\frac{\beta^2(1-\beta^2)}{u_0} \quad (3.4.42)$$

and

$$J'''(u_0) = \frac{\beta^2(1-\beta^2)(1+\beta^2)}{u_0^2} \quad (3.4.43)$$

Using Fourier series, we obtain another constraint:

$$\eta(\phi) = h_0 + h_1 \cos(\beta\phi) + \dots \Rightarrow 0 = (2h_1 h_0 + h_1 h_2) \frac{J''(u_0)}{2} + h_1^3 \frac{J'''(u_0)}{8} = \frac{h_1^3}{24\beta^2} (3\beta^2 J'''(u_0) + 5J''(u_0)^2) \quad (3.4.44)$$

which gives

$$0 = 3\beta^2 J'''(u_0) + 5J''(u_0)^2 \quad (3.4.45)$$

Substitute to obtain

$$\beta^2(1-\beta^2)(4-\beta^2) = 0 \quad (3.4.46)$$

which allows $\beta = 0$ (circular orbits), $\beta = 1$ (potential $U(r) \propto 1/r$), $\beta = 2$ (potential $U(r) \propto r^2$), as desired. $\square$

3.5 Kepler Problem

The Kepler problem consists of studying the allowed motions in a radial field proportional to the inverse square of the distance⁶, such that the potential has the form

$$U = -\frac{\beta}{r} \quad (3.5.1)$$

It is therefore possible to take the general equation that describes motion in a field $U(r)$,

$$\phi = \int \frac{M}{r^2 \sqrt{2m(E - U(r)) - \frac{M^2}{r^2}}} dr + \phi_0 \quad (3.5.2)$$

and transform it for our particular case $U = -\beta/r$. We first find that the effective potential energy is given by

$$U_{\text{eff}} = -\frac{\beta}{r} + \frac{M^2}{2mr^2} \quad (3.5.3)$$

Its minimum value is

$$U_{\text{eff,min}} = -\frac{m\beta^2}{2M^2} \quad (3.5.4)$$

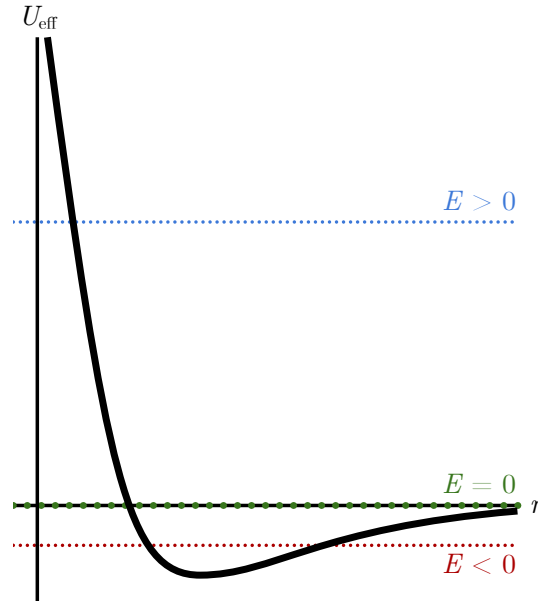


Figure 3. U_{eff} as a function of r for a potential $U(r) = -\beta/r$.

We see that for $E < 0$, the condition $T = E - U_{\text{eff}} > 0$ requires that the region in which the motion takes place be finite; $E > 0$ allows an unbounded region.

We now seek to describe the precise shape of the curve $\phi(r)$.

Proposition 3.5.1 (Newtonian/Coulombian Orbits)

Let a particle of mass m be immersed in a gravitational field of the form $U(r) = -\beta/r$. The orbits will take the form

$$\frac{\alpha}{r} = 1 + \varepsilon \cos \phi \quad (3.5.5)$$

⁶ The force varies as $1/r^2$, so the potential varies as $1/r$.

Proof

We begin with a substitution:

$$\phi = \int \frac{M/r^2}{\sqrt{2m(E + \beta/r) - M^2/r^2}} dr + \phi_0 \quad (3.5.6)$$

Let $u = 1/r$, which implies $du = -dr/r^2$, $dr = -r^2 du$ and $r = 1/u$:

$$\phi = \int \frac{M/r^2}{\sqrt{2m(E + \beta u) - M^2 u^2}} (-r^2 du) + \phi_0 \quad (3.5.7)$$

$$= - \int \frac{M}{\sqrt{-M^2 u^2 + 2m\beta u + 2mE}} du + \phi_0 \quad (3.5.8)$$

We know that

$$- \int \frac{1}{\sqrt{1-x^2}} dx = \arccos x + C \quad (3.5.9)$$

We complete the square in the denominator, making $-M^2 u^2 + 2m\beta u + 2mE$ take the form $M^2(A^2 - (u - u_0)^2)$:

$$M^2(A^2 - (u - u_0)^2) = A^2 M^2 - M^2 u_0^2 - M^2 u^2 + 2M^2 u_0 u \quad (3.5.10)$$

The coefficients of each power u^2 , u , and $u^0 = 1$ match the quantities in the expressions above:

$$2M^2 u_0 = 2m\beta \Rightarrow u_0 = \frac{m\beta}{M^2} \quad (3.5.11)$$

Thus,

$$A^2 = u_0^2 + \frac{2mE}{M^2} \quad (3.5.12)$$

Our integral becomes

$$\phi = - \int \frac{M}{\sqrt{M^2(A^2 - (u - u_0)^2)}} du + \phi_0 \quad (3.5.13)$$

$$= - \int \frac{1}{\sqrt{1 - \left(\frac{u - u_0}{A}\right)^2}} du + \phi_0 \quad (3.5.14)$$

in the desired form. Choose the origin of ϕ such that $\phi_0 = 0$. We therefore obtain

$$\phi = \arccos\left(\frac{u - u_0}{A}\right) \quad (3.5.15)$$

which in turn gives

$$\cos \phi = \frac{u - u_0}{A} \Rightarrow A \cos \phi = u - u_0 \Rightarrow A \cos \phi = \frac{1}{r} - u_0 \quad (3.5.16)$$

We rearrange:

$$1 + \frac{A}{u_0} \cos \phi = \frac{u_0^{-1}}{r} \quad (3.5.17)$$

This yields the form of a conic section:

$$\frac{\alpha}{r} = 1 + \varepsilon \cos \phi \quad (3.5.18)$$

with

$$\alpha = u_0^{-1} \quad ; \quad \varepsilon = \frac{A}{u_0} \quad (3.5.19)$$

which have the values

$$\alpha = \frac{M^2}{m\beta} \quad ; \quad \varepsilon = \sqrt{\frac{A^2}{u_0^2}} = \sqrt{1 + \frac{2mE}{M^2} \cdot \frac{M^4}{m^2\beta^2}} = \sqrt{1 + \frac{2M^2E}{m\beta^2}} \quad (3.5.20)$$

as desired. $\square$

Now, let us determine the criterion for choosing among hyperbolas, parabolas, ellipses, and circles. We know that the eccentricity of a conic section determines the choice mentioned—let us see the dependence of the eccentricity on the energy. If the energy is positive,

$$\varepsilon^2 = 1 + \frac{2M^2E}{m\beta^2} > 1 \Rightarrow \varepsilon > 1 \quad (3.5.21)$$

which corresponds to the hyperbola. If the energy is 0,

$$\varepsilon^2 = 1 + \frac{2M^2\cancel{E}}{m\beta^2} = 1 \Rightarrow \varepsilon = 1 \quad (3.5.22)$$

which corresponds to the parabola. If the energy is negative,

$$\varepsilon^2 = 1 + \frac{2M^2\cancel{E}}{m\beta^2} \in [0, 1) \Rightarrow \varepsilon \in [0, 1) \quad (3.5.23)$$

which corresponds to the ellipse (and the circle for $\varepsilon = 0$).

Consider the case of the ellipse. The minimal and maximal radii are

$$r_{\min} = \frac{\alpha}{1 + \varepsilon} \quad ; \quad r_{\max} = \frac{\alpha}{1 - \varepsilon} \quad (3.5.24)$$

The period is given by conservation of angular momentum. We have already proved that $2mf = M$ in (3.4.9). Thus, if f is πab , the area of the ellipse, we obtain

$$2mf = TM \quad (3.5.25)$$

with T the period. Expressing a and b in terms of β and E :

$$a = \frac{\alpha}{1 - \varepsilon^2} \quad ; \quad b = \sqrt{\alpha a} \quad (3.5.26)$$

and thus

$$T = 2\pi a^{3/2} \sqrt{\frac{m}{\beta}} = 2\pi \sqrt{\frac{a^3}{GM}} = \pi\beta \sqrt{\frac{m}{2E^3}} \quad (3.5.27)$$

We observe that the first/second expression gives Kepler's Third Law, by which $T^2 \propto a^3$.

3.6 Examples

3.6.1 Spatial Oscillator

Consider a particle in a region of space with a potential such that

$$U(r) = \frac{kr^2}{2} \quad (3.6.1)$$

We can derive the analogues of Kepler's Laws for such a system. Let us write the Lagrangian:

$$L = \frac{m\mathbf{r}^2}{2} - \frac{kr^2}{2} \quad (3.6.2)$$

It is convenient to observe that indeed this system is a set of coupled oscillators, such that x , y , and z are the normal coordinates. We can eliminate one by observing that angular momentum is conserved:

$$|\dot{\ell}| = \dot{p}_\theta \equiv \frac{\partial L}{\partial \theta} = 0 \Rightarrow |\ell| = \text{ct} \quad (3.6.3)$$

Thus, the motion will remain in a plane, and so we choose x and y to represent the Cartesian coordinates of this plane, such that $z = 0$. Thus, the Lagrangian is expressed as follows:

$$L = \frac{m(\dot{x}^2 + \dot{y}^2)}{2} - \frac{k(x^2 + y^2)}{2} \quad (3.6.4)$$

The Euler–Lagrange equation then gives

$$m\ddot{x} + kx = 0 \quad ; \quad m\ddot{y} + ky = 0 \quad (3.6.5)$$

and thus

$$\ddot{x} + \frac{k}{m}x = 0 \quad ; \quad \ddot{y} + \frac{k}{m}y = 0 \quad (3.6.6)$$

Thus, x and y each follow a sinusoidal oscillatory motion, described by

$$\mathbf{r} = \langle x, y \rangle = \left\langle a \cos \left(\sqrt{\frac{k}{m}} t \right), b \sin \left(\sqrt{\frac{k}{m}} t \right) \right\rangle \quad (3.6.7)$$

if we choose the time origin such that the particle is at $\langle a, 0 \rangle$ and the amplitude in y is given by b . This describes an ellipse (Kepler’s 1st Law):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (3.6.8)$$

Because angular momentum is conserved, Kepler’s Second Law also applies:

$$\dot{f} = \frac{\ell}{2m} = \text{ct} \quad (3.6.9)$$

for $f = r^2\dot{\theta}/2$, the areal velocity. The Third Law, however, is modified:

$$\tau = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \Rightarrow \tau^2 = \frac{4\pi^2 m}{k} \quad (3.6.10)$$

There is no dependence on the size of the “orbit”—this is seen immediately from mechanical similarity (see Section 2.5).

4 Oscillatory Systems

4.1 One-Dimensional Motion

We recall the Lagrangian of a simple harmonic oscillation:

$$L = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2} \quad (4.1.1)$$

The associated equation of motion is

$$m\ddot{x} + kx = 0 \quad (4.1.2)$$

which, for $\omega \equiv \sqrt{k/m}$, gives

$$\ddot{x} + \omega^2 x = 0 \quad (4.1.3)$$

The variable m denotes a mass *only* if x is the Cartesian coordinate. The solution takes the form

$$x = a \cos(\omega t + \alpha) \quad (4.1.4)$$

where a and α are constants that depend on the initial state. The energy of such a system is given by

$$E = \frac{m\dot{x}^2}{2} + \frac{kx^2}{2} = \frac{m\omega^2 a^2}{2} \quad (4.1.5)$$

where a is the amplitude of the oscillations, from the equation of motion.

4.2 Multi-Dimensional Oscillations

Suppose we have a system with s degrees of freedom, such that a stable equilibrium point occurs for

$$\{q_{i0}\} \quad (4.2.1)$$

Let

$$x_i \equiv q_i - q_{i0} \quad (4.2.2)$$

For small displacements, we use the quadratic form

$$U = \frac{1}{2} \sum_{i,k} k_{ik} x_i x_k = \frac{1}{2} \mathbf{x}^T [\mathcal{K}] \mathbf{x} \quad (4.2.3)$$

for $\mathbf{x} \equiv \langle \cdots x_i \cdots \rangle$. This form is positive definite. The kinetic energy is similarly given by a positive definite quadratic form:

$$T = \frac{1}{2} \sum_{i,k} m_{ik} \dot{x}_i \dot{x}_k = \frac{1}{2} \dot{\mathbf{x}}^T [\mathcal{M}] \dot{\mathbf{x}} \quad (4.2.4)$$

Thus,

$$L = T - U = \frac{1}{2} \sum_{i,k} m_{ik} \dot{x}_i \dot{x}_k - \frac{1}{2} \sum_{i,k} k_{ik} x_i x_k \quad (4.2.5)$$

The differential of L is

$$dL = \frac{1}{2} \left(\sum_{i,k} m_{ik} \dot{x}_i d\dot{x}_k + \sum_{i,k} m_{ik} d\dot{x}_i \dot{x}_k - \sum_{i,k} k_{ik} x_i dx_k - \sum_{i,k} k_{ik} dx_i x_k \right) \quad (4.2.6)$$

$$= \sum_{i,k} m_{ik} \dot{x}_i d\dot{x}_k - \sum_{i,k} k_{ik} dx_i x_k \quad (4.2.7)$$

Thus,

$$\frac{\partial L}{\partial \dot{x}_i} = \sum_k m_{ik} \dot{x}_k \quad ; \quad \frac{\partial L}{\partial x_i} = - \sum_k k_{ik} x_k \quad (4.2.8)$$

The equations of motion are therefore

$$\sum_k m_{ik} \ddot{x}_k = - \sum_k k_{ik} x_k \quad (4.2.9)$$

We propose the following ansatz:

$$x_k = A_k e^{i\omega t} \iff \mathbf{x} = \mathbf{A} e^{i\omega t} \quad (4.2.10)$$

Thus,

$$\sum_k (-\omega^2 m_{ik} + k_{ik}) A_k = 0 \quad (4.2.11)$$

which requires that the following determinant be 0:

$$|k_{ik} - \omega^2 m_{ik}| = 0 \quad (4.2.12)$$

This equation is called the *characteristic polynomial* of the system; it is of degree s in ω^2 ; its solutions are the natural frequencies of the system. They are all positive because the tensors k and m are positive definite.

We can use the minors of the last expression to find the normal modes $\{\mathbf{A}\}$. Let

$$D \equiv [\mathcal{K}] - \omega^2 [\mathcal{M}] \quad (4.2.13)$$

We therefore seek

$$D\mathbf{A} = \mathbf{0} \quad (4.2.14)$$

Because D is positive definite, we have

$$(\text{adj } D) D = D (\text{adj } D) = |D| \cdot I \quad (4.2.15)$$

We have chosen $|D| = 0$. Thus,

$$D (\text{adj } D) = 0 \quad (4.2.16)$$

which indicates that any column of $\text{adj } D$ belongs to the null space of D . Thus, because any normal mode $\mathbf{A}$ is also in the null space of D , the columns of $\text{adj } D$ and $\mathbf{A}$ are proportional:

$$\mathbf{A}_\alpha \propto \text{adj } D \Big|_{\omega_a^2} \quad (4.2.17)$$

We know that

$$(\text{adj } D)_{kj} = (-1)^{j+k} \Delta_{jk} \quad (4.2.18)$$

where Δ_{jk} is the (j, k) minor of D . Thus,

$$A_{k\alpha} \propto \Delta_{k\alpha} \quad (4.2.19)$$

Thus,

$$x_k = \text{Re} \sum_{\alpha=1}^s \Delta_{k\alpha} C_\alpha e^{i\omega_\alpha t} \equiv \sum_{\alpha} \Delta_{k\alpha} \Theta_\alpha \quad (4.2.20)$$

where

$$\Theta_\alpha \equiv \text{Re} (C_\alpha e^{i\omega_\alpha t}) \quad (4.2.21)$$

are the “normal coordinates” satisfying

$$\ddot{\Theta}_\alpha + \omega_\alpha^2 \Theta_\alpha = 0 \quad (4.2.22)$$

These are simple harmonic oscillators all independent of one another! The Lagrangian simplifies:

$$L = \sum_{\alpha} \frac{m_{\alpha}}{2} (\dot{\Theta}_{\alpha}^2 - \omega_{\alpha}^2 \Theta_{\alpha}^2) \quad (4.2.23)$$

Note that m_{α} is no longer a mass, but simply a coefficient (which depends on the masses). It is possible to eliminate this issue by defining

$$Q_{\alpha} \equiv \sqrt{m_{\alpha}} \Theta_{\alpha} \quad (4.2.24)$$

such that

$$L = \frac{1}{2} \sum_{\alpha} (\dot{Q}_{\alpha}^2 - \omega_{\alpha}^2 Q_{\alpha}^2) \quad (4.2.25)$$

It is also possible to simplify the forced problem by decomposing the forces F_k such that

$$f_{\alpha}(t) = \frac{1}{\sqrt{m_{\alpha}}} \sum_k F_k(t) \Delta_{k\alpha} \quad (4.2.26)$$

which give

$$\ddot{Q}_{\alpha} + \omega_{\alpha}^2 Q_{\alpha} = f_{\alpha}(t) \quad (4.2.27)$$

5 Constrained Motion

5.1 Holonomic Constraints

Consider a system with s degrees of freedom, such that there are k constraints. Let us express them with functions

$$f_k(q_i, t) = 0 \quad (5.1.1)$$

representing a point $\{q_i\}$ satisfying the constraint. We begin again from Hamilton's Principle:

$$\delta S = 0 \quad (5.1.2)$$

The new Lagrangian will be

$$L_\lambda(q, \dot{q}, t, \lambda_k) = L(q, \dot{q}, t) + \sum_k \lambda_k(t) f_k(q, t) \quad (5.1.3)$$

where λ_k are Lagrange multipliers, which allow the use of optimization via Lagrange multipliers. Hamilton's Principle gives

$$0 = \delta S = \delta \int_{t_1}^{t_2} L_\lambda(q, \dot{q}, t, \lambda_k) dt \quad (5.1.4)$$

$$= \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right) \delta q_i + \sum_{k,i} \left(\lambda_k \frac{\partial f_k}{\partial q_i} \delta q_i + f_k \delta \lambda_k \right) dt \quad (5.1.5)$$

Because this holds for all $\delta \lambda_k$ and all δq_i , we obtain an additional term in the Euler–Lagrange equation and also the constraint equation again:

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = -\lambda_k \frac{\partial f_k}{\partial q_i} \quad (5.1.6)$$

and

$$f_k = 0 \quad (5.1.7)$$

Proposition 5.1.1 (Euler–Lagrange Equation Under Holonomic Constraints)

Let a system have s degrees of freedom, such that there are k holonomic constraints. That is, the constraints are functions that depend only on time and the coordinates. The Euler–Lagrange equation, in this case, becomes

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = -\lambda_k \frac{\partial f_k}{\partial q_i} \quad (5.1.8)$$

for $f_k = 0$ the constraint functions. The constraint forces are given by

$$\mathbf{F}_i = \lambda_k \frac{\partial f_k}{\partial q_i} \hat{\mathbf{q}}_i \quad (5.1.9)$$

where $\hat{\mathbf{q}}_i$ denotes the direction of the coordinate q_i .

5.2 Non-Holonomic Constraints

We can treat certain non-holonomic constraints as holonomic ones if they contain a total time derivative.

Suppose that the constraint $f(q, \dot{q}, t)$ takes the form

$$f(q, \dot{q}, t) = \frac{\partial g}{\partial t} + \sum_i \frac{\partial g}{\partial q_i} \dot{q}_i \quad (5.2.1)$$

Let $\alpha_i \equiv \frac{\partial g}{\partial q_i}$ and $\beta = \frac{\partial g}{\partial t}$. Using Clairaut's theorem, we obtain the following constraints.

Proposition 5.2.1 (Non-Holonomic Constraints That Can Be Used for Lagrange)

A function $f(q, \dot{q}, t)$ can be modeled as a holonomic constraint if and only if

$$\exists \{\alpha_i(q, t)\}, \beta(q, t) : f = \sum_i \alpha_i(q, t) \dot{q}_i + \beta(q, t) \quad (5.2.2)$$

where

$$\frac{\partial \alpha_i}{\partial q_j} = \frac{\partial \alpha_j}{\partial q_i} \quad \text{and} \quad \frac{\partial \beta}{\partial q_i} = \frac{\partial \alpha_i}{\partial t} \quad (5.2.3)$$

In this case, one must use the partial derivative with respect to $\dot{q}_i$ and not q_i for this constraint.

In any case, if $\partial f / \partial \dot{q}_i$ is anything other than a function of q, t , it is not possible to carry out the holonomic procedure.

However, in the general case this is no longer possible. We therefore use the approach developed by d'Alembert.

Proposition 5.2.2 (d'Alembert Treatment of Non-Holonomic Constraints)

Let a system have s degrees of freedom. Let L be its Lagrangian. If the constraints f_c of this system take the form

$$f_c = \sum_i \gamma_i^c(\mathbf{q}) \dot{q}_i = 0 \quad (5.2.4)$$

for functions γ_i^c that depend only on the coordinates, it is possible to recover the equations of motion and the forces F_i acting on each coordinate using the Euler–Lagrange–d'Alembert equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \sum_c F_i^c = \sum_c \lambda_c \gamma_i^c(\mathbf{q}) \quad (5.2.5)$$

If we write in terms of matrices, for a case with four degrees of freedom, the force acting on each coordinate is given by F_i below:

$$F_i = \begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 \end{bmatrix} \begin{bmatrix} \gamma_1^1 & \gamma_2^1 & \gamma_3^1 & \gamma_4^1 \\ \gamma_1^2 & \gamma_2^2 & \gamma_3^2 & \gamma_4^2 \\ \gamma_1^3 & \gamma_2^3 & \gamma_3^3 & \gamma_4^3 \\ \gamma_1^4 & \gamma_2^4 & \gamma_3^4 & \gamma_4^4 \end{bmatrix} \begin{bmatrix} \delta_{1i} \\ \delta_{2i} \\ \delta_{3i} \\ \delta_{4i} \end{bmatrix} \quad (5.2.6)$$

with δ_{ij} the identity tensor (1 for $i = j$ and 0 otherwise).

Proposition 5.2.3 (Conservation of Energy for Non-Holonomic Constraints)

Let a system have s degrees of freedom. Let L be its Lagrangian. Let γ_i^c be the coordinate functions that can be used to describe non-holonomic constraints of the form

$$f_c = \sum_i \gamma_i^c(\mathbf{q}) \dot{q}_i = 0 \quad (5.2.7)$$

If the work done by these forces is zero, the energy of the system is conserved.

Proof

The work done by the c -th force is given by

$$W_c = \mathbf{F}^c \cdot \delta \mathbf{q} \quad (5.2.8)$$

Because $\delta q_i \approx \dot{q}_i \delta t$, and δt being arbitrary, we factor it out of the expression when we set the work $W_c = 0$:

$$\sum F_i^c \dot{q}_i = 0 \quad (5.2.9)$$

This is independent of λ_c in the expression for F_i^c , and thus

$$\sum \gamma_i^c(\mathbf{q})\dot{q}_i = 0 \quad (5.2.10)$$

We now seek to prove that $E \equiv \sum_i (\partial L / \partial \dot{q}_i) \dot{q}_i - L$ is constant. That is, if we show that its total time derivative $\dot{E}$ is 0, energy is conserved. We begin:

$$\sum_i \gamma_i^c(\mathbf{q})\dot{q}_i = 0 \quad (5.2.11)$$

This gives

$$\forall c : \lambda_c \sum_i \gamma_i^c(\mathbf{q})\dot{q}_i = 0 \quad (5.2.12)$$

Writing the sum over c does not change the value (still zero); we also change the order of summation:

$$0 = \sum_c \lambda_c \sum_i \gamma_i^c(\mathbf{q})\dot{q}_i = \sum_c \sum_i \lambda_c \gamma_i^c(\mathbf{q})\dot{q}_i = \sum_i \sum_c \lambda_c \gamma_i^c(\mathbf{q})\dot{q}_i = \sum_i \left(\sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \quad (5.2.13)$$

It is convenient to add and subtract the partial derivatives of L with respect to q_i and $\dot{q}_i$, with suitable factors:

$$0 = \sum_i \left(\sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \quad (5.2.14)$$

$$= \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i - \frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i - \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \left(\sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \right) \quad (5.2.15)$$

We rearrange, recognizing the total time derivative of L :

$$0 = \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \left(\sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \right) - \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right) \quad (5.2.16)$$

$$= \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \left(\sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \right) - \frac{dL}{dt} \quad (5.2.17)$$

We rearrange the first sum:

$$0 = \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \left(\sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \right) - \frac{dL}{dt} \quad (5.2.18)$$

$$= \sum_i \left(\frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \left(\frac{\partial L}{\partial q_i} + \sum_c \lambda_c \gamma_i^c(\mathbf{q}) \right) \dot{q}_i \right) - \frac{dL}{dt} \quad (5.2.19)$$

The Euler–Lagrange–d’Alembert equation tells us that

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i} + \sum_c \lambda_c \gamma_i^c(\mathbf{q}) \quad (5.2.20)$$

We substitute this expression and display the result as a derivative of a product, as well as the total time derivative of the whole expression:

$$0 = \sum_i \left(\frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i + \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) - \frac{dL}{dt} \quad (5.2.21)$$

$$= \sum_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) - \frac{dL}{dt} \quad (5.2.22)$$

$$= \frac{d}{dt} \left(\sum_i \left(\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) - L \right) \quad (5.2.23)$$

$$= \frac{d}{dt} (E) \quad (5.2.24)$$

$$= \dot{E} \quad (5.2.25)$$

Thus, the energy E is conserved, as desired. $\square$

5.3 Examples

5.3.1 Simple Pendulum

Let us take the trivial example of the simple pendulum. The constraint would be

$$f(q, t) = r - l = 0 \quad (5.3.1)$$

Thus, the Lagrangian takes the form

$$L = \frac{m(\dot{r}^2 + r^2\dot{\theta}^2)}{2} + mgr \cos \theta \quad (5.3.2)$$

where we note that now r is free to vary, subject to the constraint f that will be applied later. The modified Euler–Lagrange equations are thus

$$\frac{\partial L}{\partial r} - \frac{d}{dt} \frac{\partial L}{\partial \dot{r}} = -\lambda \frac{\partial f}{\partial r} \quad (5.3.3)$$

and

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = -\lambda \frac{\partial f}{\partial \theta} \quad (5.3.4)$$

We see that $\partial f / \partial r = 1$ and that $\partial f / \partial \theta = 0$. Thus,

$$-m\ddot{r} + mg \cos \theta + mr\dot{\theta}^2 = -\lambda \quad (5.3.5)$$

and

$$-mgr \sin \theta - mr^2\ddot{\theta} - 2mr\dot{r}\dot{\theta} = 0 \quad (5.3.6)$$

which simplify to give

$$\ddot{r} = \frac{\lambda}{m} + g \cos \theta + r\dot{\theta}^2 \quad (5.3.7)$$

and

$$\ddot{\theta} + \frac{g}{r} \sin \theta + \frac{2\dot{r}\dot{\theta}}{r} = 0 \quad (5.3.8)$$

We apply the constraint, which gives $\dot{r} = \ddot{r} = 0$. Thus,

$$\lambda = -mg \cos \theta - mr\dot{\theta}^2 \quad (5.3.9)$$

and

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0 \quad (5.3.10)$$

which is indeed the pendulum equation. The tension in the string is therefore

$$F_T = \lambda \frac{\partial f}{\partial r} \hat{\mathbf{r}} + \lambda \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} = -(mg \cos \theta + mr\dot{\theta}^2) \hat{\mathbf{r}} \quad (5.3.11)$$

which is indeed the sum of the centripetal force and gravity.

5.3.2 Non-Holonomic Constraint

Let two rings be in contact such that they roll without friction. Let the origin be at the center of one of the rings, with radius R . Let the position of the second ring be given in polar coordinates by r, θ . Let ϕ be the rotation angle of the second ring, with the positive direction being clockwise, and let a be its radius. The mass of each ring is concentrated at the center of the ring. The contact constraint is

$$f_1 = r - (R + a) = 0 \quad (5.3.12)$$

The no-slip (rolling) contact constraint is

$$f_2 = (R + a)\dot{\theta} + a\dot{\phi} = 0 \quad (5.3.13)$$

We seek the angle at which contact is lost.

The Lagrangian is therefore

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{ma^2\dot{\phi}^2}{2} - mgr \sin \theta \quad (5.3.14)$$

We verify that f_2 satisfies Prop. 5.2.1: it is already linear in $\dot{\theta}$ and $\dot{\phi}$.

Here, we write

$$f_2 = (R + a)\dot{\theta} + a\dot{\phi} = \alpha_2\dot{\theta} + \alpha_3\dot{\phi} \quad (5.3.15)$$

and thus we must verify that

$$\frac{\partial \alpha_i}{\partial q_j} = \frac{\partial \alpha_j}{\partial q_i} \quad (5.3.16)$$

for any pair i, j in $\{1, 2, 3\} \times \{1, 2, 3\}$. Clearly, if i or j is 1, corresponding to the coordinate r , the relation is satisfied. For $i = 2$ and $j = 3$:

$$\frac{\partial \alpha_2}{\partial \phi} = \frac{\partial \alpha_3}{\partial \theta} \Rightarrow 0 = 0 \quad (5.3.17)$$

which is indeed true. Because $\beta = 0$ and there is no explicit time dependence, we also satisfy the second condition of the proposition.

The modified Euler–Lagrange equation for r is

$$\frac{\partial L}{\partial r} - \frac{d}{dt} \frac{\partial L}{\partial \dot{r}} = -\lambda_1 \frac{\partial f_1}{\partial r} - \lambda_2 \frac{\partial f_2}{\partial r} \quad (5.3.18)$$

The partial derivatives give

$$mr\dot{\theta}^2 - mg \sin \theta - m\ddot{r} = \lambda_1 + 0 \quad (5.3.19)$$

which gives

$$m(\ddot{r} - r\dot{\theta}^2) = \lambda_1 - mg \sin \theta \quad (5.3.20)$$

We see that λ_1 corresponds to a radial force, the normal force between the rings. For θ , we obtain

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} = -\lambda_1 \frac{\partial f_1}{\partial \theta} - \lambda_2 \frac{\partial f_2}{\partial \theta} \quad (5.3.21)$$

The partial derivatives give

$$-mgr \cos \theta - \frac{d}{dt}(mr^2\dot{\theta}) = 0 - \lambda_2(R + a) \quad (5.3.22)$$

Thus,

$$m(r^2\ddot{\theta} + 2r\dot{r}\dot{\theta}) = \lambda_2(R + a) - mgr \cos \theta \quad (5.3.23)$$

We see that λ_2 corresponds to a torque due to friction, in the $\hat{\theta}$ direction, which allows rolling without slipping. For ϕ , we obtain

$$\frac{\partial L}{\partial \phi} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}} = -\lambda_1 \frac{\partial f_1}{\partial \phi} - \lambda_2 \frac{\partial f_2}{\partial \phi} \quad (5.3.24)$$

The partial derivatives give

$$-ma^2\ddot{\phi} = -a\lambda_2 \Rightarrow \lambda_2 = ma\ddot{\phi} \quad (5.3.25)$$

Now, we begin to solve the system. The first constraint gives $\dot{r} = \ddot{r} = 0$ and $r = R + a$. Thus, equation (5.3.20) becomes

$$-m(R + a)\dot{\theta}^2 + mg \sin \theta = \lambda_1 \quad (5.3.26)$$

For θ , we obtain

$$m(R + a)\ddot{\theta} + mg \cos \theta = \lambda_2 \quad (5.3.27)$$

and thus

$$m(R + a)\ddot{\theta} + mg \cos \theta = \lambda_2 = m\ddot{\phi} \quad (5.3.28)$$

Through the constraint f_2 , the last equation becomes

$$m(R + a)\ddot{\theta} + mg \cos \theta = -m(R + a)\ddot{\theta} \quad (5.3.29)$$

Thus,

$$2(R + a)\ddot{\theta} = -g \cos \theta \quad (5.3.30)$$

Multiplying both sides by $\dot{\theta}$:

$$2(R + a)\ddot{\theta}\dot{\theta} = -g \cos \theta \dot{\theta} \quad (5.3.31)$$

We integrate both sides:

$$2(R + a)\dot{\theta}^2 = -g \sin \theta + C \quad (5.3.32)$$

where C satisfies the initial conditions, which are $\dot{\theta}(0) = 0$ and $\theta(0) = \pi/2$, and thus

$$2(R + a)\dot{\theta}^2 = g(1 - \sin \theta) \quad (5.3.33)$$

Thus,

$$\lambda_1 = mg(2 \sin \theta - 1) \quad (5.3.34)$$

We seek the moment when the force associated with λ_1 becomes zero:

$$0 = mg(2 \sin \theta - 1) \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \boxed{\theta = 30^\circ} \quad (5.3.35)$$

5.3.3 Coin on a Ramp

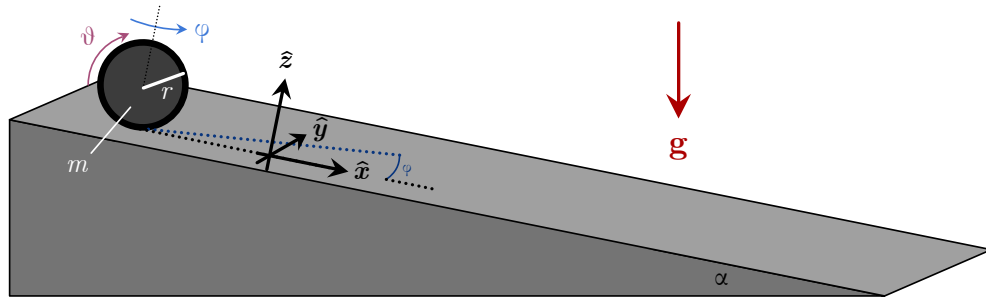


Figure 4. Coin on a Ramp

Consider a coin of mass m and radius r on a ramp at angle α . Let us take the coordinate system drawn in the figure. This coin rolls without friction; we seek to characterize its motion.

The position of the center of mass is given by

$$\mathbf{r}_{\text{CM}} = \begin{bmatrix} x \\ y \\ r \end{bmatrix} \quad (5.3.36)$$

Its time derivative is

$$\dot{\mathbf{r}}_{\text{CM}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ 0 \end{bmatrix} \quad (5.3.37)$$

Because the coin rolls without slipping, we write:

$$\mathbf{v}_{\text{contact}} = \dot{\mathbf{r}}_{\text{CM}} + \boldsymbol{\Omega} \times \mathbf{r}_{\text{contact}} = \mathbf{0} \quad (5.3.38)$$

The magnitude of the angular velocity is $|\boldsymbol{\Omega}| = \dot{\theta}$; its direction is

$$\widehat{\boldsymbol{\Omega}} = \begin{bmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{bmatrix} \quad (5.3.39)$$

where we measure ϕ from the axis $\hat{\mathbf{x}}$ to the projection of the coin onto the xy plane. The vector from the CM to the contact point is

$$\mathbf{r} = \begin{bmatrix} 0 \\ 0 \\ -r \end{bmatrix} \quad (5.3.40)$$

Thus, their cross product is

$$\boldsymbol{\Omega} \times \mathbf{r}_{\text{contact}} = |\boldsymbol{\Omega}| \cdot (\widehat{\boldsymbol{\Omega}} \times \mathbf{r}_{\text{contact}}) \quad (5.3.41)$$

$$= |\boldsymbol{\Omega}| \cdot \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & -r \end{vmatrix} \quad (5.3.42)$$

$$= |\boldsymbol{\Omega}| \cdot \begin{bmatrix} -r \cos \phi \\ -r \sin \phi \\ 0 \end{bmatrix} \quad (5.3.43)$$

The constraint is

$$\mathbf{0} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ 0 \end{bmatrix} + \dot{\theta} \cdot \begin{bmatrix} -r \cos \phi \\ -r \sin \phi \\ 0 \end{bmatrix} \iff \boxed{\begin{cases} 0 = \dot{x} - r \cos \phi \dot{\theta} \\ 0 = \dot{y} - r \sin \phi \dot{\theta} \end{cases}} \quad (5.3.44)$$

We use the d'Alembert framework. The functions γ_i^c are therefore

$$\underline{\gamma} = \begin{bmatrix} \gamma_1^1 & \gamma_2^1 & \gamma_3^1 & \gamma_4^1 \\ \gamma_1^2 & \gamma_2^2 & \gamma_3^2 & \gamma_4^2 \\ \gamma_1^3 & \gamma_2^3 & \gamma_3^3 & \gamma_4^3 \\ \gamma_1^4 & \gamma_2^4 & \gamma_3^4 & \gamma_4^4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -r \cos \phi & 0 \\ 0 & 1 & -r \sin \phi & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (5.3.45)$$

and thus the forces associated with the constraints are

$$\mathbf{F}_i = [\lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \lambda_4] \begin{bmatrix} \gamma_1^1 & \gamma_2^1 & \gamma_3^1 & \gamma_4^1 \\ \gamma_1^2 & \gamma_2^2 & \gamma_3^2 & \gamma_4^2 \\ \gamma_1^3 & \gamma_2^3 & \gamma_3^3 & \gamma_4^3 \\ \gamma_1^4 & \gamma_2^4 & \gamma_3^4 & \gamma_4^4 \end{bmatrix} \begin{bmatrix} \delta_{1i} \\ \delta_{2i} \\ \delta_{3i} \\ \delta_{4i} \end{bmatrix} \quad (5.3.46)$$

where δ_{ij} is the identity tensor. In our case, the only nonzero functions are

$$\boxed{\gamma_1^1 = 1} \quad ; \quad \boxed{\gamma_3^1 = -r \cos \phi} \quad ; \quad \boxed{\gamma_2^2 = 1} \quad ; \quad \boxed{\gamma_3^2 = -r \sin \phi} \quad (5.3.47)$$

We see quite clearly that this constraint cannot be reduced to a holonomic constraint. To express it in holonomic form, each constraint must take the form

$$f = \sum \alpha_i(q, t) \dot{q}_i + \beta(q, t) \quad (5.3.48)$$

with the following conditions on α and β satisfied:

$$\frac{\partial \alpha_i}{\partial q_j} = \frac{\partial \alpha_j}{\partial q_i} \quad ; \quad \frac{\partial \beta}{\partial q_i} = \frac{\partial \alpha_i}{\partial t} \quad (5.3.49)$$

In our case, $\alpha_i = \gamma_i^1$ and $\beta = 0$. Take $i = 3$ and $j = 4$:

$$\frac{\partial}{\partial \phi}(-r \cos \phi) \stackrel{?}{=} \frac{\partial}{\partial \theta}(0) \quad (5.3.50)$$

$$\Rightarrow r \sin \phi \stackrel{?}{=} 0 \quad (5.3.51)$$

which is clearly not always true. Thus, the constraints are non-holonomic, and we proceed with the d'Alembert framework.

We compute the Lagrangian and the equations of motion. The kinetic energy is given by

$$T = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) + \frac{mr^2}{2} \frac{\dot{\theta}^2}{2} + \frac{mr^2}{4} \frac{\dot{\phi}^2}{2} \quad (5.3.52)$$

in which we used the *Perpendicular Axis Theorem*, which tells us that the moment of inertia of a planar object about the axis normal to the plane is the sum of the moments of inertia about the other two axes. Thus, by the symmetry of the coin, I_{xx} and I_{zz} are identical, and thus $I_{yy} = I_{xx} + I_{zz}$ which gives

$$I_{zz} = \frac{I_{yy}}{2} = \frac{mr^2}{4} \quad (5.3.53)$$

The gravitational potential energy is

$$U_g = -mg(\mathbf{r}_{\text{CM}} \cdot \hat{\mathbf{g}}) = -mg(x \sin \alpha - r \cos \alpha) \quad (5.3.54)$$

Thus,

$$L = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) + \frac{mr^2}{4} \dot{\theta}^2 + \frac{mr^2}{8} \dot{\phi}^2 + mgx \sin \alpha \quad (5.3.55)$$

where we eliminated the constant (and thus total time derivative) $-mgr \cos \alpha$. We also observe that $\dot{x}^2 + \dot{y}^2$ can be simplified using the constraints:

$$\dot{x}^2 + \dot{y}^2 = r^2 \cos^2 \phi \dot{\theta}^2 + r^2 \sin^2 \phi \dot{\theta}^2 = r^2 \dot{\theta}^2 \quad (5.3.56)$$

Thus, we can express the Lagrangian in terms of only x , $\dot{\phi}$, and $\dot{\theta}$:

$$\boxed{L = \frac{3mr^2}{4} \dot{\theta}^2 + \frac{mr^2}{8} \dot{\phi}^2 + mgx \sin \alpha} \quad (5.3.57)$$

We apply the Euler–Lagrange–d'Alembert equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = \sum_c \lambda_c \gamma_i^c(\mathbf{q}) \implies \begin{cases} \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = \lambda_1 \\ \frac{d}{dt} \frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = \lambda_2 \\ \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = -\lambda_1 r \cos \phi - \lambda_2 r \sin \phi \\ \frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}} - \frac{\partial L}{\partial \phi} = 0 \end{cases} \quad (5.3.58)$$

The nonzero partial derivatives are

$$\frac{\partial L}{\partial x} = mg \sin \alpha \quad ; \quad \frac{\partial L}{\partial \dot{\theta}} = \frac{3mr^2}{2} \dot{\theta} \quad ; \quad \frac{\partial L}{\partial \dot{\phi}} = \frac{3mr^2}{4} \dot{\phi} \quad (5.3.59)$$

The equations become

$$\begin{cases} \lambda_1 = -mg \sin \alpha \\ \lambda_2 = 0 \\ \ddot{\theta} = -\frac{2}{3mr} (\lambda_1 \cos \phi + \lambda_2 \sin \phi) \\ \ddot{\phi} = 0 \end{cases} \quad (5.3.60)$$

We substitute λ_1 and λ_2 :

$$\boxed{\ddot{\theta} + \frac{2g \sin \alpha}{3r} \cos \phi = 0} \quad ; \quad \boxed{\ddot{\phi} = 0} \quad (5.3.61)$$

From the relation $\ddot{\phi} = 0$, we integrate to obtain $\phi(t) = \dot{\phi}(0)t + \phi(0)$. Suppose the coin satisfies $\dot{\phi}(0) = \omega$, $\theta(0) = \phi(0) = \dot{\theta}(0) = 0$. Thus, $\boxed{\phi(t) = \omega t}$ which gives

$$\ddot{\theta} = -\frac{2g \sin \alpha}{3r} \cos \omega t \quad (5.3.62)$$

This gives

$$\theta = \frac{2g \sin \alpha}{3r\omega^2} \cos \omega t + c_1 t + c_2 \quad (5.3.63)$$

where c_1 and c_2 satisfy $\theta(0) = \dot{\theta}(0) = 0$, which implies

$$\boxed{\theta(t) = \frac{2g \sin \alpha}{3r\omega^2} (1 - \cos \omega t)} \quad (5.3.64)$$

We can therefore recover $x(t)$ and $y(t)$ from the constraints, such that

$$\dot{x} = r \cos(\omega t) \cdot \frac{d}{dt} \left(\frac{2g \sin \alpha}{3r\omega^2} (1 - \cos \omega t) \right) \quad (5.3.65)$$

$$= \frac{g \sin \alpha}{3\omega} \sin(2\omega t) \quad (5.3.66)$$

$$\Rightarrow x(t) = \frac{g \sin \alpha}{6\omega^2} \cos(2\omega t) + c_3 \quad (5.3.67)$$

where c_3 satisfies $x(0) = 0$:

$$\boxed{x(t) = \frac{g \sin \alpha}{6\omega^2} (1 - \cos(2\omega t))} \quad (5.3.68)$$

Similarly, $y(t)$ is found from its constraint:

$$\dot{y} = r \sin(\omega t) \cdot \frac{d}{dt} \left(\frac{2g \sin \alpha}{3r\omega^2} (1 - \cos \omega t) \right) \quad (5.3.69)$$

$$= \frac{2g \sin \alpha}{3\omega} \sin^2(\omega t) \quad (5.3.70)$$

$$= \frac{g \sin \alpha}{3\omega} (1 - \cos(2\omega t)) \quad (5.3.71)$$

$$\Rightarrow y(t) = \frac{g \sin \alpha}{3\omega^2} \left(\omega t - \frac{\sin(2\omega t)}{2} \right) + c_4 \quad (5.3.72)$$

where c_4 satisfies $y(t) = 0$, which is trivially the case, so

$$\boxed{y(t) = \frac{g \sin \alpha}{3\omega^2} \left(\omega t - \frac{1}{2} \sin 2\omega t \right)} \quad (5.3.73)$$

6 The Hamiltonian

6.1 Hamilton's/Canonical Equations

We begin with a mathematical tool. A *Legendre transform* allows us to eliminate a dependence of a function by exchanging it for another variable. For example, take a function $F(x, y)$. Its differential is

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \quad (6.1.1)$$

Let $u \equiv \frac{\partial F}{\partial x}$ and $v \equiv \frac{\partial F}{\partial y}$. Thus,

$$dF = u dx + v dy \quad (6.1.2)$$

Let $G = F - vy$. We then observe that

$$dG = u dx + v dy - v dy - y dv = u dx - y dv \quad (6.1.3)$$

Thus indeed $G = G(x, v)$ and

$$\frac{\partial G}{\partial y} = \frac{\partial F}{\partial y} - v = v - v = 0 \quad (6.1.4)$$

and therefore G no longer depends on y .

We will use this method to define the Hamiltonian.

Definition 6.1.1 (The Hamiltonian)

Let L be the Lagrangian of a system with s degrees of freedom. Then,

$$H(q, p, t) \equiv \sum p_i \dot{q}_i - L(q, \dot{q}, t) \quad (6.1.5)$$

where $p_i \equiv \partial L / \partial \dot{q}_i$.

Proposition 6.1.2 (The Hamiltonian as a Legendre Transform of the Lagrangian)

The Hamiltonian is a Legendre transform of the Lagrangian, such that the dependence on the velocities $\dot{q}_i$ is eliminated.

Proof

Trivially the case; we show that $\partial H / \partial \dot{q}_i = 0$:

$$\frac{\partial H}{\partial \dot{q}_i} = \sum p_i - \sum \frac{\partial L}{\partial \dot{q}_i} = \sum p_i - \sum p_i = 0 \quad (6.1.6)$$

as desired. □

Proposition 6.1.3 (Hamilton's Equations/Canonical Equations)

For a system with Hamiltonian H , the *canonical equations* are

$$\frac{\partial H}{\partial p_i} = \dot{q}_i \quad ; \quad \frac{\partial H}{\partial q_i} = -\dot{p}_i \quad (6.1.7)$$

We also observe that $\partial H / \partial t = -\partial L / \partial t$.

Proof

We begin from Hamilton's Principle, according to which

$$S = \int_{t_1}^{t_2} L dt = \int_{t_1}^{t_2} \left(\sum_i p_i \dot{q}_i - H(q, p, t) \right) dt \quad (6.1.8)$$

Thus, the variation must be zero:

$$\delta S = 0 = \int_{t_1}^{t_2} \left(\sum_i \delta p_i \dot{q}_i + p_i \delta \dot{q}_i - \frac{\partial H}{\partial q_i} \delta q_i - \frac{\partial H}{\partial p_i} \delta p_i \right) dt \quad (6.1.9)$$

We recognize that

$$\frac{d}{dt}(p_i \delta q_i) = p_i \delta \dot{q}_i + \dot{p}_i \delta q_i \quad (6.1.10)$$

and thus

$$\delta S = 0 = \int_{t_1}^{t_2} \left(\sum_i \delta p_i \left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right) - \delta q_i \left(\dot{p}_i + \frac{\partial H}{\partial q_i} \right) \right) dt \quad (6.1.11)$$

from which we deduce

$$\frac{\partial H}{\partial p_i} = \dot{q}_i \quad ; \quad \frac{\partial H}{\partial q_i} = -\dot{p}_i \quad (6.1.12)$$

as desired. $\square$

We can also write these equations in a compact way. Let $\boldsymbol{\eta}$ be a vector such that

$$[\boldsymbol{\eta}]_i = q_i \quad ; \quad [\boldsymbol{\eta}]_{n+i} = p_i \quad (6.1.13)$$

for $i \leq n$, where n is the number of degrees of freedom of the system in question. Thus,

$$\dot{\boldsymbol{\eta}} = \boldsymbol{J} \frac{\partial H}{\partial \boldsymbol{\eta}} \quad (6.1.14)$$

where $\boldsymbol{J}$ is a block matrix, defined below:

$$\boldsymbol{J} = \begin{bmatrix} \mathbf{0} & \boldsymbol{I}_n \\ -\boldsymbol{I}_n & \mathbf{0} \end{bmatrix} \in \mathbb{M}_{2n \times 2n} \quad (6.1.15)$$

This notation is called “symplectic.”

6.2 Poisson Bracket

We begin with the definition.

Definition 6.2.1 (Poisson Bracket)

Let F and G be two functions that depend on the generalized coordinates and generalized momenta q and p . The Poisson bracket of these two functions is given by

$$[F, G] = \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial G}{\partial q_i} \right) \quad (6.2.1)$$

We immediately deduce a few properties.

Proposition 6.2.2 (Properties of the Poisson Bracket)

Let F, G, H be functions. Let a, b be constants. Then,

- $[a, b] = 0$
- $[F, a] = 0$
- $[F, G] = -[G, F]$
- $[F + G, H] = [F, H] + [G, H]$

Concretely for the coordinates, we observe that for a set of coordinates following the Legendre transform, we have the following proposition.

Proposition 6.2.3 (Poisson Brackets for Canonical Coordinates)

Let $\{q, p\}$ be a set of canonical coordinates and canonical momenta. Then,

$$[q_i, q_j] = [p_i, p_j] = 0 \quad ; \quad [q_i, p_j] = \delta_{ij} \quad (6.2.2)$$

Proof

Clearly the coordinates are independent of one another, and the momenta are similar in this respect. Thus, the first two brackets are 0. The third is

$$[q_i, p_j] = \sum_{\ell} \left(\frac{\partial q_i}{\partial q_{\ell}} \frac{\partial p_j}{\partial p_{\ell}} - \frac{\partial q_i}{\partial p_{\ell}} \frac{\partial p_j}{\partial q_{\ell}} \right) = \sum_{\ell} (\delta_{i\ell} \delta_{j\ell} - 0) = \delta_{ij} \quad (6.2.3)$$

as desired. $\square$

Proposition 6.2.4 (Total Time Derivative with the Hamiltonian)

Let F be a function that depends on the coordinates and the momenta. Then,

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + [F, H] \quad (6.2.4)$$

Proof

We write it explicitly, substituting the canonical equations:

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \sum_i \left(\frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial p_i} \dot{p}_i \right) \quad (6.2.5)$$

$$= \frac{\partial F}{\partial t} + \sum_i \left(\frac{\partial F}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i} \frac{\partial H}{\partial q_i} \right) \quad (6.2.6)$$

$$= \frac{\partial F}{\partial t} + [F, H] \quad (6.2.7)$$

as desired. $\square$

Proposition 6.2.5 (Poisson Brackets with the Hamiltonian)

Let H be the Hamiltonian of a system described with coordinates q and momenta p . Then,

$$[q_i, H] = \dot{q}_i \quad ; \quad [p_i, H] = \dot{p}_i \quad (6.2.8)$$

Proof

This can be proved directly, or by using the previous proposition. $\square$

Proposition 6.2.6 (Conserved Quantity)

Let F be a function that depends on the coordinates and the momenta with $\partial F / \partial t = 0$. Then,

$$[F, H] = 0 \iff F \text{ is a conserved quantity.} \quad (6.2.9)$$

Proof

Use the total time derivative. $\square$

Property 6.2.7 (Torque with the Poisson Bracket)

Let the Hamiltonian of a system be such that

$$H = \frac{1}{2m} (p_x^2 + p_y^2 + p_z^2) + U(x, y, z) \quad (6.2.10)$$

Let the angular momentum ℓ be given by $\mathbf{r} \times \mathbf{p}$. We can recover the torque that would change the component ℓ_z using the Poisson bracket:

$$\frac{d\ell_z}{dt} = [l_z, H] = |\mathbf{r} \times \mathbf{F}|_z \quad (6.2.11)$$

Proof

We observe that the terms of the bracket with x and y will be the only nonzero ones:

$$\frac{d\ell_z}{dt} = [l_z, H] = \frac{\partial l_z}{\partial x} \frac{\partial H}{\partial p_x} - \frac{\partial l_z}{\partial p_x} \frac{\partial H}{\partial x} + \frac{\partial l_z}{\partial y} \frac{\partial H}{\partial p_y} - \frac{\partial l_z}{\partial p_y} \frac{\partial H}{\partial y} \quad (6.2.12)$$

$$= p_y \frac{p_x}{m} + y \frac{\partial U}{\partial x} - p_x \frac{p_y}{m} - x \frac{\partial U}{\partial y} \quad (6.2.13)$$

$$= -yF_x + xF_y \quad (6.2.14)$$

$$= |\mathbf{r} \times \mathbf{F}|_z \quad (6.2.15)$$

which is indeed the expected result. $\square$

Proposition 6.2.8 (Jacobi Identity)

Let F , G , and H be three functions. Then,

$$[H, [F, G]] + [F, [G, H]] + [G, [H, F]] = 0 \quad (6.2.16)$$

Proof

We do not prove this proposition here. See Section §42 of Landau. $\square$

Proposition 6.2.9 (Constructing a Conserved Quantity)

Let F , G be two functions. Let H be the Hamiltonian of a system. Suppose that

$$[F, H] = 0 = [G, H] \quad (6.2.17)$$

Then F and G are conserved. It follows that

$$\frac{d}{dt}[F, G] = 0 \quad (6.2.18)$$

as well.

Proof

We know that

$$\frac{d}{dt}[F, G] = [[F, G], H] \quad (6.2.19)$$

Applying the Jacobi Identity, we obtain

$$0 = [[F, G], H] + [F, [G, H]] + [G, [H, F]] = [[F, G], H] + \cancel{[F, 0]} + \cancel{[G, 0]} = [[F, G], H] = 0 \quad (6.2.20)$$

which is indeed the desired result. $\square$

Proposition 6.2.10 (Poisson Bracket of Angular Momenta)

Let ϵ_{ijk} be the permutation tensor. Then,

$$[\ell_i, \ell_j] = \epsilon_{ijk} \ell_k \quad (6.2.21)$$

for $\{\ell_i\}$ the angular momenta.

Proof

We prove for $i \rightarrow x$, $j \rightarrow y$, and $k \rightarrow z$:

$$[l_x, l_y] = [yp_z - zp_y, zp_x - xp_z] \quad (6.2.22)$$

$$= [yp_z, zp_x] - [yp_z, xp_z] - [zp_y, zp_x] + [zp_y, xp_z] \quad (6.2.23)$$

$$= -yp_x + p_yx \quad (6.2.24)$$

$$= \ell_z \quad (6.2.25)$$

Thus,

$$[\ell_x, \ell_y] = \epsilon_{xyz} \ell_z \quad (6.2.26)$$

If we permute xyz into the correct order, the result does not change. If we invert one of the orders, the permutation tensor will ensure that the signs reflect the effects of Poisson brackets with reversed arguments. $\square$

Property 6.2.11 (Possible Conservations of Angular Momenta)

It is only possible to conserve one component of angular momentum, or all three.

Proof

Using the previous proposition and the construction of a conserved quantity, the conclusion follows immediately. $\square$

Theorem 6.2.12 (Runge–Lenz Vector)

Consider the motion of a particle under the influence of a central field. We observe that of the 6 degrees of freedom in $\mathbf{r}$ and $\mathbf{p}$, two are eliminated by the direction of the angular momentum, and two others are eliminated by conservation of energy and the magnitude of the angular momentum, leaving r, θ as degrees of freedom. However, in Newtonian orbits, there is no precession. Thus, we seek to eliminate one more degree of freedom. This conserved quantity is called the Runge–Lenz vector:

$$\mathbf{A} \equiv (\mathbf{p} \times \boldsymbol{\ell}) - \mu\beta\hat{\mathbf{r}} \quad (6.2.27)$$

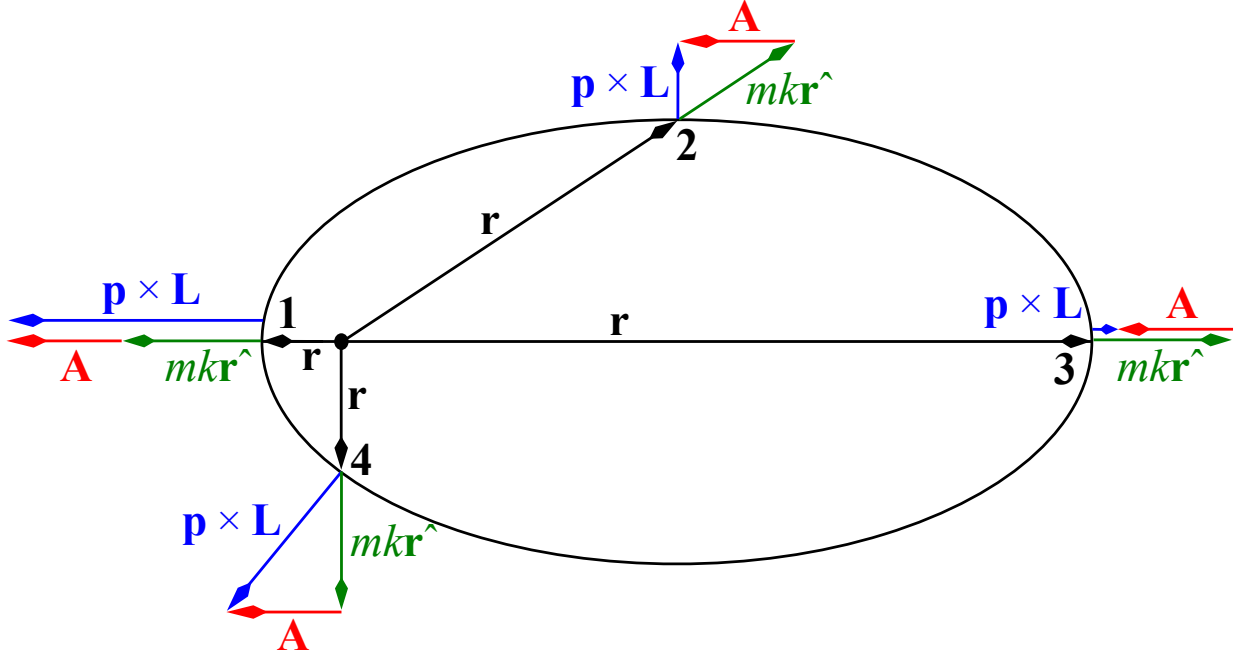
where $\beta \equiv Gm_1m_2$ is the numerator of the gravitational potential and $\mu = m_1m_2/(m_1 + m_2)$ is the reduced mass. Thus, the only quantity needed to deduce the position and all characteristics of a particle in such a field is the angle θ . This implicitly demonstrates that Newtonian orbits are closed.

Proof

This is proved by computing the Poisson bracket for $\mathbf{p} \times \boldsymbol{\ell}$ and that of $x/r = [\hat{\mathbf{r}}]_x$, $y/r = [\hat{\mathbf{r}}]_y$ with H , observing that the quantities differ by a factor $\mu\beta$, which yields

$$\dot{\mathbf{A}} = \begin{bmatrix} [(\mathbf{p} \times \boldsymbol{\ell})_x, H] - \mu\beta \left[\frac{x}{r}, H \right] \\ [(\mathbf{p} \times \boldsymbol{\ell})_y, H] - \mu\beta \left[\frac{y}{r}, H \right] \\ 0 \end{bmatrix} = \mathbf{0} \quad (6.2.28)$$

This vector points from the focal point to perihelion. Since its direction is constant, the only remaining degree of freedom is θ . $\square$

Figure 5. Runge-Lenz Vector $\mathbf{A}$.**Proposition 6.2.13 (Elliptic Orbits with Runge-Lenz)**

Consider a region with potential $U(r) = \beta/r$. Let $\mathbf{A}$ be the Runge-Lenz vector, given by

$$\mathbf{A} = (\mathbf{p} \times \boldsymbol{\ell}) - \mu\beta\hat{\mathbf{r}} \quad (6.2.29)$$

It is therefore clear that the trajectories of a particle of mass μ in such a medium are conic sections of the form

$$\hat{\mathbf{r}} \cdot \mathbf{A} + \mu\beta = \ell^2/r \quad (6.2.30)$$

Proof

We take the dot product of $\mathbf{A}$ with $\hat{\mathbf{r}}$:

$$\hat{\mathbf{r}} \cdot \mathbf{A} = \hat{\mathbf{r}} \cdot (\mathbf{p} \times \boldsymbol{\ell}) - \hat{\mathbf{r}} \cdot \mu\beta\hat{\mathbf{r}} \quad (6.2.31)$$

which gives

$$\boxed{\hat{\mathbf{r}} \cdot \mathbf{A} + \mu\beta = \ell^2/r} \quad (6.2.32)$$

Theorem 6.2.14 (Fradkin Tensor)

Consider the Hamiltonian for a spatial oscillator:

$$H = \frac{\mathbf{p}^2}{2m} + \frac{k\mathbf{r}^2}{2} \quad (6.2.33)$$

The additional conserved quantity is the tensor Q_{ij} given by

$$Q_{ij} = \frac{p_i p_j}{2m} + \frac{k x_i x_j}{2} - \frac{H}{3} \delta_{ij} \quad (6.2.34)$$

Proof

We write the Poisson bracket for the ij component of the tensor:

$$\dot{Q}_{ij} = [Q_{ij}, H] = \frac{1}{2m}[p_i p_j, H] + \frac{k}{2}[x_i x_j, H] - \frac{1}{3}[H, H]\delta_{ij} \quad (6.2.35)$$

The last term vanishes. The first gives

$$[p_i p_j, H] = -k p_i x_j - k x_i p_j \quad (6.2.36)$$

The second gives

$$[x_i x_j, H] = x_i \frac{p_j}{m} + \frac{p_i}{m} x_j \quad (6.2.37)$$

Thus,

$$\dot{Q}_{ij} = -\frac{k}{2m}(p_i x_j + x_i p_j) + \frac{k}{2m}(x_i p_j + x_j p_i) = 0 \quad (6.2.38)$$

as desired. We write the tensor explicitly, defining

$$x(t) = a \cos(\omega t) \quad ; \quad y(t) = b \cos(\omega t + \Delta\phi) \quad ; \quad E_{x,y} = \frac{p_{x,y}^2}{2m} + \frac{k(x,y)^2}{2} \quad ; \quad \Rightarrow E = E_x + E_y \quad (6.2.39)$$

Thus,

$$\mathbf{Q} = \begin{pmatrix} E_x - \frac{E}{3} & \sqrt{E_x E_y} \cos \Delta\phi & 0 \\ \sqrt{E_x E_y} \cos \Delta\phi & E_y - \frac{E}{3} & 0 \\ 0 & 0 & -\frac{E}{3} \end{pmatrix} \quad (6.2.40)$$

The angular momentum is given by

$$\ell = 2\sqrt{E_x E_y} \sin \Delta\phi \quad (6.2.41)$$

6.3 Canonical Transformations

We can now characterize transformations of coordinates and momenta.

Definition 6.3.1 (Canonical Transformation)

Let q, p be a set of coordinates and momenta used to describe a system. A transformation

$$q_i, p_i \longrightarrow Q_i, P_i \quad (6.3.1)$$

is said to be *canonical* if

$$[Q_i, Q_j]_{q,p} = [P_i, P_j]_{q,p} = 0 \quad ; \quad [Q_i, P_j]_{q,p} = \delta_{ij} \quad (6.3.2)$$

Proposition 6.3.2 (Invariance of Poisson Brackets Under a Canonical Transformation)

Let F, G be two functions that depend on the coordinates q and momenta p . Let Q, P be a canonical transformation of q, p . Then,

$$[F, G]_{q,p} = [F, G]_{Q,P} \quad (6.3.3)$$

Property 6.3.3

If Q, P is a canonical transformation of q, p ,

$$\dot{Q}_i = \frac{\partial H}{\partial P_i} \quad ; \quad \dot{P}_i = -\frac{\partial H}{\partial Q_i} \quad (6.3.4)$$

Proof

This follows immediately from the proposition on the total derivative. $\square$

We will explore what the consequences of canonical transformations are. If we perform a canonical transformation from $\{p, q\}$ to $\{P, Q\}$, where h, l are the Hamiltonian and the Lagrangian of the system in $\{p, q\}$, and H, L are the corresponding ones for $\{P, Q\}$. For the equations of motion not to change,

$$p\dot{q} - h = P\dot{Q} - H + \frac{d}{dt}G(q, Q, t) \quad (6.3.5)$$

In this case, the variational principle will not change the result. We perform another Legendre transform, such that

$$G = -QP + F(q, P, t) \quad (6.3.6)$$

We take the total time derivative:

$$\frac{dG}{dt} = -G\dot{P} - \dot{Q}P + \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q}\dot{q} + \frac{\partial F}{\partial P}\dot{P} \quad (6.3.7)$$

Thus,

$$p\dot{q} - h = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q}\dot{q} + \frac{\partial F}{\partial P}\dot{P} - Q\dot{P} - H \quad (6.3.8)$$

The variational procedure therefore gives

$$\dot{q} : p = \frac{\partial F}{\partial q} \quad ; \quad \dot{P} : Q = \frac{\partial F}{\partial P} \quad (6.3.9)$$

It must also be that

$$h = H - \frac{\partial F}{\partial t} \quad (6.3.10)$$

Proposition 6.3.4 (Generating Canonical Transformations)

Let l, h be the Lagrangian and the Hamiltonian of the system before the transformation. For any “nice” function $F(q, P, t)$, the canonical transformation is obtained from

$$p = \frac{\partial F}{\partial q} \quad ; \quad Q = \frac{\partial F}{\partial P} \quad ; \quad H = h + \frac{\partial F}{\partial t} \quad (6.3.11)$$

Proof

Proved above. □

Definition 6.3.5 (Classes of Canonical Transformations)

Let a canonical transformation use a function $F(q, P, t)$. This transformation is said to be “infinitesimal” if

$$F \equiv qP + \epsilon f(q, P) \quad (6.3.12)$$

The new coordinates and momenta will be

$$P = p - \epsilon \frac{\partial f}{\partial q} \quad ; \quad Q = q + \epsilon \frac{\partial f}{\partial P} \quad (6.3.13)$$

A transformation can also multiply a coordinate by a factor and the momentum by its inverse:

$$F = \alpha qP \Rightarrow P = \frac{p}{\alpha} \quad ; \quad Q = \alpha q \quad (6.3.14)$$

Property 6.3.6 (Advancing a System in Time)

Take $F = qP + h(q, P)\delta t$ as the generating function of a canonical transformation. In this case,

$$P = p(t + \delta t) \quad ; \quad Q = q(t + \delta t) \quad (6.3.15)$$

and thus the new coordinate and the new momentum represent the state of the system at a time $t + \delta t$.

Proof

Using the infinitesimal transformation, the result follows immediately. □

Property 6.3.7 (Advancing a System in Space)

Take $F = qP + P\delta q$ as the generating function of a canonical transformation. In this case,

$$P = p \quad ; \quad Q = q + \delta q \quad (6.3.16)$$

and thus the momentum does not change, but it advances the coordinate.

Proof

Using the infinitesimal transformation, the result follows immediately. □

Property 6.3.8 (Rotation of a System)

Take $F = xP_x + yP_y + zP_z + (yP_x - xP_y)d\varphi$ as the generating function of a canonical transformation. In this case,

$$P_z = p_z \quad ; \quad Z = z \quad (6.3.17)$$

and thus neither the z momentum nor the z coordinate changes, but x and y change as

$$X = x + yd\varphi \quad ; \quad Y = y - xd\varphi \quad ; \quad P_x = p_x + p_yd\varphi \quad ; \quad P_y = p_y - p_xd\varphi \quad (6.3.18)$$

Proof

Using the infinitesimal transformation, the result follows immediately. □

Proposition 6.3.9 (Types of Generating Functions)

There are four types of generating functions:

$$\begin{aligned}
 \bullet F_1(q, Q) & : p_i = \frac{\partial F_1}{\partial q_i}, P_i = -\frac{\partial F_1}{\partial Q_i} & \bullet F_3(p, Q) & : q_i = \frac{\partial F_3}{\partial p_i}, P_i = -\frac{\partial F_3}{\partial Q_i} \\
 \bullet F_2(q, P) & : p_i = \frac{\partial F_2}{\partial q_i}, Q_i = \frac{\partial F_2}{\partial P_i} & \bullet F_4(p, P) & : q_i = \frac{\partial F_4}{\partial p_i}, Q_i = \frac{\partial F_4}{\partial P_i}
 \end{aligned}$$

Proof

We recall the fact that Lagrangians that differ by a total time derivative:

$$L' = L + \frac{df}{dt} \quad (6.3.19)$$

We represent L' with $Q, \dot{Q}$:

$$L'(Q, \dot{Q}, t) = L(q, \dot{q}, t) \quad (6.3.20)$$

The Hamiltonians are

$$h \equiv p\dot{q} - L \quad ; \quad H = P\dot{Q} - L' \quad (6.3.21)$$

Thus,

$$L = p\dot{q} - h = P\dot{Q} - H + \frac{dF}{dt} = L' \quad (6.3.22)$$

We can multiply by dt to obtain the differential dF in full:

$$dF = p\dot{q}dt - hdt - P\dot{Q}dt + Hdt \quad (6.3.23)$$

Now, we choose two variables on which F can depend: one new, in $\{P, Q\}$, and one old, in $\{p, q\}$. The first case is q, Q :

$$dF = p\dot{q}dt - hdt - P\dot{Q}dt + Hdt \quad (6.3.24)$$

$$= pdq - hdt - PdQ + Hdt \quad (6.3.25)$$

From this, we immediately read off the differential equations:

$$\frac{\partial F}{\partial q} = p \quad ; \quad \frac{\partial F}{\partial Q} = -P \quad (6.3.26)$$

If F depends on q, P , we obtain

$$dF = p\dot{q}dt - hdt - P\dot{Q}dt + Hdt \quad (6.3.27)$$

$$= pdq - hdt - PQ + QdP + Hdt \quad (6.3.28)$$

The result is correct:

$$p_i = \frac{\partial F_2}{\partial q_i}, Q_i = \frac{\partial F_2}{\partial P_i} \quad (6.3.29)$$

Equivalently, we recover the others. □

6.4 Liouville's Theorem

We now think in terms of phase space.

Definition 6.4.1 (Phase Space)

Phase space is a $2s$ -dimensional space, such that each point in the space is defined by the generalized coordinates and generalized momenta. The progression of a system in time can be represented in phase space using a *phase trajectory*.

The unit of volume in this space is given by the product of differentials:

$$d\Gamma = \prod dq_i \prod dp_i \quad (6.4.1)$$

Theorem 6.4.2 (Liouville)

Let $\{q, p\}$ be a set of coordinates and momenta describing a system. Suppose that the transformation $\{q, p\} \rightarrow \{Q, P\}$ is canonical. Then, the Jacobian of the transformation from one phase space to the other is 1; that is, there will be no change in the size of the unit volume. This holds as long as the forces are divergence-free in momentum space.

Proof

We seek to show that

$$\frac{\partial(Q_1, \dots, Q_s, P_1, \dots, P_s)}{\partial(q_1, \dots, q_s, p_1, \dots, p_s)} = 1 \quad (6.4.2)$$

We add an intermediate step:

$$D \equiv \frac{\partial(Q_1, \dots, Q_s, P_1, \dots, P_s)}{\partial(q_1, \dots, q_s, p_1, \dots, p_s)} = \frac{\partial(Q_1, \dots, Q_s, P_1, \dots, P_s)}{\partial(q_1, \dots, q_s, P_1, \dots, P_s)} \bigg/ \frac{\partial(q_1, \dots, q_s, p_1, \dots, p_s)}{\partial(q_1, \dots, q_s, P_1, \dots, P_s)} \quad (6.4.3)$$

In the first term, the momenta are constant, and in the second term, the coordinates are constant. Thus,

$$D = \left\{ \frac{\partial(Q_1, \dots, Q_s)}{\partial(q_1, \dots, q_s)} \right\}_{P=\text{ct}} \bigg/ \left\{ \frac{\partial(p_1, \dots, p_s)}{\partial(P_1, \dots, P_s)} \right\}_{q=\text{ct}} \quad (6.4.4)$$

We recall the generating function $F(q, P)$, such that

$$\frac{\partial Q_i}{\partial q_k} = \frac{\partial^2 F}{\partial q_k \partial P_i} \quad (6.4.5)$$

Equivalently,

$$\frac{\partial p_i}{\partial P_k} = \frac{\partial^2 F}{\partial q_i \partial P_k} \quad (6.4.6)$$

Thus,

$$D = \frac{\left| \frac{\partial^2 F}{\partial q_k \partial P_i} \right|}{\left| \frac{\partial^2 F}{\partial q_i \partial P_k} \right|} = 1 \quad (6.4.7)$$

because the matrices whose determinants we compute differ only by an interchange of rows and/or columns. Thus, the theorem is proved. $\square$

The consequence of this fact is that the *density of states in an ensemble of identical states for different initial conditions along each possible trajectory in state space* is equivalent to an incompressible flow.

6.5 Action–Angle Coordinates

For periodic motions in phase space, it is sometimes convenient to perform a particular canonical transformation that allows us to extract the periodicity very simply.

Definition 6.5.1 (Action–Angle Canonical Coordinate Transformation)

Let $H(\mathbf{q}, \mathbf{p})$ be the Hamiltonian of a system that is periodic in phase space. The generating function W is of the second type, such that

$$W : W(\mathbf{J}, \mathbf{q}) : \quad p_k = \frac{\partial W}{\partial q_k}, \quad w_k = \frac{\partial W}{\partial P_k} \quad (6.5.1)$$

$$H(\mathbf{q}, \mathbf{p}) \longrightarrow H(\mathbf{w}, \mathbf{J}) \quad (6.5.2)$$

and

$$J_k = \oint p_k dq_k \quad (6.5.3)$$

We observe that this quantity $J_k = \oint p_k dq_k$ is in fact the area enclosed by the periodicity in phase space projected onto the $p_k q_k$ plane. The path of integration is implicitly determined by the energy. Consequently, the momenta J_k are conserved (they are integrals of motion). Thus,

$$\frac{d}{dt} J_k = 0 = \frac{\partial H}{\partial w_k} \quad (6.5.4)$$

and

$$w_k = \frac{\partial W}{\partial J_k} \quad (6.5.5)$$

Thus, K cannot vary with w_k . Therefore, the dynamics come from Hamilton's equations:

$$\frac{d}{dt} w_k = \frac{\partial H}{\partial J_k} = \nu_k(\mathbf{J}) \quad (6.5.6)$$

which is a constant, because all $\mathbf{J}$ are constant. Thus, we anticipate that

$$w_k = \nu_k(\mathbf{J})t + \beta_k \quad (6.5.7)$$

for β_k an integration constant. Now suppose that the initial generalized coordinate q_k has gone through one full period in a time T . The change Δw_k of the new variable is given by

$$\Delta w_k = \oint \frac{\partial w_k}{\partial q_k} dq_k = \oint \frac{\partial^2 W}{\partial J_k \partial q_k} dq_k = \frac{d}{dJ_k} \oint \frac{\partial W}{\partial q_k} dq_k = \frac{d}{dJ_k} \oint p_k dq_k = \frac{dJ_k}{dJ_k} = 1 \quad (6.5.8)$$

Thus,

$$\nu_k = \frac{1}{T} \quad (6.5.9)$$

which is therefore the frequency of the motion.

Proposition 6.5.2 (Period of an Oscillatory Motion in Phase Space)

The period of an oscillatory motion of the k -th coordinate of a system with conserved energy E in phase space is given by

$$\tau_k = \frac{\partial J_k}{\partial H} \quad (6.5.10)$$

for

$$J_k = \oint p_k dq_k \quad (6.5.11)$$

Proof

We found that

$$\nu_k = \frac{\partial H}{\partial J_k} \quad (6.5.12)$$

and thus the period is

$$\tau_k = \frac{1}{\nu_k} = \frac{\partial J_k}{\partial H} = \boxed{\frac{\partial J_k}{\partial H}} \quad (6.5.13)$$

as desired. □

6.6 Examples

6.6.1 Simple Spring

Let us take the simple example of the spring. The Lagrangian is

$$L = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2} \quad (6.6.1)$$

The momentum p is

$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x} \quad (6.6.2)$$

Thus, the Hamiltonian is

$$H = p\dot{x} - L = p \cdot \frac{p}{m} - \frac{p^2}{2m} + \frac{kx^2}{2} = \boxed{\frac{p^2}{2m} + \frac{kx^2}{2}} \quad (6.6.3)$$

which is indeed the familiar energy. The equations of motion are

$$\frac{\partial H}{\partial p} = \dot{x} = \frac{p}{m} \quad (6.6.4)$$

and

$$\frac{\partial H}{\partial x} = -\dot{p} = kx \quad (6.6.5)$$

Thus,

$$m\ddot{x} = \dot{p} = -kx \Rightarrow \boxed{m\ddot{x} + kx = 0} \quad (6.6.6)$$

which is indeed the expected result.

6.6.2 Coupled Springs

We want to see how to use canonical transformations in a more complicated example. Consider two coupled pendulums with a spring. The Lagrangian was

$$L = \frac{1}{m}\ell^2 2(\dot{\theta}_1^2 + \dot{\theta}_2^2) - \frac{mg\ell}{2}(\theta_1^2 + \theta_2^2) - \frac{k\ell}{2}(\theta_1 - \theta_2)^2 \quad (6.6.7)$$

The momenta are

$$p_1 = \frac{\partial L}{\partial \dot{\theta}_1} = m\ell^2 \dot{\theta}_1 \quad ; \quad p_2 = \frac{\partial L}{\partial \dot{\theta}_2} = m\ell^2 \dot{\theta}_2 \quad (6.6.8)$$

Thus, the Hamiltonian for this conservative system is

$$H = K + U = \frac{p_1^2 + p_2^2}{2m\ell^2} + \frac{mg\ell}{2}(\theta_1^2 + \theta_2^2) + \frac{k\ell^2}{2}(\theta_2 - \theta_1)^2 \quad (6.6.9)$$

We propose the following change of variables:

$$Q_+ = \frac{\ell}{\sqrt{2}}(\theta_1 + \theta_2) \quad ; \quad Q_- = \frac{\ell}{\sqrt{2}}(\theta_1 - \theta_2) \quad (6.6.10)$$

The new momenta will be

$$P_+ = \frac{1}{\ell\sqrt{2}}(p_1 + p_2) \quad ; \quad P_- = \frac{1}{\ell\sqrt{2}}(p_1 - p_2) \quad (6.6.11)$$

We verify that the transformation is canonical:

$$[Q_+, P_+] = \frac{\ell}{\ell\sqrt{2}\sqrt{2}}[\theta_1 + \theta_2, p_1 + p_2] \quad (6.6.12)$$

$$= \frac{1}{2}([\theta_1, p_1] + [\theta_2, p_1] + [\theta_1, p_2] + [\theta_2, p_2]) \quad (6.6.13)$$

$$= \frac{1}{2}(1 + 0 + 0 + 1) \quad (6.6.14)$$

$$= 1 \quad (6.6.15)$$

Likewise,

$$[P_+, P_-] = [Q_+, Q_-] = 0 \quad (6.6.16)$$

We observe that

$$Q_+^2 + Q_-^2 = \frac{\ell^2}{2}(\theta_1^2 + \theta_2^2 + 2\theta_1\theta_2 + \theta_1^2 + \theta_2^2 - 2\theta_1\theta_2) = \ell^2(\theta_1^2 + \theta_2^2) \quad (6.6.17)$$

Equivalently,

$$P_+^2 + P_-^2 = \frac{1}{\ell^2}(p_1^2 + p_2^2) \quad (6.6.18)$$

Thus, we rewrite the Hamiltonian:

$$H = \frac{p_1^2 + p_2^2}{2m\ell^2} + \frac{mg\ell}{2}(\theta_1^2 + \theta_2^2) + \frac{k\ell^2}{2}(\theta_2 - \theta_1)^2 \quad (6.6.19)$$

$$= \frac{P_+^2 + P_-^2}{2m} + \frac{mg}{2\ell}(Q_+^2 + Q_-^2) + kQ_-^2 \quad (6.6.20)$$

$$= \left(\frac{P_+^2}{2m} + \frac{mg}{2\ell}Q_+^2 \right) + \left(\frac{P_-^2}{2m} + \frac{1}{2} \left(\frac{mg}{\ell} + 2k \right) Q_-^2 \right) \quad (6.6.21)$$

$$= \left(\frac{P_+^2}{2m} + \frac{m}{2} \frac{g}{\ell} Q_+^2 \right) + \left(\frac{P_-^2}{2m} + \frac{m}{2} \left(\frac{g}{\ell} + \frac{2k}{m} \right) Q_-^2 \right) \quad (6.6.22)$$

This is the sum of two simple harmonic oscillators, with

$$\boxed{\omega_+ = \sqrt{\frac{g}{\ell}}} \quad ; \quad \boxed{\omega_- = \sqrt{\frac{g}{\ell} + \frac{2k}{m}}} \quad (6.6.23)$$

which were indeed the results obtained from the normal modes.

6.6.3 Damped Oscillator (II)

We return to the Lagrangian of the damped harmonic oscillator:

$$L = e^{\gamma t} \left(\frac{1}{2} m \dot{x}^2 - \frac{1}{2} k x^2 \right) \quad (6.6.24)$$

The generalized momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = m \dot{x} e^{\gamma t} \quad (6.6.25)$$

We observe that this is not a true momentum! The Hamiltonian is

$$H = p\dot{x} - L = p \cdot \frac{p}{m}e^{-\gamma t} - e^{\gamma t} \left(\frac{p^2}{2m}e^{-2\gamma t} - \frac{1}{2}kx^2 \right) = \frac{p^2}{2m}e^{-\gamma t} + \frac{1}{2}kx^2e^{\gamma t} \neq E \quad (6.6.26)$$

which is not the energy! The canonical equations give

$$\frac{\partial H}{\partial p} = \dot{x} = \frac{p}{m}e^{-\gamma t} \quad ; \quad \frac{\partial H}{\partial x} = -\dot{p} = kxe^{\gamma t} \quad (6.6.27)$$

Thus,

$$-kxe^{\gamma t} = m \frac{d}{dt} (\dot{x}e^{\gamma t}) = m\ddot{x}e^{\gamma t} + m\dot{x}\gamma e^{\gamma t} \quad (6.6.28)$$

which gives

$$\boxed{\ddot{x} + \gamma\dot{x} + \frac{k}{m}x = 0} \quad (6.6.29)$$

which is indeed the expected result. We now propose a canonical transformation such that

$$P = pe^{-\gamma t/2} \quad ; \quad X = xe^{\gamma t/2} \quad (6.6.30)$$

In this case, we must find the function $F(x, P, t)$:

$$F = e^{\gamma t/2} xP \quad (6.6.31)$$

such that

$$p = \frac{\partial F}{\partial x} = e^{\gamma t/2} P \quad ; \quad X = \frac{\partial F}{\partial P} = e^{\gamma t/2} x \quad ; \quad H = h + \frac{\partial F}{\partial t} = h + \frac{xP\gamma}{2}e^{\gamma t/2} = \frac{\gamma}{2}XP \quad (6.6.32)$$

Thus, the new Hamiltonian becomes

$$H = \frac{P^2}{2m} + \frac{1}{2}kX^2 + \frac{\gamma}{2}XP \quad (6.6.33)$$

which resembles the undamped oscillator. The canonical equations give

$$\frac{\partial H}{\partial P} = \dot{X} = \frac{P}{m} + \frac{\gamma}{2}X \quad ; \quad \frac{\partial H}{\partial X} = \dot{P} = kX + \frac{\gamma}{2}P \quad (6.6.34)$$

Thus,

$$\ddot{X} = \frac{\dot{P}}{m} + \frac{\gamma}{2}\dot{X} = \frac{1}{m} \left(-kX - \frac{\gamma}{2}P \right) + \frac{\gamma}{2} \left(\frac{P}{m} + \frac{\gamma}{2}X \right) = - \left(\frac{k}{m} - \left(\frac{\gamma}{2} \right)^2 \right) X \quad (6.6.35)$$

and thus

$$\ddot{X} + \left(\frac{k}{m} - \left(\frac{\gamma}{2} \right)^2 \right) X = 0 \quad (6.6.36)$$

This is an SHO with frequency $\omega^2 = \frac{k}{m} - \left(\frac{\gamma}{2} \right)^2$. We recover the form of x :

$$X(t) = A \cos \omega t + B \sin \omega t \quad \implies \quad x(t) = e^{-\gamma t/2} (A \cos(\omega t) + B \sin(\omega t)) \quad ; \quad \omega = \sqrt{\frac{k}{m} - \left(\frac{\gamma}{2} \right)^2} \quad (6.6.37)$$

exactly the expected result.

6.6.4 Action–Angle for an SHO

The SHO indeed has Hamiltonian

$$H = \frac{p^2}{2m} + \frac{kx^2}{2} = E \quad (6.6.38)$$

In phase space, this is an ellipse with semi-axes $a = \sqrt{2mE}$ and $b = \sqrt{2E/k}$. Thus, the area is $\pi ab = 2\pi E \sqrt{m/k}$. The period is therefore

$$\tau = \frac{\partial}{\partial E} \oint p \, dq = \boxed{2\pi \sqrt{\frac{m}{k}}} \quad (6.6.39)$$

which is indeed the familiar result.

6.6.5 Action–Angle for a Cycloid

Consider a particle of mass m that moves along the trajectory defined by

$$x(\phi) = R(\phi + \sin \phi) \quad ; \quad y(\phi) = -R \cos \phi \quad (6.6.40)$$

The velocities are

$$\dot{x} = R(\dot{\phi} + \dot{\phi} \cos \phi) \quad ; \quad \dot{y} = R\dot{\phi} \sin \phi \quad (6.6.41)$$

from which we construct the Lagrangian:

$$L = T - U = \frac{mR^2\dot{\phi}^2}{2} ((1 + \cos \phi)^2 + \sin^2 \phi) + mgR(\cos \phi - 1) = 2mR^2\dot{\phi}^2 \cos^2 \frac{\phi}{2} - 2mgR \sin^2 \frac{\phi}{2} \quad (6.6.42)$$

The canonical momentum is given by

$$p = \frac{\partial L}{\partial \dot{\phi}} = 4mR^2\dot{\phi} \cos^2 \frac{\phi}{2} \quad (6.6.43)$$

and the Hamiltonian is therefore

$$H = p\dot{\phi} - L = \frac{p^2}{8mR^2 \cos^2(\phi/2)} + 2mgR \sin^2 \frac{\phi}{2} \quad (6.6.44)$$

This is indeed the energy, because there is no explicit time dependence. Thus, evaluated at the initial point $\phi(0) = \phi_0$, $\dot{\phi}(0) = 0$, we have

$$H = E = 2mgR \sin^2 \frac{\phi_0}{2} \quad (6.6.45)$$

which gives

$$p^2 = 16m^2R^3g \cos^2 \frac{\phi}{2} \left(\sin^2 \frac{\phi_0}{2} - \sin^2 \frac{\phi}{2} \right) \quad (6.6.46)$$

We observe that the integral would be done more efficiently by taking the integral from $0 \rightarrow \phi_0$ four times:

$$J = \oint p \, dq = 4 \int_0^{\phi_0} p \, d\phi = 16mR^{3/2}\sqrt{g} \int_0^{\phi_0} \cos \frac{\phi}{2} \sqrt{\sin^2 \frac{\phi_0}{2} - \sin^2 \frac{\phi}{2}} \, d\phi \quad (6.6.47)$$

We substitute

$$z = \sin \frac{\phi}{2} \Rightarrow dz = \frac{d\phi}{2} \cos \frac{\phi}{2} \quad ; \quad z_0 = \sin \frac{\phi_0}{2} \quad (6.6.48)$$

to obtain

$$J = 16mR^{3/2}\sqrt{g} \int_0^{z_0} 2\sqrt{z_0^2 - z^2} \, dz = 16mR^{3/2}\sqrt{g} \left(z_0^2 \arcsin(z/z_0) + z\sqrt{z_0^2 - z^2} \right) \Big|_0^{z_0} \quad (6.6.49)$$

We evaluate:

$$J = 8\pi mR^{3/2}\sqrt{g}z_0^2 = 8\pi mR^{3/2}\sqrt{g} \sin^2 \frac{\phi_0}{2} = 4\pi E \sqrt{\frac{R}{g}} \quad (6.6.50)$$

This immediately gives the period:

$$\tau = \frac{\partial J}{\partial E} = \boxed{4\pi \sqrt{\frac{R}{g}}} \quad (6.6.51)$$

which, like the pendulum for small oscillations, has a period that does not depend on the amplitude.

7 Particle Scattering

7.1 Elastic Collisions

Elastic collisions have two essential characteristics:

- Total momentum is conserved.
- Energy is conserved.

Consider two particles of masses m_1 and m_2 traveling along a line, toward one another. There are two important reference frames to consider: the frame $\mathbf{C}$, that of the center of mass, and the frame $\mathbf{L}$, that of a laboratory where the center of mass has a nonzero velocity (in general).

In the $\mathbf{C}$ frame, the momenta of the particles will be equal and opposite before the collision. After the collision, conservation of momentum shows that the new momenta will also be equal and opposite (but not necessarily the same as those before the collision). Conservation of energy in fact allows us to see that the magnitude does not change either.

From conservation of momentum:

$$\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}'_1 + \mathbf{p}'_2 \quad (7.1.1)$$

In the $\mathbf{C}$ frame:

$$\mathbf{p}_1 = -\mathbf{p}_2 \quad (7.1.2)$$

and thus

$$\mathbf{p}'_1 + \mathbf{p}'_2 = \mathbf{0} \Rightarrow \mathbf{p}' \equiv \mathbf{p}'_1 = -\mathbf{p}'_2 \quad (7.1.3)$$

Conservation of energy gives

$$\frac{p^2}{2m_1} + \frac{p^2}{2m_2} = \frac{(p')^2}{2m_1} + \frac{(p')^2}{2m_2} \Rightarrow p = p' \quad (7.1.4)$$

Thus, the momenta do not change even in magnitude. The only degree of freedom that remains is the direction of the trajectory after the collision. Let $\hat{\mathbf{n}}_0$ be the unit vector describing the direction of $\mathbf{p}'_1$, the momentum of the first mass after the collision. The angle between $\mathbf{n}_0$ and the vector $\mathbf{p}_1$, the direction of the first mass before the collision, is called χ , the “scattering angle.”

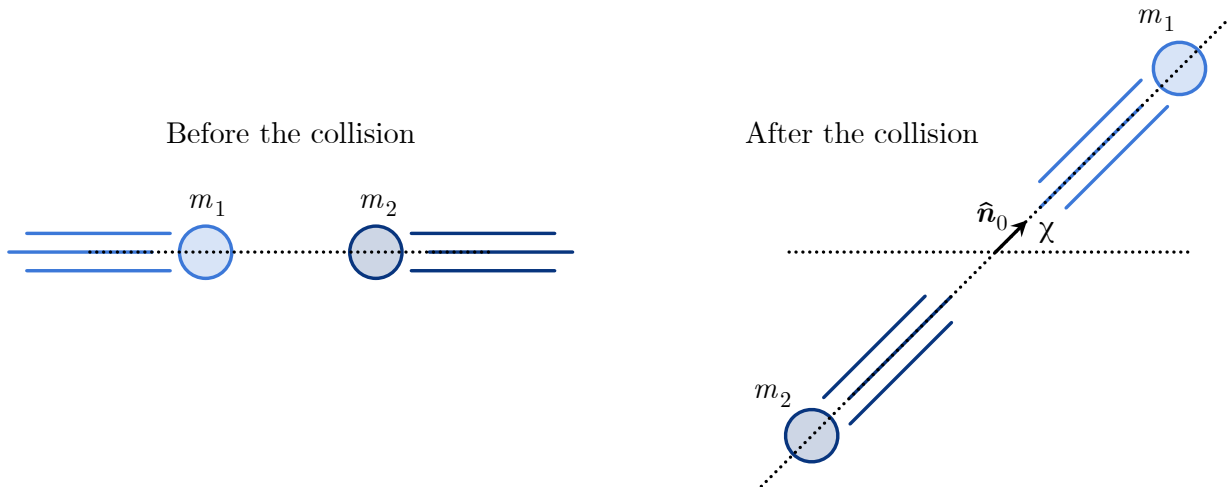


Figure 6. Elastic collision in the $\mathbf{C}$ frame.

7.2 Scattering

Scattering is a “collision” of a particle with a field. Let $U(r)$ be a central potential.

Definition 7.2.1 (Impact Parameter)

The *impact parameter* b is the (minimum) distance between the initial trajectory of the particle and the center of the potential $U(r)$.

Definition 7.2.2 (Scattering Angle)

The *scattering angle* χ is the angle between the initial trajectory and the new trajectory.

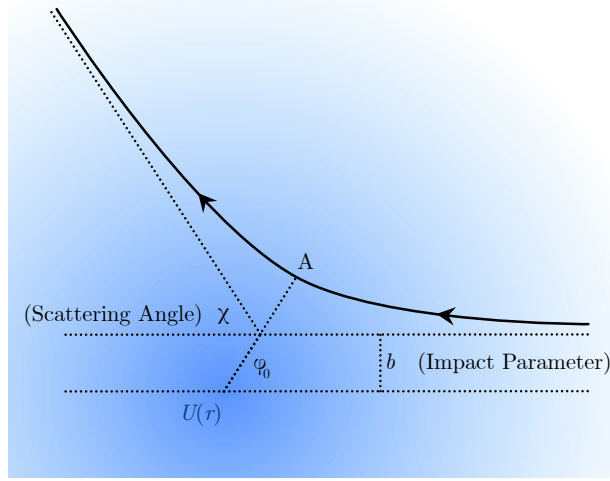


Figure 7. Scattering in a Central Field

Geometry tells us that $\chi = |\pi - 2\phi_0|$. We recall the equation that described the angle swept under the influence of a central field (3.4.17):

$$\phi = \int \frac{\ell/r^2}{\sqrt{2m(E - U(r)) - \ell^2/r^2}} dr \quad (7.2.1)$$

In this case, we integrate from $r_{\min}$ to ∞ . We substitute the values $E = mv_\infty^2/2$ and $\ell = mbv_\infty = Eb/v_\infty$:

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{b/r^2}{\sqrt{1 - b^2/r^2 - U/E}} dr \quad (7.2.2)$$

Thus,

$$\chi = \pi - 2 \int_{r_{\min}}^{\infty} \frac{b/r^2}{\sqrt{1 - b^2/r^2 - U/E}} dr \quad (7.2.3)$$

This is not very easy to use, and usually we express b as a function of χ . We define a more interesting quantity, the (*scattering*) *cross section* σ .

Definition 7.2.3 (Cross Section)

The (*scattering*) *cross section* σ is given by

$$d\sigma \equiv \frac{dN}{n} \quad (7.2.4)$$

where dN is the number of particles scattered between an angle χ and one of $\chi + d\chi$, and n is the total number of particles sent in. The units of σ are those of area.

Let us express this quantity using b and χ . The geometric correspondence for dN is the number of particles lying between two circles of radii b and $b + db$, multiplied by the total number of particles that were sent:

$$dN = 2\pi b db \cdot n \Rightarrow \boxed{d\sigma = 2\pi b db} \quad (7.2.5)$$

Definition 7.2.4 (Scattering Rate)

The scattering rate is given by the product of σ with the *luminosity*, which is the number of particles per unit time and per unit area. It is written Γ or w in the literature.

If we know the bounds on b , we can simply integrate to obtain

$$\boxed{\sigma = \pi b^2 \Big|_{b_{\min}}^{b_{\max}}} \quad (7.2.6)$$

Suppose we want to find the distribution in 3D. It is difficult to work with angles in 3D, so we introduce the *solid angle* Ω , which resembles a plane angle but represents a spherical cap (which would have area $r^2\Omega$). Solid angle is expressed in *steradians*, such that there are 4π steradians in a sphere. To convert spherical coordinates from an angle to a solid angle, we write

$$d\Omega = \sin \chi d\chi d\varphi \rightarrow 2\pi \sin \chi d\chi \quad (7.2.7)$$

with φ the azimuth. We integrate to confirm the total solid angle:

$$\Omega = \int d\Omega = \int_0^{2\pi} \int_0^\pi \sin \chi d\chi d\varphi = 2\pi \cdot 2 = \boxed{4\pi} \quad (7.2.8)$$

as desired. Thus, by the chain rule,

$$d\sigma = 2\pi b db = 2\pi b \left| \frac{db}{d\chi} \right| d\chi = \boxed{\frac{b(\chi)}{\sin \chi} \left| \frac{db}{d\chi} \right| d\Omega} \quad (7.2.9)$$

We note that if $db/d\chi$ has several “branches,” i.e. it can assume several values, we take their sum when evaluating the derivative.

7.3 Rutherford Formula

Consider the particular potential $U = \alpha/r$; we can evaluate the integral in (7.2.2):

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{b/r^2}{\sqrt{1 - b^2/r^2 - U/E}} dr \quad (7.3.1)$$

$$= \int_{r_{\min}}^{\infty} \frac{b/r^2}{\sqrt{1 - b^2/r^2 - k/r}} dr \quad (7.3.2)$$

where $k \equiv \alpha/E$. The minimal radius $r_{\min}$ occurs where the square root vanishes. Thus,

$$r_{\min} = \frac{k + \sqrt{k^2 + 4b^2}}{2} \quad (7.3.3)$$

We substitute $u = 1/r$:

$$\phi_0 = \int_{1/r_{\min}}^0 \frac{-b}{\sqrt{1 - ku - b^2u^2}} = b \int_0^{r_{\min}} du \quad (7.3.4)$$

We complete the square: let $A = 1 + k^2/4b^2$ and let $\beta = b/\sqrt{A}$, which give

$$\phi_0 = \frac{b}{\sqrt{A}} \int_0^{1/r_{\min}} \frac{du}{\sqrt{1 - \beta^2(u + k/2b^2)^2}} \quad (7.3.5)$$

Let $w = \beta(u + k/2b^2)$. Thus,

$$\phi_0 = \frac{b}{\sqrt{A}\beta} \arcsin w \Big|_0^{1/r_{\min}} = \boxed{\arccos \left(\frac{\alpha/2E}{\sqrt{b^2 + (\alpha/2E)^2}} \right)} \quad (7.3.6)$$

We substitute $\phi_0 = \frac{1}{2}(\pi - \chi)$:

$$b^2 = \frac{\alpha^2}{m^2 v_{\text{infty}}^4} \tan^2 \phi_0 \Rightarrow b^2 = \frac{\alpha^2}{m^2 v_{\infty}^4} \cot^2 \frac{\chi}{2} \quad (7.3.7)$$

We can now evaluate the angular distribution:

$$d\sigma = \pi \left(\frac{\alpha}{m v_{\infty}^2} \right)^2 \frac{\cos \frac{\chi}{2}}{\sin^3 \frac{\chi}{2}} d\chi \quad (7.3.8)$$

In terms of solid angle,

$$\boxed{\frac{d\sigma}{d\Omega} = \left(\frac{\alpha}{2m v_{\infty}^2} \right)^2 \left(\sin^4 \frac{\chi}{2} \right)^{-1}} \quad (7.3.9)$$

Proposition 7.3.1 (Rutherford Formula)

Let a beam be scattered by a potential $U(r) = \alpha/r$. The angular distribution is

$$\frac{d\sigma}{d\Omega} = \left(\frac{\alpha}{2m v_{\infty}^2} \right)^2 \left(\sin^4 \frac{\chi}{2} \right)^{-1} \quad (7.3.10)$$

The cross section diverges! If we integrate only from $\chi_{\min} \leftrightarrow b_{\max}$ to $\pi \leftrightarrow 0$, we simply obtain

$$\sigma = \pi b_{\max}^2 \quad (7.3.11)$$

We can obtain this result in a simpler way. We recall that in a $1/r$ potential, the motions are given by

$$\frac{\beta}{r} = 1 + \epsilon \cos \varphi \quad (7.3.12)$$

with

$$\beta = \frac{\ell^2}{mk} \quad ; \quad \epsilon = \sqrt{1 + \frac{2E\ell^2}{mk}} > 1 \quad (7.3.13)$$

where the last inequality comes from $E > 0$. For a repulsive potential,

$$-\frac{\beta}{r} = 1 + \epsilon \cos(\phi + \alpha) \quad (7.3.14)$$

with α the incidence angle relative to the normal from the center of the potential, such that the minimal distance occurs at α . We want to express α using the other known quantities:

$$\alpha = \arccos \left(\frac{1}{\epsilon} - \frac{\beta}{r\epsilon} - \phi \right) \quad (7.3.15)$$

In the limit $\phi \rightarrow \pi$ and $r \rightarrow \infty$, we have

$$\frac{1}{\cos^2 \alpha} = \epsilon^2 \quad (7.3.16)$$

We evaluate ϵ in this limit:

$$\epsilon = \sqrt{1 + \frac{2E\ell^2}{mk}} = \sqrt{1 + \frac{2mv_\infty^2 \cdot m^2 v_\infty^2 b^2}{mk}} = \sqrt{1 + \frac{2m^2 v_\infty^4 b^2}{k}} \quad (7.3.17)$$

Let $\gamma = k/2E$. Thus,

$$\epsilon = \sqrt{1 + \frac{b^2}{\gamma^2}} \Rightarrow 1 + \frac{b^2}{\gamma^2} = \frac{1}{\cos^2 \alpha} \quad (7.3.18)$$

and thus

$$b^2 = \gamma^2 \left(\frac{1}{\cos^2 \alpha} - 1 \right) = \gamma^2 \tan^2 \alpha = \gamma^2 \tan^2 \left(\frac{\pi}{2} - \frac{\chi}{2} \right) = \gamma^2 \cot^2 \frac{\chi}{2} \quad (7.3.19)$$

We thus have $b(\chi)$. The rest follows:

$$\frac{d\sigma}{d\Omega} = \frac{b(\chi)}{\sin \chi} \left| \frac{db}{d\chi} \right| = \frac{\gamma^2 \cot^2 \frac{\chi}{2}}{\sin \chi} \left| -\frac{1}{\sin^2 \frac{\chi}{2}} \frac{1}{2} \right| = \frac{\gamma^2 \cot \chi/2}{\sin \chi \cdot 2 \sin^2 \chi/2} = \frac{\gamma^2}{4 \sin^4 \frac{\chi}{2}} \quad (7.3.20)$$

In terms of k ,

$$\frac{d\sigma}{d\Omega} = \frac{k^2}{(4E)^4 \sin^4 \frac{\chi}{2}} \quad (7.3.21)$$

which is indeed Rutherford's formula.

7.4 Mean Distance Between Collisions

An essential characteristic of the rate of a process is the mean distance traveled between collisions, which gives the time between collisions.

Proposition 7.4.1 (Mean Distance Between Collisions)

Suppose we observe a scattering process due to central potentials (or at least symmetric about the azimuth). Let σ be the cross section of a scattering process. Let ρ be the density of central potentials that can scatter particles. Suppose that the time spent scattering (each event) is very small compared to the time between scattering events. The mean distance between collisions is given by

$$\langle \ell \rangle = \frac{1}{\sigma \rho} \quad (7.4.1)$$

Proof

If the time spent scattering is negligible compared to the time between scattering events, it is convenient to model these events using a Poisson process, because each event is random, at a constant rate, and memoryless. Consider a cylinder of length ℓ and cross-sectional area σ . The volume is therefore $\sigma\ell$. Thus, the number of potentials in this volume is $\rho\sigma\ell$, assuming the distribution is uniform.

For a Poisson process, the probability of k events in a system with rate λ is given by

$$P_k = \frac{\lambda^k e^{-\lambda}}{k!} \quad (7.4.2)$$

If we want no events, the rate corresponds to the number of potentials $N = \rho\sigma\ell$ and $k = 0$. Thus, the probability that a particle travels a distance L at least as large as ℓ is given by

$$P_0(L \geq \ell) = e^{-\rho\sigma\ell} \quad (7.4.3)$$

The cumulative distribution function (CDF) is, by definition,

$$\text{CDF}(\ell) \equiv P_0(L < \ell) = 1 - e^{-\rho\sigma\ell} \quad (7.4.4)$$

We take the derivative to obtain the probability density function (PDF):

$$f(\ell) \equiv \text{PDF}(\ell) \equiv \frac{d}{d\ell} \text{CDF}(\ell) = \frac{d}{d\ell} (1 - e^{-\rho\sigma\ell}) = \rho\sigma e^{-\rho\sigma\ell} \quad (7.4.5)$$

The *mean* distance is therefore the mean value of the PDF:

$$\lambda = \langle \ell \rangle = \int_0^\infty \ell f(\ell) d\ell = \int_0^\infty \ell \rho \sigma e^{-\rho \sigma \ell} d\ell \quad (7.4.6)$$

We recognize the integral as having the form $\int_0^\infty a x e^{-ax} dx$, which is evaluated by parts:

$$\int_0^\infty a x e^{-ax} dx = \left. \frac{a}{-a} x e^{-ax} \right|_0^\infty + \frac{a}{a} \int_0^\infty e^{-ax} dx = \frac{1}{a} \quad (7.4.7)$$

Thus,

$$\boxed{\lambda} = \int_0^\infty \ell \rho \sigma e^{-\rho \sigma \ell} d\ell = \frac{1}{\sigma \rho} = \boxed{\frac{1}{\rho \pi R^2}} \quad (7.4.8)$$

as desired. $\square$

An intuitive justification comes from considering a similar cylinder of volume $L\sigma$. The number of potentials is

$$N = \rho L \sigma \quad (7.4.9)$$

Thus, on average, the number of scattering events will be N . Therefore, for a mean distance without collisions λ , N will be 1:

$$\boxed{\lambda = \frac{1}{\rho \sigma}} \quad (7.4.10)$$

7.5 Examples

7.5.1 Hard Sphere

Consider a sphere of radius R such that the potential is infinite inside ($U(r \mid r < R) = \infty$) and zero outside ($U = 0$). Find (a) the total cross section and (b) the angular distribution of the scattering.

We begin by considering the allowed impact parameters b . If the free trajectory passes more than R from the center of the sphere, the trajectory will not be changed, and thus there will be no scattering. Thus,

$$\sigma = \pi b^2 \Big|_0^R = \boxed{\pi R^2} \quad (7.5.1)$$

Now we find the angular distribution. Intuition tells us that the distribution will be isotropic, such that $d\sigma/d\Omega$ is constant and does not depend on Ω or χ . Consider a trajectory as shown below:

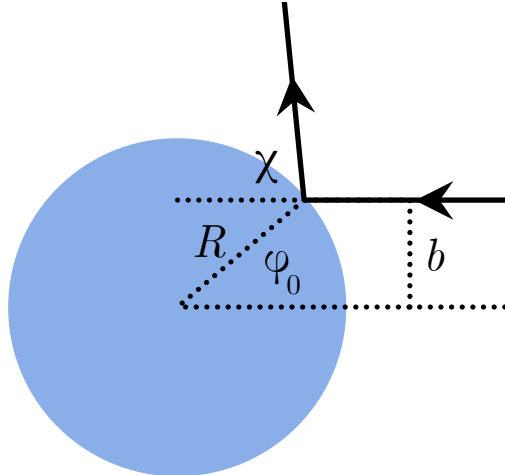


Figure 8. Hard-sphere scattering.

We recall that $\chi = \pi - 2\varphi_0$. From the sketch,

$$b = R \sin \varphi_0 \Rightarrow \varphi_0 = \frac{\pi - \chi}{2} \Rightarrow b(\chi) = R \sin \left(\frac{\pi - \chi}{2} \right) = R \cos \frac{\chi}{2} \quad (7.5.2)$$

We take the derivative:

$$\frac{db}{d\chi} = \frac{d}{d\chi} \left(R \cos \frac{\chi}{2} \right) = -\frac{R}{2} \sin \frac{\chi}{2} \quad (7.5.3)$$

which gives

$$d\sigma = \frac{b(\chi)}{\sin \chi} \left| \frac{db}{d\chi} \right| d\Omega = \frac{R \cos \chi/2}{\sin \chi} \frac{R}{2} \sin \frac{\chi}{2} d\Omega = \frac{R^2}{2} \frac{\cancel{\sin \chi}}{\cancel{2 \sin \chi}} d\Omega \quad (7.5.4)$$

which gives

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{R^2}{4}} \quad (7.5.5)$$

As expected, it is constant and thus the scattering is isotropic.

7.5.2 Hard Sphere and a Nontrivial Particle

Suppose that the scattered particle in fact has radius r (the sphere has radius R). This situation can be modeled easily as a trivial particle and a hard sphere with increased radius: $R' = r + R$. Thus, the cross section is

$$\boxed{\sigma = \pi(r + R)^2} \quad (7.5.6)$$

and the distribution is

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{(r + R)^2}{4}} \quad (7.5.7)$$

8 Rigid Body Motion

8.1 Angular Velocity

We begin with the definition of a rigid body.

Definition 8.1.1 (Rigid Body)

Let there be a collection of particles each of mass dm . These form a *rigid body* if the distances between the particles are fixed, that is,

$$|\mathbf{r}_i - \mathbf{r}_j| = \text{ct} \quad (8.1.1)$$

for any pair $\{i, j\}$ of points.

Given this constraint, a rigid body has 6 degrees of freedom: 3 translational coordinates, and 3 angles about three orthogonal axes. We consider two reference frames: an inertial one outside (in general) the body, and a frame centered on the body that moves with it.

We can decompose the motion of the particles in a body into translational motion with the center of mass, and a rotation about the center of mass:

$$d\mathbf{r} = d\mathbf{R} + d\boldsymbol{\phi} \times \mathbf{r} \quad (8.1.2)$$

such that

$$\mathbf{v} = \mathbf{V} + \boldsymbol{\Omega} \times \mathbf{r} \quad (8.1.3)$$

with $\mathbf{V}$ the velocity of the center of mass, $\boldsymbol{\Omega} \equiv d\boldsymbol{\phi}/dt$ the rate of change of angle with respect to the body frame, and $\mathbf{r}$ the distance of a particular point from the center of mass.

Proposition 8.1.2 (Invariance of Angular Velocity)

Let a body have angular velocity $\boldsymbol{\Omega}$ and translational velocity $\mathbf{V}$. After a Galilean transformation from this reference frame to another such that the new translational velocity is $\mathbf{V}'$, the angular velocity is invariant.

8.2 Inertia Tensor

Let us write the kinetic energy of a rigid body.

$$T = \sum \frac{dm v_i^2}{2} = \sum \frac{dm}{2} (\mathbf{V} + \boldsymbol{\Omega} \times \mathbf{r})^2 = \sum \frac{dm}{2} V^2 + \sum dm \mathbf{V} \cdot \boldsymbol{\Omega} \times \mathbf{r} + \sum \frac{dm}{2} (\boldsymbol{\Omega} \times \mathbf{r})^2 \quad (8.2.1)$$

The second term rearranges so that it becomes $\mathbf{V} \times \boldsymbol{\Omega} \sum dm \mathbf{r}$. We can choose the origin such that $\sum dm \mathbf{r} = \mathbf{0}$. Thus,

$$T = \frac{\mu V^2}{2} + \frac{1}{2} \sum dm (\Omega^2 r^2 - (\boldsymbol{\Omega} \cdot \mathbf{r})^2) \quad (8.2.2)$$

The second term represents rotational kinetic energy. We define the *inertia tensor* by rearranging T_{rot} :

$$T_{\text{rot}} = \frac{1}{2} \sum dm (\Omega_i^2 x_i^2 - \Omega_i x_i \Omega_k x_k) \quad (8.2.3)$$

$$= \frac{1}{2} \sum dm (\Omega_i \Omega_k \delta_{ik} x_\ell^2 - \Omega_i \Omega_k x_i x_k) \quad (8.2.4)$$

$$= \frac{1}{2} \Omega_i \Omega_k \sum dm (x_\ell^2 \delta_{ik} - x_i x_k) \quad (8.2.5)$$

where the inertia tensor $\mathbf{I}$ is

$$\mathbf{I}: I_{ik} = \sum dm (x_\ell^2 \delta_{ik} - x_i x_k) \quad (8.2.6)$$

which gives

$$T = \frac{\mu V^2}{2} + \frac{I_{ik} \Omega_i \Omega_k}{2} \quad (8.2.7)$$

In a Cartesian system, we can write the tensor explicitly:

$$\mathbf{I} = \begin{bmatrix} \sum dm(y^2 + z^2) & -\sum dmx y & -\sum dmx z \\ -\sum dmy x & \sum m(x^2 + z^2) & -\sum dmy z \\ -\sum dmz x & -\sum dmz y & \sum dm(x^2 + y^2) \end{bmatrix}_{\{\hat{x}, \hat{y}, \hat{z}\}} \quad (8.2.8)$$

This tensor being symmetric, it is always possible to diagonalize it.

Definition 8.2.1 (Principal Axes of a Body)

The choice of axes such that the tensor is diagonalized are called the *principal axes* of the body. The tensor takes the form

$$\mathbf{I} = \begin{bmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{bmatrix} \quad (8.2.9)$$

Proposition 8.2.2 (Additivity of the Inertia Tensor)

The inertia tensor is **additive**: inertia tensors add when we consider two bodies (one simply needs a shared origin).

If we have decomposed the angular velocity along the principal axes, we can write

$$T_{\text{rot}} = \frac{1}{2}(I_1 \Omega_1 + I_2 \Omega_2 + I_3 \Omega_3) \quad (8.2.10)$$

8.3 Angular Momentum

We define it:

Definition 8.3.1 (Angular Momentum of a Rigid Body)

The angular momentum of a body is given by

$$\mathbf{M} = \sum m \mathbf{r} \times (\boldsymbol{\Omega} \times \mathbf{r}) = \sum m(r^2 \boldsymbol{\Omega} - \mathbf{r}(\mathbf{r} \cdot \boldsymbol{\Omega})) \quad (8.3.1)$$

Proposition 8.3.2 (Angular Momentum of a Rigid Body with the Inertia Tensor)

We can write the vector $\mathbf{M}$ as

$$\mathbf{M} = \mathbf{I} \boldsymbol{\Omega} \quad (8.3.2)$$

Proof

We write the definition in tensor form:

$$\mathbf{M} = \sum m(r^2 \boldsymbol{\Omega} - \mathbf{r}(\mathbf{r} \cdot \boldsymbol{\Omega})) \Rightarrow M_i = \sum m(x_\ell^2 \Omega_i - x_i x_k \Omega_k) = \Omega_k \sum m(x_\ell^2 \delta_{ik} - x_i x_k) = \Omega_k I_{ik} \quad (8.3.3)$$

which indeed gives the desired equation. $\square$

Of course, if the angular velocity happens to lie along one of the principal axes of the body, we have

$$\mathbf{M} \parallel \boldsymbol{\Omega} \quad (8.3.4)$$

which is the trivial case of a point moving in a plane. In the case where $\mathbf{M} \not\parallel \boldsymbol{\Omega}$, we can observe *precession*, given by

$$\Omega_{\text{prec}} = \frac{M}{I_1} \quad (8.3.5)$$

for a body mainly rotating about I_3 and with $\mathbf{M}$ in the plane defined by the first and third principal axes.

8.4 Stability

We can therefore write conservation of energy and conservation of angular momentum. Take a rigid body with principal moments of inertia I_1, I_2, I_3 . Let the components of the angular momentum vector along the principal axes be M_1, M_2, M_3 . Thus,

$$E = \frac{M_1^2}{2I_1} + \frac{M_2^2}{2I_2} + \frac{M_3^2}{2I_3} \quad (8.4.1)$$

$$M = M_1^2 + M_2^2 + M_3^2 \quad (8.4.2)$$

The components M_i must satisfy these two equations, which are an ellipsoid and a sphere, respectively.

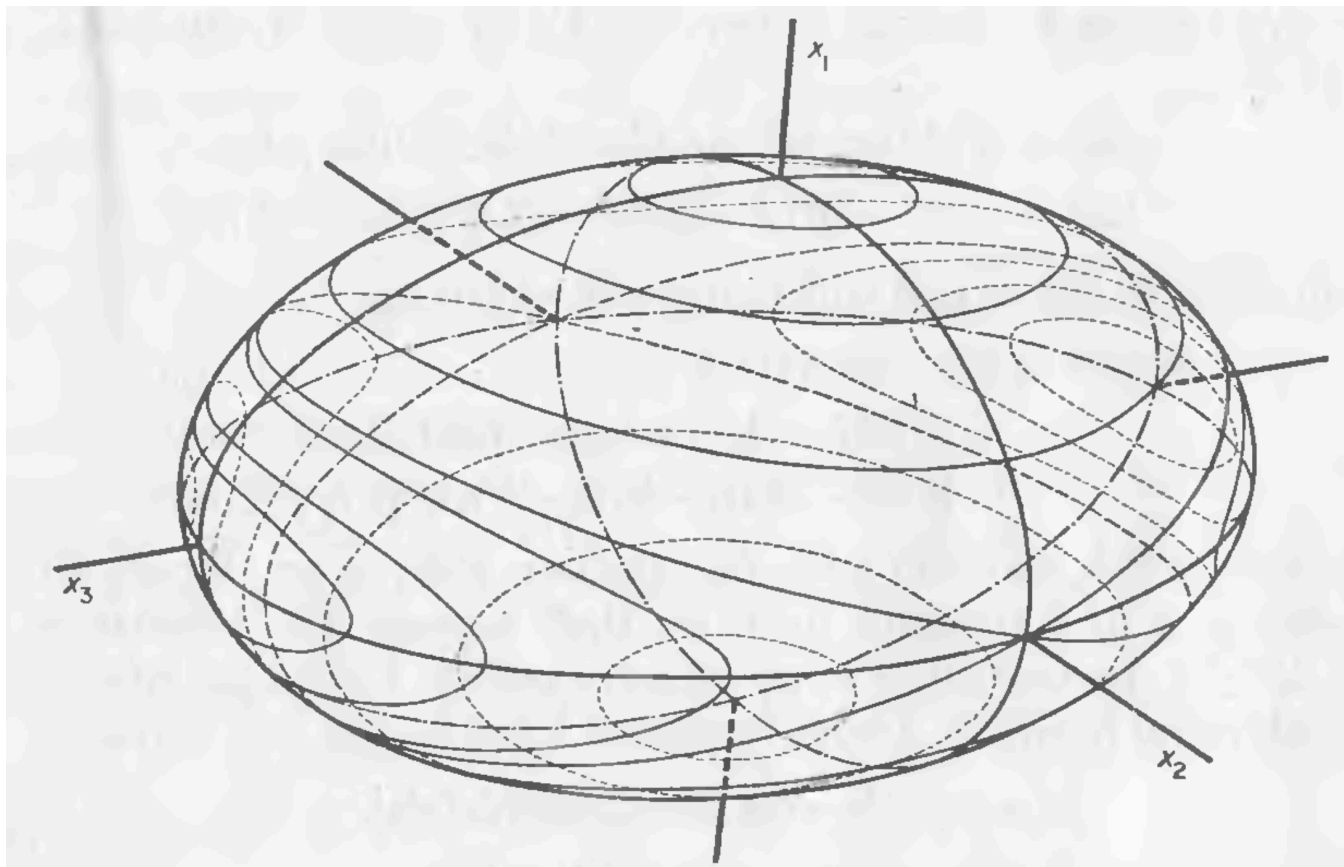


Figure 9. Possible Angular Momenta of an Asymmetric Rigid Body, for $I_1 < I_2 < I_3$.

We see that a perturbation about the x_1 and x_3 axes allows a small precession about that axis, and thus they are called “stable.” By contrast, about the I_2 axis, no such precession is allowed, due to the geometry of the intersection of a sphere with an ellipsoid. Thus, that axis is called “unstable.” We also observe that if the rigid body is slightly elastic, the only stable principal axis will be the one with the largest moment of inertia, because it is the state of lowest energy.

8.5 Equations of Motion

We have six degrees of freedom, so one equation comes from the time derivative of the body’s momentum:

$$\frac{d}{dt}(\mu \mathbf{V}) = \dot{\mathbf{p}} = \mathbf{F} = \int d\mathbf{f} \quad (8.5.1)$$

which is a sum over all the small forces $d\mathbf{f}$ on each element of the body. We also have the time derivative of the angular momentum $\mathbf{M}$:

$$\dot{\mathbf{M}} = \mathbf{K} \equiv \int \mathbf{r} \times d\mathbf{f} \quad (8.5.2)$$

The term $\mathbf{r} \times \mathbf{f}$ is called the “moment” of the force $\mathbf{f}$, and thus $\mathbf{K}$ is the “torque” (also called moment more generally) on the body.

Let $\mathbf{R}$ be the position of the body’s center of mass. Given a potential U , we have

$$\mathbf{F} = -\frac{\partial U}{\partial \mathbf{R}} \quad ; \quad \mathbf{K} = -\frac{\partial U}{\partial \boldsymbol{\phi}} \quad (8.5.3)$$

8.6 Euler Angles

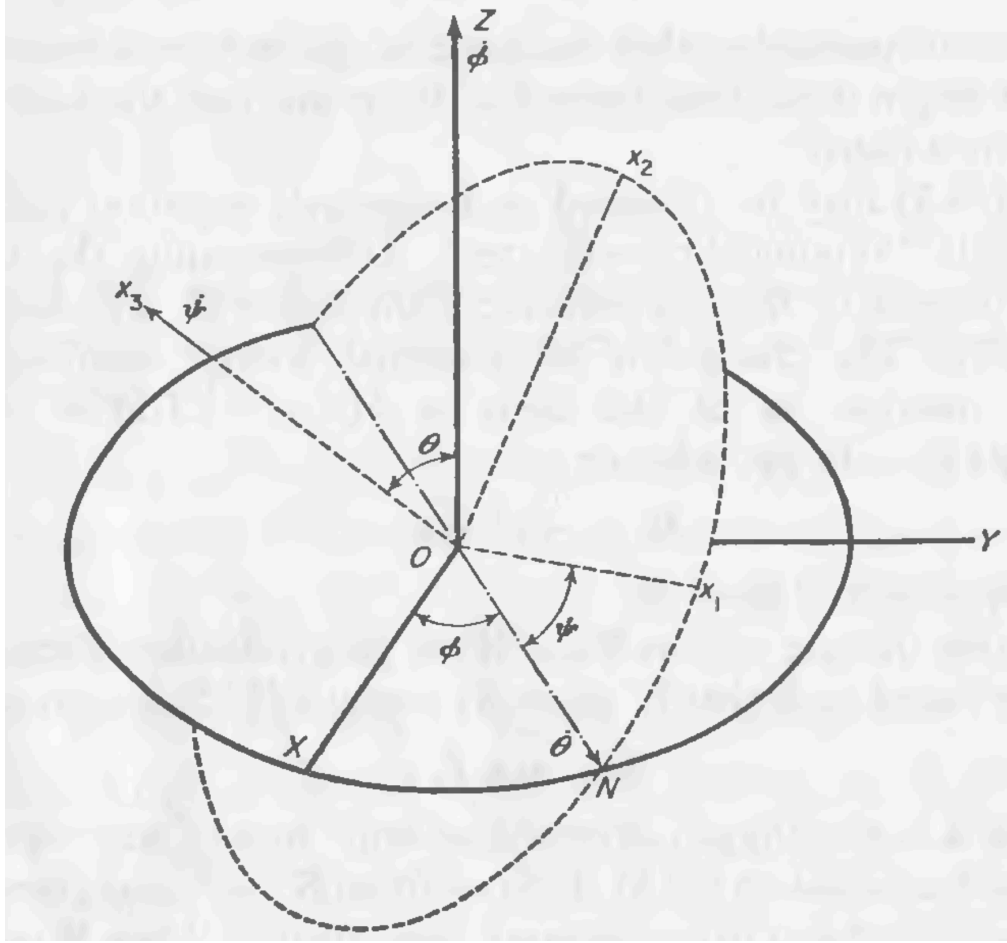


Figure 10. Euler Angles.

We define three angles that represent the orientation of the body. The angle θ represents the angle between the global z axis and the body’s x_3 axis. The angle ϕ represents the angle between the intersection of the x_1x_2 plane and the xy plane relative to the x axis. The angle ψ represents the angle between the x_1 axis and the intersection of x_1x_2 with xz .

Expressing $\boldsymbol{\Omega}$ along x_1 to x_3 , we have

$$\begin{cases} \Omega_1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi \\ \Omega_2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi \\ \Omega_3 = \dot{\phi} \cos \theta + \dot{\psi} \end{cases} \quad (8.6.1)$$

If x_1 to x_3 are the principal axes of the body, the kinetic energy is read off immediately.

8.7 Euler’s Equations

Let $\mathbf{A}$ be an arbitrary vector. Let $d\mathbf{A}/dt$ be its time derivative with respect to the fixed reference frame. If the vector $\mathbf{A}$ is invariant in the body frame, it can be expressed as

$$\frac{d\mathbf{A}}{dt} = \boldsymbol{\Omega} \times \mathbf{A} \quad (8.7.1)$$

See Section §9 in LL. Thus,

$$\frac{d\mathbf{A}}{dt} = \frac{d'\mathbf{A}}{dt} + \boldsymbol{\Omega} \times \mathbf{A} \quad (8.7.2)$$

where d' denotes differentiation in the body frame. This gives

$$\frac{d'\mathbf{P}}{dt} + \boldsymbol{\Omega} \times \mathbf{P} = \mathbf{F} \quad ; \quad \frac{d'\mathbf{M}}{dt} + \boldsymbol{\Omega} \times \mathbf{M} = \mathbf{K} \quad (8.7.3)$$

We write these in components:

$$\begin{cases} F_1 = \mu \left(\frac{dV_1}{dt} + \Omega_2 V_3 - \Omega_3 V_2 \right) \\ F_2 = \mu \left(\frac{dV_2}{dt} + \Omega_3 V_1 - \Omega_1 V_3 \right) \\ F_3 = \mu \left(\frac{dV_3}{dt} + \Omega_1 V_2 - \Omega_2 V_1 \right) \end{cases} \quad (8.7.4)$$

For the torque:

$$\begin{cases} K_1 = I_1 \dot{\Omega}_1 + (I_3 - I_2) \Omega_2 \Omega_3 \\ K_2 = I_2 \dot{\Omega}_2 + (I_1 - I_3) \Omega_3 \Omega_1 \\ K_3 = I_3 \dot{\Omega}_3 + (I_2 - I_1) \Omega_1 \Omega_2 \end{cases} \quad (8.7.5)$$

In free rotation, with $\mathbf{K} = 0$,

$$\begin{cases} 0 = \dot{\Omega}_1 + (I_3 - I_2) \Omega_2 \Omega_3 / I_1 \\ 0 = \dot{\Omega}_2 + (I_1 - I_3) \Omega_3 \Omega_1 / I_2 \\ 0 = \dot{\Omega}_3 + (I_2 - I_1) \Omega_1 \Omega_2 / I_3 \end{cases} \quad (8.7.6)$$

Suppose we rotate about the principal axis x_2 . We can find a second-order equation:

$$\ddot{\Omega}_2 + \Omega_2 \Omega_1^2 \left(\frac{(I_1 - I_3)(I_1 - I_2)}{I_2 I_3} \right) = 0 \quad (8.7.7)$$

This equation describes stable motion if $(I_1 - I_3)(I_1 - I_2) > 0$ and unstable motion if $(I_1 - I_3)(I_1 - I_2) < 0$.

9 Mathematical Tools

9.1 Calculus of Variations

We will look in detail at the mathematics behind the derivation of the Euler–Lagrange equation.

Definition 9.1.1 (Functional)

Let $f(t)$ be a function $\mathbb{R} \rightarrow \mathbb{R}$. Let $J[f]$ be a relation that returns a scalar given a function f . J is thus called a “functional.”

The functionals of interest are usually definite integrals:

$$J[f] = \int_{t_1}^{t_2} g(t, f) dt \quad (9.1.1)$$

In the case of the action of a one-dimensional system, we have

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad (9.1.2)$$

We want to compute the *variation of the action*. Take a function $\eta(t)$ such that $\eta(t_1) = \eta(t_2) = 0$. The *variation* of q is given by

$$\delta q = \epsilon \eta(t) \quad (9.1.3)$$

for ϵ an arbitrary small constant. Define a function of ϵ :

$$\Phi(\epsilon) = S[q + \epsilon \eta] \quad (9.1.4)$$

Thus,

$$\frac{\partial \Phi}{\partial \epsilon} = \int_{t_1}^{t_2} \frac{d}{d\epsilon} (L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t)) dt \quad (9.1.5)$$

The total derivative is

$$\frac{d}{d\epsilon} (L(q', \dot{q}', t)) = \frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} + \frac{\partial L}{\partial \epsilon} \quad (9.1.6)$$

Thus,

$$\frac{\partial \Phi}{\partial \epsilon} = \int_{t_1}^{t_2} \frac{\partial L}{\partial q} \eta + \frac{\partial L}{\partial \dot{q}} \dot{\eta} dt = \int_{t_1}^{t_2} \eta \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) dt + \left(\eta \frac{\partial L}{\partial \dot{q}} \right) \Big|_{t_1}^{t_2} \quad (9.1.7)$$

If we want $\frac{\partial \Phi}{\partial \epsilon} = 0$ for any η , we must have

$$\boxed{\frac{\partial L}{\partial q} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}}} \quad (9.1.8)$$

as desired.

9.2 Examples, Applications

We can apply this calculus of variations to prove a fairly obvious but nevertheless nontrivial fact.

Proposition 9.2.1 (Linearity of the Shortest Path)

Let there be two points in space A and B at points $\mathbf{r}_A$ and $\mathbf{r}_B$. The shortest path between these points is the straight line defined as passing through A and B .

Proof

Consider a plane containing A and B as the xy plane. We can therefore describe possible paths as functions $y = f(x)$ such that $y_A = f(x_A)$ and $y_B = f(x_B)$. The differential path length dl traversed by the function f is

$$dl = dx\sqrt{1 + (f')^2} \quad (9.2.1)$$

Define the functional $S[f]$ such that

$$S[f] = \int_{x_A}^{x_B} \sqrt{1 + (f')^2} \, dx \quad (9.2.2)$$

We want to minimize this functional. Thus, we seek functions f that satisfy

$$\Phi(\epsilon) \equiv S[f + \epsilon\eta] : \Phi'(0) = 0 \quad (9.2.3)$$

for $\eta(x_A) = \eta(x_B) = 0$. We know that this constraint leads to the Euler–Lagrange equation:

$$\frac{\partial}{\partial f} \left(\sqrt{1 + (f')^2} \right) = \frac{d}{dt} \frac{\partial}{\partial f'} \left(\sqrt{1 + (f')^2} \right) \quad (9.2.4)$$

We see that the left-hand side is zero because there is no dependence on f . The right-hand side becomes

$$0 = \frac{d}{dt} \left(\frac{2f'}{2\sqrt{1 + (f')^2}} \right) \quad (9.2.5)$$

We can therefore argue that this quantity has a constant value—let us call it k . Thus,

$$\frac{f'}{\sqrt{1 + (f')^2}} = k \Rightarrow f' = k\sqrt{1 + (f')^2} \Rightarrow (f')^2 = k^2(1 + (f')^2) \quad (9.2.6)$$

Thus,

$$(f')^2 = \frac{k^2}{1 - k^2} \Rightarrow f' = \sqrt{\frac{k^2}{1 - k^2}} = ct \quad (9.2.7)$$

which is also constant. Thus, the slope $f'(x)$ is constant, and therefore f is a straight line. Because we constrained f such that it must pass through A and B , we have proved the desired fact. $\square$

9.3 Elliptic Integrals

9.3.1 Example

Suppose that the equation of motion of a system is given by

$$\ddot{x} = -A \sin x \quad (9.3.1)$$

Multiply by $\dot{x}$

$$\ddot{x}\dot{x} = -A\dot{x} \sin x \quad (9.3.2)$$

Integrate in time:

$$\frac{\dot{x}^2}{2} - A \cos x = ct \quad (9.3.3)$$

Adding a constant allows the expression to remain constant:

$$\frac{\dot{x}^2}{2} + A(1 - \cos x) = ct \quad (9.3.4)$$

We observe that this is the energy of the system. We are interested in the “libration” regime (also called the oscillatory one). Thus, the maximum value of the energy is found for $\dot{x} = 0$ and $x = x_0$:

$$\frac{\dot{x}^2}{2} + A(1 - \cos x) = A(1 - \cos x_0) \Rightarrow \dot{x}^2 = 2A(\cos x - \cos x_0) \quad (9.3.5)$$

We recall that $\cos x = 1 - 2\sin^2(x/2)$. Thus,

$$\dot{x}^2 = 4A \left(\sin^2 \frac{x_0}{2} - \sin^2 \frac{x}{2} \right) \quad (9.3.6)$$

We define the value of the elliptic modulus, or eccentricity, k , such that

$$k = \sin \frac{x_0}{2} \in (0, 1) \quad (9.3.7)$$

Thus,

$$\dot{x}^2 = 4A \left(k^2 - \sin^2 \frac{x}{2} \right) \quad (9.3.8)$$

Let s be shorthand for $\sin x/2$. It follows that

$$\dot{s} = \frac{1}{2} \cos \frac{x}{2} = \frac{1}{2} \sqrt{1 - s^2} \dot{x} \quad (9.3.9)$$

which leads to

$$\dot{x} = \frac{2\dot{s}}{\sqrt{1 - s^2}} \quad (9.3.10)$$

This gives

$$\frac{4\dot{s}^2}{1 - s^2} = 4A(k^2 - s^2) \Rightarrow \dot{s}^2 = A(1 - s^2)(k^2 - s^2) \quad (9.3.11)$$

We separate variables:

$$\dot{s} = \sqrt{A(1 - s^2)(k^2 - s^2)} \Rightarrow \frac{ds}{\sqrt{(1 - s^2)(k^2 - s^2)}} = \sqrt{A} dt \quad (9.3.12)$$

To bring this expression into the standard elliptic form, one must perform a change of variables given by $s = k \sin \phi$ (which is equivalent to $\sin \phi = s/k$). Thus,

$$ds = k \cos \phi d\phi \quad (9.3.13)$$

which gives

$$1 - s^2 = 1 - k^2 \sin^2 \phi \quad ; \quad k^2 - s^2 = k^2(1 - \sin^2 \phi) = k^2 \cos^2 \phi \quad (9.3.14)$$

The product under the square root is

$$\sqrt{(1 - s^2)(k^2 - s^2)} = k \cos \phi \sqrt{1 - k^2 \sin^2 \phi} \quad (9.3.15)$$

Thus,

$$\frac{ds}{\sqrt{(1 - s^2)(k^2 - s^2)}} = \sqrt{A} dt \Rightarrow \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \sqrt{A} dt \quad (9.3.16)$$

Integrating gives

$$\int^{\phi(t)} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \sqrt{A}(t - t_0) \quad (9.3.17)$$

The left-hand side is an *incomplete elliptic integral of the first kind*, denoted $F(\phi(t), k)$. Thus,

$$F(\phi(t), k) = \sqrt{A}(t - t_0) \quad (9.3.18)$$

The *Jacobi amplitude function*, written $\text{am}(u, k)$, is the inverse of F with respect to ϕ , such that

$$u = F(\phi, k) \iff \phi = \text{am}(u, k) \quad (9.3.19)$$

This gives

$$\phi(t) = \text{am}(\sqrt{A}(t - t_0), k) \quad (9.3.20)$$

We substitute $s = k \sin \phi$. The *Jacobi elliptic sine* is defined by

$$\text{sn}(u, k) \equiv \sin(\text{am}(u, k)) \quad (9.3.21)$$

Thus,

$$s(t) = k \sin \phi(t) = k \text{sn}(\sqrt{A}(t - t_0), k) \quad (9.3.22)$$

We can finally write the function in terms of t using $s = \sin(x/2)$:

$$\boxed{x(t) = 2 \arcsin(k \text{sn}(\sqrt{A}(t - t_0), k)) \quad ; \quad k = \sin \frac{x_0}{2} \in (0, 1)} \quad (9.3.23)$$

The period of this function is

$$\boxed{T = \frac{4}{\sqrt{A}} K(k)} \quad (9.3.24)$$

where $K(k) = F(\pi/2, k)$ is the complete elliptic integral of the first kind, because $\text{sn}(u, k)$ has a real period of $4K(k)$. The derivative $\dot{x}$ can be represented using the cn function:

$$\boxed{\dot{x} = 2k\sqrt{A} \text{cn}(\sqrt{A}(t - t_0), k)} \quad (9.3.25)$$

where $\text{cn}(u, k) = \cos(\text{am}(u, k))$.

We observe that in the limit $k \rightarrow 0$, which corresponds to $x_0 \rightarrow 2k$, the solution tends to that of the SHO:

$$x(t) = 2k \sin(\sqrt{A}(t - t_0)) = x_0 \sin(\omega t - \varphi) \quad (9.3.26)$$

with $\omega \equiv \sqrt{A}$, the frequency of the approximate SHO described by (9.3.1).

9.4 Linear Algebra

Certain identities are useful.

Proposition 9.4.1 (The Determinant and the Trace)

Let $\mathbf{A}$ be a square matrix.

$$\det |e^{\mathbf{A}t}| = e^{\text{tr}(\mathbf{A})t} \quad (9.4.1)$$

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